

## Moralizing Technology



# *Moralizing Technology*

Understanding and Designing the Morality of Things

PETER-PAUL VERBEEK

The University of Chicago Press  
Chicago and London

Peter-Paul Verbeek is professor of philosophy of technology at the University of Twente, the Netherlands, extraordinary professor (Socrates chair) of philosophy of human enhancement at Delft University of Technology, and chairman of the Young Academy, a division of the Royal Dutch Academy of Arts and Sciences. He is the author of *What Things Do: Philosophical Reflections on Technology, Agency, and Design*.

The University of Chicago Press, Chicago 60637  
The University of Chicago Press, Ltd., London  
© 2011 by The University of Chicago  
All rights reserved. Published 2011.  
Printed in the United States of America

This research was made possible by the Netherlands Organization for Scientific Research (NWO VENI grant, project “Technology and the Matter of Morality”).

20 19 18 17 16 15 14 13 12 11      1 2 3 4 5

ISBN-13: 978-0-226-85291-1 (cloth)  
ISBN-13: 978-0-226-85293-5 (paper)  
ISBN-10: 0-226-85291-1 (cloth)  
ISBN-10: 0-226-85293-8 (paper)

Library of Congress Cataloging-in-Publication Data  
Verbeek, Peter-Paul, 1970–

Moralizing technology : understanding and designing the morality of things /  
Peter-Paul Verbeek.  
p. cm.

Includes bibliographical references and index.

ISBN-13: 978-0-226-85291-1 (cloth : alk. paper)

ISBN-10: 0-226-85291-1 (cloth : alk. paper)

ISBN-13: 978-0-226-85293-5 (pbk. : alk. paper)

ISBN-10: 0-226-85293-8 (pbk. : alk. paper) 1. Technology—Moral and ethical aspects. I. Title.

T14.V473 2011

174'.96—dc22

2010052462

Ⓢ This paper meets the requirements of ANSI/NISO Z39.48-1992 (Permanence of Paper).

# *Contents*

*Preface* vii

|                                       |     |
|---------------------------------------|-----|
| 1 Mediated Morality                   | 1   |
| 2 A Nonhumanist Ethics of Technology  | 21  |
| 3 Do Artifacts Have Morality?         | 41  |
| 4 Technology and the Moral Subject    | 66  |
| 5 Morality in Design                  | 90  |
| 6 Moral Environments: An Application  | 120 |
| 7 Morality beyond Mediation           | 139 |
| 8 Conclusion: Accompanying Technology | 153 |

*Notes* 167

*References* 171

*Index* 179



## *Preface*

When my wife and I entered the ultrasound examination room several years ago, our eyes immediately fell on the artwork on the wall. It was a colorful and almost roguish serigraph depicting an owl, made by a Dutch artist who “suffers” from Down syndrome. Coincidentally, we had just bought a print of the same serigraph.

We were rather ambivalent about having this ultrasound scan made. While it was exciting that we would actually see our unborn, it also forced us to see the child as a potential patient about whose life we would have to make a decision if anything appeared to be wrong. And because we did not want to be charged with this responsibility, we decided we only wanted the scan to determine the age of the unborn, keeping out the available tests for Down syndrome and spina bifida. Seeing the serigraph, therefore, was a great reassurance. Because we happened to know the background of the artist, the presence of this particular artwork at this place clearly expressed that in this practice sonograms were not made to automatically stigmatize congenital “defects” as “abnormal” or even “undesirable.”

Indeed, we managed to keep the tests at bay. But nonetheless, the ultrasound scan had fundamentally shaped our experience of the unborn child. Even though no test was done, we could not help examining the facial expression of the woman making the sonogram, anxiously scanning for any signs that something was wrong. The mere availability of testing possibilities had made us feel responsible for *not* testing and accepting the “risks” connected with that. The decision not to be put in the position of having to make a decision appeared to be a decision as well. It was the owl image that permitted us to realize that the framework of choice and responsibility was not the only one available here and that we could frame the situation differently.

This experience echoed, in a remote way, a discussion in the philosophy and ethics of technology. In 1995, Dutch philosopher Hans Achterhuis published an article in which he made a plea for what he called “the moralization of devices.” Following upon Bruno Latour’s diagnosis that morality is not only to be found in humans but also in things, Achterhuis stated that human beings should stop permanently moralizing each other and start moralizing technology instead. If we all agree that it is better not to shower too long, to buy a ticket before we enter the metro, or to turn off the light when we leave specific places in our houses, why not delegate the responsibility for performing these actions to water-saving showerheads, turnstiles, and time switches? When the spirit is willing, technologies can be used to strengthen the flesh. Achterhuis’s ideas evoked a lot of discussion, with respect to both the possibilities and the desirability of a “material morality.” Critics—abusively, as I will demonstrate in this book—argued that such behavior-influencing technologies would threaten human freedom and dignity, resulting in moral laziness and a technocratic society.

When returning home from the ultrasound examination, I realized my wife and I had experienced an aspect of the moral significance of technology which had remained underdeveloped in this discussion about morality and technology. Even though the technology in the ultrasound practice clearly had moral significance, it did not directly steer our behavior. Rather, it helped to shape our experience of our unborn child and the interpretive frameworks that guided our actions and decisions. By establishing a very specific form of contact between the fetus and us, this technology had not simply granted us a “peek into the womb”; it had reorganized the relations between our unborn child and ourselves.

The moral significance of technology has been a fascinating field of inquiry for me ever since, both because of the philosophical challenges it generates and because of the potential to make analyses of this “material morality” fruitful for design practices. How to conceptualize the moral significance of things, when morality is usually seen as an exclusively human affair? How to understand the moral character of actions that are induced by technologies rather than autonomous decisions? And how to develop a framework that helps designers to deal with this morality of their designs in a responsible way?

Many people have been at my side while I was working on this study. First of all, I would like to thank Hans Achterhuis for the inspiration springing from both his work and his personality. The title of this book—*Moralizing Technology*—should be read as a tribute to his ideas on the “moralization

of devices” which were the sparks that fired my fascination with the moral dimensions of materiality. I would also like to thank Petran Kockelkoren and Steven Dorrestijn for all the engaged and congenial discussions about mediation, morality, art, and design, and for their critical reviews of earlier versions of this book. Steven’s idiosyncratic reading of Foucault’s ethical work was a major source of inspiration for me, as were Petran’s ideas about mediation and technology. Without Petra Bruulsema’s sensible look over my shoulder, in both my texts and my working activities, I doubt I would have ever finished this book. Finn Olesen was a great host and discussion partner when I was a guest professor at Aarhus University, where I wrote part of the book. Richard Heersmink was so kind as to go through the manuscript carefully and critically. Thank you all.



# Mediated Morality

## Introduction

Our daily lives have become intricately interwoven with technologies. Cars enable us to travel long distances, mobile phones help us to communicate, medical devices make it possible to detect and cure diseases.<sup>1</sup> Life has become unthinkable without sophisticated technology. Contrary to what many people intuitively think, these technologies are not simply neutral instruments that facilitate our existence. While fulfilling their function, technologies do much more: they give shape to what we do and how we experience the world. And in doing so they contribute actively to the ways we live our lives (cf. Verbeek 2005b).

Cars, for instance, do not only take us from A to B. They also lengthen the radius enclosing our most frequent social contacts. They help to determine how far we live from where we work. And they organize how we design cities and neighborhoods. Mobile phones make it easy to contact each other but also introduce new norms of contact and new styles of communication. By making it possible to detect specific diseases, medical diagnostic devices do not simply produce images of the body but also generate complicated responsibilities, especially in the case of antenatal diagnostics and in situations of unbearable and endless suffering.

This active contribution of technologies to our daily lives has an important moral dimension. First of all, the quality of their contributions to our existence can be assessed in moral terms. Some roles played by technology can be called “good” and other roles “bad”—even if it is not possible to blame technologies for the “bad.” And second, by helping to shape human actions and experiences, technologies also participate in our ways of doing ethics. Speed bumps, to use a favorite example of Bruno Latour, help us make the

moral decision not to drive too fast near a school. Ultrasound scans help us to ask and answer moral questions about the lives of unborn children. Energy-saving lightbulbs take over part of our environmental conscience. Coin locks on supermarket pushcarts remind us to return each cart neatly to its place (Akkerman 2002). Turnstiles tell us to buy a ticket before boarding a train (Achterhuis 1995). Current developments in information technology show this moral significance more explicitly. With the development of ambient intelligence and persuasive technology, technologies start to interfere openly with our behavior, interacting with people in sophisticated ways and subtly persuading them to change their behavior, as I will discuss extensively in the final chapter of this book.

Even though the fact usually remains unnoticed, technologies appear to have moral significance. Latour even states that those who complain about the alleged moral decay of our culture are simply looking in the wrong direction. Rather than looking only to humans, we should start to recognize that nonhuman entities are bursting with morality. This is a challenging observation. Mainstream ethical theory, after all, does not leave much room for such a moral dimension of material objects. Ethics is commonly considered to be an exclusively human affair. The claim that technological artifacts can have morality immediately raises the suspicion that one adheres to a backward form of animism, which equips things with spirit. Material objects do not have minds or consciousness, they lack free will and intentionality and cannot be held responsible for their actions; therefore they cannot be fully fledged parts of the moral community, the argument goes. At the same time, though, technologies *do* help to shape our existence and the moral decisions we take, which undeniably gives them a moral dimension. The time has come, therefore, to develop an ethical framework to conceptualize this moral relevance of technology. How can we do justice to the moral dimensions of material objects?

Further, addressing the moral significance of technology is not only a challenge for ethical theory. It also has important implications for *doing* ethics. Both the use and the design of technology involve ethical questions that are closely related to the moral character of technological artifacts. How can users deal with the ways in which technologies mediate moral decisions and help to attribute responsibilities and instill norms? How can designers anticipate the future moral roles of their designs, or even “build in” specific forms of morality? Is it desirable at all that designers get to play such a role? How can designers and users of technology bear moral responsibility for technologically mediated actions? What forms of moral discourse could accompany the use and design of moral technologies?

## Ethics and Technology

Technologies and ethics have always had a complicated relationship. While many technologies have obviously relieved humanity from misery and toil—like penicillin, agricultural equipment, surgical instruments, heating systems for buildings—many others have received negative evaluations. Nuclear weapons, for instance, have caused destruction and suffering to such a degree that it is hardly possible to see any beneficial aspects to them. Even the birth control pill, which is widely used and has played a tremendous role in the emancipation process—not only for women but also for gays and lesbians, because of its disconnection of sex and reproduction (cf. Mol 1997)—is still contested in some conservative religious circles because it interferes with the allegedly “natural” course of things.

In philosophy, various approaches to the ethics of technology have developed, which differ radically from each other. In its early days, ethical approaches to technology took the form of *critique* (cf. Swierstra 1997). Rather than addressing specific ethical problems related to actual technological developments, ethical reflection on technology consisted in criticizing the phenomenon of “Technology” itself. Classical approaches in the philosophy and ethics of technology were rooted in fear regarding the ongoing fusion of technology and culture and aimed to protect humanity from technology’s alienating powers. They saw the technologization of society as a threat to human authenticity and to the meaningfulness of reality. People would come to exist only as cogs in the machine of a technologized society, reduced to the function they have in the apparatus of mass production (cf. Jaspers 1951), while reality would have meaning only as a heap of raw materials available to the human will to power (cf. Heidegger 1977b). Technology was approached not in terms of specific artifacts that help to shape our everyday lives but as a monolithic phenomenon that is hostile to the human world.

Gradually, however, philosophers developed the field of “ethics of technology,” seeking increased understanding of and contact with actual technological practices and developments. Rather than placing itself outside or even against the realm of technology, ethics now came to address actual ethical problems related to technology. Applied subfields emerged, like biomedical ethics, ethics of information technology, and ethics of nanotechnology. Those who work in these subfields investigate specific moral problems that are connected to the design, use, and social impact of technologies. Moreover, ethics became more interested in the *process* of technology development. Subfields like engineering ethics and ethics of design came into being, explicitly directed at the practice of technology development. Over the past

decades, applied ethics has seen an explosion of journals directed at specific domains of technology, ranging from ethics of information technology to “nano-ethics” and from bioethics to engineering ethics.

There are good arguments, though, that the current connection between ethics and technology does not yet go far enough. Paradoxical as it may seem, many ethical approaches to technology still have too little contact with technology itself and its social and cultural roles. Quite often the ethics of technology takes a position toward technology that is just as externalist as that of the early critique of technology. At the basis of both approaches is a radical separation between the realms of technology and society. Engineering ethics, for example, focuses strongly on issues of safety and risk: the realm of society needs to be protected against the risks generated in the realm of technology, and engineers have to blow the whistle when they discover immoral practices or negative consequences of specific innovations. Often-cited case studies concern the roles of engineers in the development of the exploding space shuttle *Challenger* and the Ford Pinto with a gas tank that ruptured in collisions at 25 mph (Birsch and Fielder 1994). Much of computer ethics, to give another example, focuses on issues of privacy, also approaching technology as a potential intruder in the realm of human beings. Technologies are approached here in a merely instrumentalist way: they fulfill a function, and if they fail to do this in a morally acceptable way, the whistle should be blown. The central focus of ethics is to make sure that technology does not have detrimental effects in the human realm and that human beings control the technological realm in morally justifiable ways.

What remains out of sight in this externalist approach is the fundamental intertwining of these two domains. The two simply cannot be separated. Humans are technological beings, just as technologies are social entities. Technologies, after all, play a constitutive role in our daily lives. They help to shape our actions and experiences, they inform our moral decisions, and they affect the quality of our lives. When technologies are used, they inevitably help to shape the context in which they function. They help specific relations between human beings and reality to come about and co-shape new practices and ways of living. To use the example of the cell phone again: this is not just a functional instrument that helps us to talk to other people wherever we are and wherever they are. Once they fulfill this function, cell phones directly help to generate new ways of communicating and interacting. They create new ways of dealing with appointments; long-term planning becomes less necessary if everybody can be reached everywhere anytime. They generate new styles of communication, especially through texting functionality, which even gave rise to a new “language” (Crystal 2008). And they help to redefine

the boundary between public and private by inviting people to have private conversations in public, because the presence of the person with whom one is communicating appears to be nearer than the presence of the persons in one's immediate environment.

The moral relevance of technology is closely related to this active contribution of technologies to human practices and experiences. On the one hand, a concrete instance of technological mediation can be assessed in moral terms: it can be morally good or bad. Langdon Winner's analysis of some low-hanging overpasses on parkways in Long Island (New York) giving access to the beach is a good example here. Architect Robert Moses deliberately built these overpasses so low that buses cannot use the parkways, implicitly limiting access to the beach for African Americans who could not afford cars of their own. On the other hand, the phenomenon of technological mediation lays bare how technologies also contribute to the moral actions and decisions of human beings. Technologies contribute actively to how humans do ethics. A good example here is genetic diagnostic tests for hereditary forms of breast cancer. Such tests focus on mutations in the breast cancer genes BRCA1 and BRCA2, which can predict the probability that somebody will develop this form of cancer. Carriers of such mutations (mostly women, but men can also develop breast cancer) are presented with the choice to do nothing and run a high risk of developing breast cancer; to undergo regular testing so that cancer can be detected at an early stage; or to have a preventive double mastectomy (cf. Boenink 2007).

The discovery of such mutations, therefore, transforms healthy people into potential patients. Moreover, this form of genetic testing translates a congenital defect into a preventable form of suffering; by choosing to have your breasts amputated, you can prevent any development of breast cancer. When this technology is used, therefore, it organizes a situation of choice. This choice is complicated, because it involves a new category that is introduced by this new technology: between health and illness, genetic testing introduces the area of being "not-yet-ill." The very fact that this technology makes it possible to *know* that it is very likely that a person will become ill, added to the possibility of preventively removing organs, makes this person responsible for his or her own disease. Thus the technology of genetic testing creates a moral dilemma and also suggests ways to deal with this dilemma.

This example shows that medical technologies can mediate the moral decisions that both medical doctors and patients make, by organizing situations of choice and suggesting the choice that should be made. Such technological mediations have at least as much ethical relevance as preventing disasters or finding responsible ways to deal with risks. By mediating our actions and

experiences, technologies help to shape the quality of our lives and of our moral actions and decisions. To deal adequately with the moral relevance of technology, therefore, the ethics of technology should incorporate the phenomenon of technological mediation.

This requires that ethical theory broaden its scope. Rather than approaching ethics and technology as belonging to two radically separate domains, one human and the other nonhuman, we should keep the interwoven character of the two spheres at the center (cf. Latour 1993). It is a mistake to locate ethics exclusively in the “social” realm of the human and technology exclusively in the “material” realm of the nonhuman. Technologies are social too, if only because they contribute to moral decisions—and human beings belong to the material realm too, since our lives are shaped in close interactions with the technologies we are using. Only by crossing the divide between these spheres can the ethical dimensions and relevance of technology be understood.

Crossing this divide is not an easy task, though. Taking seriously the moral relevance of technological artifacts requires that ethical theory move beyond its classical assumption that morality necessarily is a solely human affair, because technologies lack consciousness, rationality, freedom, and intentionality. How can we morally assess the impact of technologies on the quality of our lives? And how can we do justice to the manifold ways in which technological artifacts actively mediate moral practices and decisions?

### Technological Mediation

In order to understand and analyze the moral significance of technologies, we need to first get a clearer picture of the mediating roles that technologies play in our daily lives. During recent decades, philosophy of technology has increasingly paid attention to the impact of technological artifacts on the life-world of human beings (Borgmann 1984; Winner 1986; Ihde 1990; Ihde 1993; Ihde 1998; Latour 1992b; Latour 1999). As opposed to classical approaches, which were mainly focused on understanding the conditions of “Technology” taken as a monolithic phenomenon, the philosophy of technology has started to approach technology in terms of the actual material objects that help to shape human actions and experiences.

Various authors have analyzed specific aspects of the social and cultural roles of technologies. The work of the North American philosopher Don Ihde, for example, focuses on the perceptual and hermeneutic implications of technology by analyzing how specific perceptual technologies help to shape how reality can be experienced and interpreted. To mention a few other contemporary philosophers of technology: the German American phi-

philosopher Albert Borgmann analyzes how use of technological devices affects the quality of human engagement with reality; the French philosopher and anthropologist Bruno Latour has studied the hybrid character of human-technology associations and their implications for understanding society; and the US political philosopher Langdon Winner has investigated the political relevance of technological artifacts.

As I have explained elsewhere (Verbeek 2005b), the positions that have developed can be augmented and integrated into a philosophy of “technological mediation.” The philosophical analysis of technological mediation—particularly the “postphenomenological” approach in this field—will prove to be an important key to understanding the moral significance of technology. For that reason, it merits a separate introduction here.<sup>2</sup>

#### HUMAN-TECHNOLOGY RELATIONS

Phenomenology—in my elementary definition—is the philosophical analysis of the structure of the relations between human beings and their lifeworld. From such a perspective, the central idea in the philosophy of mediation is that technologies play an actively mediating role in the relationship between human beings and reality. Technological mediation can be studied without reverting to the classical fear that technology will determine society, but also without marginalizing the role of technology to mere instrumentality. Rather, it focuses on the mutual shaping of technology and society.

A good starting point for understanding technological mediation is Martin Heidegger’s classical analysis of the role of tools in the everyday relation between humans and their world. According to Heidegger (1927), tools should be understood as “connections” or “linkages” between humans and reality. Heidegger indicates the way in which tools are present to human beings when they are used as “readiness-to-hand.” Tools that are used for doing something typically withdraw from people’s attention; for example, the attention of a person who hammers a nail into a wall is not directed at the hammer but at the nail. People’s involvement with reality takes place *through* the ready-to-hand artifact. Only when it breaks down does it require attention for itself again. The artifact is then, in Heidegger’s words, “present-at-hand” and is no longer able to facilitate a relationship between a user and his or her world.

Even though ready-to-hand artifacts recede from people’s attention, they do play a constitutive role in the human-world relations that arise around them. When a technological artifact is used, it facilitates people’s involvement with reality, and in doing so it coshapes how humans can be present in their world and their world for them. In this sense, things-in-use can be

understood as *mediators* of human-world relationships. Technological artifacts are not neutral intermediaries but actively coshape people's being in the world: their perceptions and actions, experience and existence.

Don Ihde and Bruno Latour offer concepts for gaining a closer understanding of this mediating role of technologies. In order to develop this understanding, I have distinguished between two perspectives of mediation: one that focuses on perception and another one on praxis. Each of these perspectives approaches the human-world relationship from a different side. The hermeneutic or "experience-oriented" perspective starts from the side of the world and directs itself at the ways reality can be interpreted and be present for people. The main category here is *perception*. The pragmatic or "praxis-oriented" perspective approaches human-world relations from the human side. Its central question is how human beings act in their world and shape their existence. The main category here is *action*.

#### MEDIATION OF EXPERIENCE

The central hermeneutic question for a "philosophy of mediation" is how artifacts mediate human experiences and interpretations of reality. Ihde's philosophy of technology is a good starting point for answering this question, because of its focus on the technological mediation of perception. Ihde elaborates Heidegger's tool analysis into analysis of the relationships between humans and technological artifacts (Ihde 1990). He discerns several relationships human beings can have with technologies; two of these can be considered relations of mediation.<sup>3</sup>

First, Ihde discerns the "embodiment relation," which is his equivalent to Heidegger's "readiness-to-hand." In the embodiment relation, technologies are "incorporated" by their users, establishing a relationship between humans and their world *through* the technological artifact. This embodiment relation occurs, for instance, when one is looking through a pair of glasses; the artifact is not perceived itself, but it helps to perceive the environment. Technological artifacts become extensions of the human body here, as it were. Second, Ihde discerns the "hermeneutic relation." In this relation, technologies provide access to reality not because they are "incorporated," but because they provide a representation of reality, which requires interpretation (hence the name "*hermeneutic relation*"—hermeneutics being the study of interpretation). A thermometer, for instance, establishes a relationship between humans and reality in terms of temperature. Reading a thermometer does not result in a direct sensation of heat or cold but gives a value that requires interpretation in order to tell something about reality.

Ihde shows that technologies, when mediating our sensory relationship with reality, transform what we perceive. According to Ihde, the transformation of perception always has a structure of amplification and reduction. Mediating technologies amplify specific aspects of reality while reducing other aspects. When one is looking at a tree through an infrared camera, for instance, most aspects of the tree that are visible to the naked eye get lost, but at the same time a new aspect of the tree becomes visible: one can now see whether it is healthy or not. Ihde calls this transforming capacity of technology “technological intentionality”: technologies have “intentions,” and thus they are not neutral instruments but play an active role in the relationship between humans and their world.

These intentionalities are not fixed properties of artifacts, however: they obtain their shape within the relationship humans have with these artifacts. Within different relationships technologies can have different “identities.” The telephone and the typewriter, for instance, were developed not as communication and writing technologies but as equipment to help the blind and the hard of hearing hear and write. In their use context they were interpreted quite differently, however. Ihde calls this phenomenon *multistability*: a technology can have several “stabilities,” depending on the way it is embedded in a use context. Technological intentionalities, therefore, are always dependent on the specific stabilities that come about.

Ihde’s analysis of the transformation of perception has important hermeneutic implications. In fact, it shows that mediating artifacts help to determine how reality can be present for and interpreted by people. Technologies help to shape what counts as “real.” This hermeneutic role of things has important ethical consequences, since it implies that technologies can actively contribute to the moral decisions human beings make. Medical imaging technologies like MRI (magnetic resonance imaging) and ultrasound are good examples of this. Such technologies make visible aspects or parts of the human body, or of a living fetus in the womb, which cannot be seen without them. But the specific way in which these technologies represent what they “see” helps to shape how the body or a fetus is perceived and interpreted, and what decisions are made. In this way, technologies fundamentally shape people’s experience of disease, pregnancy, or their unborn child.

#### MEDIATION OF PRAXIS

Within the praxis perspective, the central question is how artifacts mediate people’s actions and the way they live their lives. While perception, from a phenomenological point of view, consists in the way the world is present for

humans, praxis can be seen as the way humans are present in their world. The work of Latour offers many interesting concepts for analyzing how artifacts mediate action (e.g., Latour 1992b; 1994). Latour points out that what humans do is in many cases coshaped by the things they use. Actions are the results not only of individual intentions and the social structures in which human beings find themselves (the classical agency-structure dichotomy) but also of people's material environment. The concept introduced by Latour and Akrich to describe the influence of artifacts on human actions is "script." Like the script of a movie or a theater play, artifacts prescribe how their users are to act when they use them. A speed bump, for instance, has the script "Slow down when you approach me," a plastic coffee cup "Throw me away after use."

This influence of artifacts on human actions has a specific character. When scripts are at work, things mediate action as material things, not as immaterial signs. A traffic sign makes people slow down because of what it signifies, not because of its material presence in the relation between humans and world. And we discard a plastic coffee cup not because its user's manual tells us to do so but because it simply is physically not able to withstand being cleaned several times. The influence of technological artifacts on human actions can be of a nonlingual kind. Artifacts are able to exert influence *as material things*, not only as *signs* or *carriers of meaning*.

As is the case with perception, in the mediation of action *transformations* occur. Following Latour, within the domain of action these transformations can be indicated as "translations" of "programs of action." Latour attributes programs of actions to all entities—human and nonhuman. When an entity enters a relationship with another entity, the original programs of action of both are translated into a new one. When somebody's action program is to "prepare meals quickly," and this program is added to that of a microwave oven ("quickly heat small portions of food"), the action program of the resulting, "composite" actor might be "regularly eat instant meals individually."

In the translation of action, a similar structure can be discerned as in the transformation of perception. Just as in the mediation of perception some aspects of reality are amplified and others are reduced, in the mediation of action one could say that specific actions are "invited" while others are "inhibited." The scripts of artifacts suggest specific actions and discourage others. This invitation-inhibition structure is context dependent, just like the amplification-reduction structure of perception; Ihde's concept of multi-stability also applies within the context of the mediation of action. The telephone, for instance, has had a major influence on the separation of our geographical and social contexts by making it possible to maintain social re-

TABLE 1. Experience and praxis

| <i>Experience</i>                         | <i>Praxis</i>             |
|-------------------------------------------|---------------------------|
| Mediation of perception                   | Mediation of action       |
| Technological intentionality              | Script                    |
| Transformation of perception              | Translation of action     |
| Amplification and reduction               | Invitation and inhibition |
| Delegation: <i>deliberate inscription</i> |                           |
| Multistability: <i>context-dependence</i> |                           |

lationships outside our immediate living environment. But it could have this influence only because it is used as a communication technology, not as the hearing aid it was originally supposed to be.

An important difference with respect to the mediation of perception, however, is the nature of the human-technology relations from which mediations of actions arise. Artifacts mediate action not only from a ready-to-hand position but also from being present-at-hand. A gun, to mention an unpleasant example, mediates action from a ready-to-hand position, translating “express my anger” or “take revenge” into “kill that person” (cf. Latour 1999). A speed bump, however, cannot be “embodied.” It will never be ready-to-hand; it exerts influence on people’s actions from a present-at-hand position.

Together, the concepts used to understand the role of technologies in the relation between humans and reality form a “vocabulary for technological mediation,” which helps to make visible the active role of technologies in their use contexts. Technological artifacts mediate perception by means of technological intentionalities: their “directedness” in organizing perception. They mediate action by means of scripts, which prescribe how to act when using the artifact. Technological mediation is context-dependent, and always entails a translation of action and a transformation of perception. The translation of action has a structure of invitation and inhibition; the transformation of perception a structure of amplification and reduction. Table 1 summarizes this vocabulary.

#### MEDIATION AND MORALITY

The philosophy of mediation usually takes a *descriptive* point of view. Until now, its main ambition has been to *analyze* the role of technology in the lifeworld. The time is ripe, however, to augment this descriptivist orientation—which is characteristic of many contemporary approaches within the

philosophy of technology (cf. Light and Roberts 2000)—with a *normative* approach. The mediating role of technologies, after all, can have a distinctly moral dimension. By helping to shape our practices and the interpretations on the basis of which we make decisions, technologies can play an explicit and active role in our moral actions.

As I will elaborate in chapter 3, the question of the moral significance of technological artifacts is not entirely new. Actually, it has been playing a role on the backbenches of the philosophy of technology for quite some time now. Langdon Winner's example of the bridges in New York dates from 1980. Six years later, Bruno Latour argued that artifacts are bearers of morality, as they help people to make all kinds of moral decisions. In 1988 he delivered a lecture in the Netherlands, "Safety Belt: The Missing Masses of Morality," in which he said it is about time that we stop complaining about the alleged moral decline of our society. Such lamentations show a lack of understanding of our daily world. Morality should not be looked for only among humans but also among things, Latour claims. Once we are able to see the moral charge of matter, we see a society that is swarming with morality.

Many cars, for instance, will not start or will produce an irritating sound until the driver is wearing his or her seatbelt. And the moral decision of how fast one drives is often delegated to speed bumps in the road with the script "Slow down before reaching me." According to Latour, such cars and bumps embody morality. Designers delegated to them the responsibility of seeing to it that drivers wear their safety belts and do not drive too fast. Moral decisions are often not made exclusively by human beings but are shaped in interaction with the technologies they use (Latour 1988; 1992b).

Analogously to Winner's claim that artifacts have politics, therefore, it is worth investigating to what extent artifacts have morality, given their active role in moral action and decision making. If ethics is about the question of "how to act" and technologies help to answer this question, technologies appear to have moral significance; at least they help us to do ethics. This is quite a radical step, though. A few centuries ago the Enlightenment, with Kant as its major representative, brought about a turnover hitherto unequaled in ethics by moving the source of morality from God to humans. Do contemporary analyses of the social and cultural role of technology now urge us to move the source of morality one place further along—considering morality not a solely human affair but also a matter of *things*?

Such a question challenges ethical theory. After all, how should we understand such a material form of morality? Is the conclusion that things mediate human actions reason sufficient to lead us to actually consider technologies to be moral agents, and if so, to what extent? In ethical theory, to qualify as a

moral agent requires at least the possession of intentionality and some degree of freedom. In order to be held morally accountable for an action, an agent needs to have the intention to act in a specific way and the freedom to realize this intention. Both requirements seem problematic with respect to artifacts, which, lacking a mind, do not have intentionality, let alone any form of autonomy.

Moreover, within the predominant ethical frameworks it is difficult not only to assign moral agency to inanimate objects but also to consider behavior resulting from technological mediation “moral actions.” After all, to what extent can these actions be considered *moral actions* when humans make certain moral decisions because technology influences them to do so? Steered behavior is different from moral action. Further, to what extent does it make sense to attribute moral responsibility to artifacts when a morally wrong situation occurs as a result of technological mediation?

The ethics of technology, therefore, seems to find itself in a paradoxical situation. If it holds on to a strictly humanist interpretation of intentionality, it fails to take into account the moral relevance of technological artifacts. And if it adheres to predominant conceptions in which moral agency requires a high degree of autonomy, there can be no such a thing as an “ethics of technology.” Such an ethics could then exist only if technologies were neutral instruments, not mediating human actions and interpretations—which would throw out the baby with the bathwater, because it would imply a denial of the phenomenon of technological mediation and its moral implications altogether. At the same time, an ethical theory that aims to take seriously the notion of technological mediation and the active moral role of things cannot entirely reject the notions of intentionality and autonomy either, since some degree of human intentionality and autonomy is needed to maintain the idea of responsibility.

In order to find a way out of this deadlock, I will defend the thesis that ethics should be approached as a matter of human-technological associations. When taking the notion of technological mediation seriously, claiming that technologies are human agents would be as inadequate as claiming that ethics is a solely human affair. The isolation of human subjects from material objects, which keeps us from approaching ethics as a hybrid rather than a human affair, is deeply entrenched in our metaphysical schemes (cf. Latour 1993). According to this metaphysical scheme, human beings are active and intentional while material objects are passive and instrumental. Human behavior can be assessed in moral terms—good or bad—but a technological artifact can be assessed only in terms of its functionality (functioning well or poorly).

If the ethics of technology is to take seriously the mediating roles of technology in society and in people's everyday lives, it must move beyond the modernist subject-object dichotomy that forms its metaphysical roots. Rather than separating or purifying "humans and nonhumans"—concepts I gratefully borrow from Latour—the ethics of technology needs to hybridize them. In this book I will elaborate a "postphenomenological" way to do this, building upon Don Ihde's philosophy of technology, Bruno Latour's Actor-Network Theory, and Michel Foucault's work on power and ethics.

### Postphenomenology

In recent decades the philosophy of technological mediation, which I sketched above, has been an important construction site for a new branch of phenomenology. Primarily inspired by the work of Ihde, phenomenological philosophy of technology broke away from its one-dimensional opposition to science and technology as second-order and alienating ways to relate to reality (Ihde 1990). By developing analyses of the structure of the relations between humans and technologies, and by investigating the actual roles of technologies in human experience and existence, phenomenology came to analyze technology as a constitutive part of the lifeworld rather than a threat to it. The new phenomenological approach that came into being often calls itself "postphenomenological," because of its opposition to some aspects of "classical" phenomenology, as I will elaborate below.

Postphenomenology aims to revive the phenomenological tradition in a way that overcomes the problems of classical phenomenology. These problems mainly concern what Ihde calls its "foundational" character (Ihde 1998, 113–26). Classical phenomenology explicitly defined itself as an alternative to science. As opposed to the scientific goal to *analyze* reality, phenomenology aimed to *describe* it (Merleau-Ponty 1962, viii–x). This claim to provide a "more authentic" way of accessing reality has become highly problematic in light of developments in twentieth-century philosophy—extensive analyses of the mediated character and contextuality of such claims.

The fact that classical phenomenology failed to take the locality and context dependence of human knowledge into account is understandable when the context in which it developed is taken into account (cf. Verbeek 2005b, 106–8). Phenomenology presented itself as a philosophical method that sought to describe "reality itself," since it opposed itself to the positivist worldview arising from modern natural science, which claims to describe reality as it actually is. But beside developing an alternative route to "authentic reality"—claiming to describe, not analyze, reality—classical phenomenol-

ogy actually started to develop highly interesting accounts of the relations between humans and reality. Maurice Merleau-Ponty analyzed this relation primarily in terms of perception, Edmund Husserl in terms of consciousness, and Martin Heidegger in terms of being-in-the-world. It is therefore more in accordance with the actual history of phenomenology to see phenomenology as a philosophical movement that seeks to analyze *the relations between human beings and their world* rather than as a *method* for describing reality.

Redefining phenomenology along these lines, Ihde developed a “nonfoundational” phenomenological approach which he calls “postphenomenological.” Ihde maintains the central phenomenological idea that human-world relations need to be understood in terms of “intentionality,” the directedness of human beings toward their world. As we saw above, however, Ihde shows that in our technological culture this intentionality relation is most often technologically mediated. Virtually all human perceptions and actions are mediated by technological devices, ranging from eyeglasses and television sets to cell phones and automobiles. And these technological mediations do not so much take us to “the things themselves” that classical phenomenology was longing for as help to construct what is real to us. Many mediated perceptions, after all, do not have counterparts in everyday reality. Radio telescopes, for instance, detect forms of radiation that are invisible to the human eye and need to be “translated” by the device before astronomers can perceive and interpret it. There is no “original” perception here that is mediated by a device; the mediated perception itself is the “original.” Phenomenological investigations of this type of mediation cannot possibly aim to return to “the things themselves” but rather aim to clarify the structure of technological mediation and its hermeneutic implications.

The postphenomenological approach makes it possible to move beyond the modernist subject-object dichotomy in two distinct ways. First of all, Ihde shows the necessity of thinking in terms of *human-technology associations* rather than approaching human subjects and technological objects as separate entities. If the fundamental intertwinement of humans and technologies is not taken into account, the relations between human beings and reality cannot be understood. Second, human-world relationships should not be seen as relations between preexisting subjects who perceive and act upon a preexisting world of objects, but rather as sites where both the objectivity of the world and the subjectivity of those who are experiencing it and existing in it are *constituted* (Verbeek 2005b, 111–13). What the world “is” and what subjects “are” arise from the interplay between humans and reality; the world that humans experience is “interpreted reality,” and human existence is “situated subjectivity.” Postphenomenology closes the gap between subject

and object not by linking subject and object via the bridge of intentionality but by claiming that they actually constitute each other. In the mutual relation between humans and reality a specific “objectivity” of the world arises, as well as a specific “subjectivity” of human beings.

This focus on the mediating role of technology in the constitution of subjectivity and objectivity makes postphenomenology directly relevant to an ethical approach of technological artifacts. By investigating how technological mediations help to constitute specific realities and specific subjectivities, postphenomenology is the approach par excellence by which to analyze the moral relevance of technology. A good example here, which I will elaborate more extensively further on in this book, is obstetric ultrasound. This technology is not simply a functional means to make visible an unborn child in the womb. It actively helps to shape the way the unborn child is humanly experienced, and in doing so it informs the choices his or her expecting parents make. Because of its ability to make visible the fetus in terms of medical norms, for instance, it constitutes the fetus as a possible patient and, in some cases, its parents as makers of decisions about the life of their unborn child.

In this way postphenomenology moves beyond the predominating modernist understanding of the relations between subjects and objects in ethics, in which subjects are active and intentional and objects are passive and mute. It shows not only that human intentionalities can be operative “through” technologies but also that in many cases “intentionality” needs to be located in human-technology associations—and therefore partly in artifacts as well—and the resulting intentionality cannot always be reduced to what was explicitly delegated to the technology by its designers or users. Moreover, the postphenomenological approach shows that we cannot hold on to the autonomy of the human subject as a prerequisite for moral agency; rather, we need to replace the “prime mover” status of the human subject with technologically mediated intentions. In our technological culture, humans and technologies do not have separate existences anymore but help to shape each other in myriad ways.

This hybrid character of humans and technologies does not easily fit our conceptual frameworks. As Aaron Smith states, the lack of a human prime mover makes it difficult to attribute responsibility for actions that occur (Smith 2003). But rather than accepting his conclusion that “when we look to very complicated situations the human prime mover is concealed and difficult to find, but it is always there” (Smith 2003, 193), I contend that hanging on to the prime-mover status of human beings fails to take seriously the moral importance of technology. As the ultrasound case will show, moral intentions come about on the basis of technological mediations of the relations

between humans and reality, and are always properties of human-technology *associations* rather than of “prime movers.” Adequate moral reflection about technology, therefore, requires us to broaden the perspective of ethical theory and the ethics of technology. We need to investigate how to rethink the status of both objects and subjects in moral theory in order to do justice to the hybrid character of human-technology associations.

For rethinking the status of the object in moral theory, the work of Latour will be an important starting point. His work, like phenomenology and postphenomenology, explicitly aims to think in a amodern way, moving beyond the subject-object distinction. Latour wanted to make visible nonhuman forms of agency and to clarify the moral roles of technological artifacts. The work of Michel Foucault will subsequently play a crucial role in helping us rethink the status of the subject in moral theory. Foucault developed an ethical approach in which the concept of *subject constitution* is central: ethics, for him, is ultimately about the question what kind of subject we want to be. Moreover, Foucault does not approach the subject as an autonomous being but as a product of power relations and of influences exerted upon it, with which it explicitly develops a free relation. From the postphenomenological approach, technological mediation can be seen as an important source of subject constitution, and this makes it possible to apply Foucault’s ethical approach directly to technology—focusing on the central question of what kind of mediated moral subjects we aspire to be.

### Outline of the Book

This book investigates the moral dimensions of technologies along several lines. Chapter 2 will set out the contours of the approach I will follow in order to “moralize technology.” By analyzing the example of obstetric ultrasound and its moral implications, I will argue that a nonhumanist approach is needed in ethics in order to do justice to the moral dimensions of objects. The humanist focus of mainstream ethics makes it virtually impossible to attribute a more-than-instrumental role to technologies, while the example of obstetric ultrasound makes clear that technologies do play an active role in moral decision making. In critical discussion with the positions of Peter Sloterdijk, Martin Heidegger, and Bruno Latour, I will articulate a amodern perspective on ethics in which moral agency becomes a matter of human-technology hybrids rather than an exclusively human affair.

Chapter 3 will deal with the status of the object in ethical theory. First, I will discuss existing accounts of the moral relevance of technological artifacts, ranging from authors like Langdon Winner and Luciano Floridi to

Bruno Latour and Albert Borgmann. After this, I will discuss the possibility of analyzing technologies in terms of moral agency. What does technological mediation imply for the conceptualization of moral agency? Can and should artifacts be seen as moral agents? And if so, how? In the past millennia, the community of moral agents has been expanded more than once; after including slaves and women, we are now faced with the question whether material things should be included too (Swierstra 1999; Verbeek 2006e). In order to answer this question, I will analyze several positions in ethical theory with regard to their implicit and explicit definitions of and requirements for moral agency. After this, I will develop a notion of moral agency that does include material entities and at the same time recognizes and articulates the differences between human and nonhuman elements of moral agency.

After this, chapter 4 will discuss the status of the subject in ethical theory. As we saw, within the predominant ethical frameworks it is difficult not only to assign moral agency to inanimate objects but also to consider behavior resulting from technological mediation as “moral actions.” Therefore it is necessary to articulate a definition of moral agency that not only includes objects but also does justice to the mediated character of actions and decisions of the subject. An analysis of the relations between the early and the late work of Foucault will form the backbone of my approach here. Foucault’s early work focuses on the forces and structures that shape the subject. In his analysis of the panopticon in *Surveiller et punir*, Foucault shows that human intentions are not “authentic” but result from structures of power that can also be present materially. The later Foucault, however, started elaborating a new perspective on ethics. Here he investigated how, amid these structures of power, human beings can constitute themselves as (moral) subjects (Foucault [1984] 1990; [1984] 1992). Humans are not only the objects of power but also subjects that create their own existence against the background of and in confrontation with these structures.

This shift makes Foucault’s work highly important for the ethics of technology, because it makes it possible to articulate a redefinition of ethics beyond the concept of the autonomous moral agent. Foucault’s late work permits us to redefine the concept of autonomy in such a way that it is in line with the phenomenon of technological mediation. Therefore I will investigate to what extent Foucault’s analysis of subject constitution and his association with classical Greek ethics could form the basis for a new ethical framework that is compatible with technological mediation.

Chapter 5 inventories the implications of the moral significance of technology for the ethics of design and the responsibility of designers. By designing technologies that will inevitably play mediating roles in the lives of

their users, engineers implicitly “materialize morality.” Technology design appears to be “ethics by other means”—to give yet another variation on Von Clausewitz’s famous dictum.<sup>4</sup> I will investigate how the ethics of design can be augmented so that this implicit moral decision making by engineers could happen in a more explicit way. This approach would overcome the predominant focus within engineering ethics on disaster cases that could have been prevented by engineers’ responsible behavior (whistleblowing). By incorporating the notion of mediation, designers can anticipate the moral dimensions of their design or even explicitly design morality into technology. The chapter will discuss and bring together several methods and approaches to anticipate, assess, and design the morality of technologies.

Chapter 6 applies the perspectives, concepts, and approaches developed in the book to the emerging field of ambient intelligence and persuasive technology. The miniaturization of electronic devices and new possibilities for wireless communication between appliances have made it possible to develop so-called *smart environments*. Such environments register what is happening around them and are able to react to this in intelligent ways—hence the term *ambient intelligence*. Persuasive technologies add to this intelligence the ability to explicitly influence the behavior of users in specific directions, effectively persuading people to behave differently. These technologies explicitly embody many of the themes elaborated in this book: the moral significance of technology, the mediated character of morality, and the hybrid character of agency and morality. This will give me a chance, after discussing these technologies in more detail, to “apply” the theory I have developed, focusing on the ethical aspects of designing such technologies. How can the perspective developed in this book help us to understand the moral dimension of technologies? And what does the perspective imply for the work of designers?

The seventh chapter reflects on the outcomes and the reach of the analysis of moral mediation developed in this book. In order to do justice to the moral significance of technologies, the book has developed a nonhumanist ethical approach in which morality is not an exclusively human affair but a matter of human-technology associations. Yet current technological developments seem to take one step further toward blurring the boundaries between the human and the technological. Brain implants, tissue engineering, and genetic modification actually have the potential to change human nature. The ambition of self-constitution, the focus of chapter 4, here takes a radical shape. Rather than taking us beyond humanism as too narrow an approach for understanding morality, these technologies seem to take us beyond the human being itself. Chapter 7 takes up several human-technology relations that go

beyond the concept of mediation and explores their implications for the relations between technology and morality. In this way, both the potential and the limitations of the approach of moral mediation become better visible.

Chapter 8, to conclude, places the approach of moral mediation in the broader context of the philosophy and ethics of technology. First, it discusses the “ethics of the good life” as a framework that could inform moral decisions about the design, implementation, and use of technologies. Second, it argues that an adequate elaboration of this ethics of the good life requires that the philosophy of technology move beyond the two turns it has made over the past decades. After the empirical turn and the ethical turn in philosophy of technology, we need one more turn, one that integrates moral reflection with close empirical research into actual technological developments. This way the ethics of technology can develop from an externally oriented form of “technology assessment” into a form of “technology accompaniment” that raises and answers ethical questions in close relation to actual technologies and technological developments.

## A Nonhumanist Ethics of Technology

### Introduction

Ever since the Enlightenment, ethics has had a humanist character. Not “the good life” but the individual person, taken as the fountainhead of moral decisions and practices, now holds the central place in ethical reflection.<sup>1</sup> Yet however much our high-technological culture is a product of the Enlightenment, this culture reveals the limits of the Enlightenment in ever more compelling ways. Not only have the ideals of manipulability and the positivist slant of Enlightenment thinking been mitigated substantially during the past decades, but so has the humanist position that originated from it. The world in which we live, after all, is increasingly populated not only by human beings but also by technological artifacts that help to shape the ways we live. Technologies have come to mediate human practices and experiences in myriad ways.

As stated in chapter 1, this technologically mediated character of our daily lives has important ethical implications. Ethics is about the questions of “how to act” and “how to live”—and in our technological culture, these questions are not answered exclusively by human beings. By helping to shape the experiences and practices of human beings, technologies also provide answers to the central ethical questions, albeit in a material way. Artifacts are morally charged; they mediate moral decisions, shape moral subjects, and play an important role in moral agency.

A good example of such a “morally charged” technology—which will function as a connecting thread through this book—is obstetric ultrasound. This technology has come to play a pervasive role in practices around pregnancy, especially in antenatal diagnostics and, consequently, in moral decisions regarding abortion. Decisions about abortion, after an ultrasound scan (and subsequent amniocentesis) have shown that the unborn child is suffering

from a serious disease, are not taken autonomously by human beings—as fountainheads of morality—but in close interaction with these technologies that open up specific interpretations and actions and generate specific situations of choice.

Because of the humanist orientation of the established frameworks of ethical theory, this moral role of technology is hard to conceptualize. Human agency is approached as an exclusively human affair: since it requires intentions and freedom, objects are not seen as having moral relevance, let alone moral agency. Moreover, such ethicists argue, human behavior that is steered or provoked by technology cannot be called “moral action.” In order to do justice to the moral relevance of technology, therefore, the humanist foundations of ethics need to be broadened. To be sure, humanism as an ideological movement has brought forth a set of values whose importance cannot be overestimated. My focus here is on the *metaphysics* of humanism, which is rooted in modernism, and its radical separation of the human subject and nonhuman objects. Analyzing the metaphysical basis of mainstream ethics will make room for an alternative approach that is needed to think about, assess, and help shape the moral relevance of technological artifacts.

In their modernity critiques, authors like Latour (1993) and Heidegger ([1947] 1976; 1977a) elaborated the thesis that this rigid separation makes it virtually impossible to see the many ways in which subjects and objects are actually interwoven. And taking into account this interwoven character is crucial to understanding our technological culture, in which human decisions and practices are increasingly shaped in interaction with technologies. Against the modern, Enlightened image of the autonomous moral subject, therefore, I will articulate an amodern, heteronomous moral subject whose actions are always closely interwoven with the material environment in which they play out.

In order to do this, I will discuss a critique of humanism that has caused a great deal of controversy: Peter Sloterdijk’s “Rules for the Human Zoo” (“Regeln für den Menschenpark,” 1999). Sloterdijk’s text is a reply to Heidegger’s “Letter on Humanism” (Heidegger [1947] 1976), which was addressed to the Frenchman Jean Beaufret. Beaufret had asked Heidegger to clarify the relations between his philosophy and existentialism, which was rapidly gaining importance and which Sartre had declared a form of humanism. Heidegger, however, did not take the side of Sartre—which could have helped him in his process of rehabilitation and de-Nazification (cf. Safranski 1999)—but instead distanced himself radically from humanism, which for him was a too narrowly modernist approach to humanity. In “Rules for the Human Zoo”

Sloterdijk takes up this critique of humanism and radicalizes it in such a way that, fifty years after Heidegger's text, he came to be associated with the same fascism that Heidegger could not shake off.

This chapter can be read as a reply to Sloterdijk's "response to the Letter on Humanism." In order to clear the path, I will first investigate the humanist character of contemporary ethics and its supporting modernist ontology. Second, I will elaborate the moral relevance of nonhuman reality by discussing the mediating role of technology in moral practices and decisions. After this, I will critically engage with Sloterdijk's "posthumanist" position. I will dispel all associations with fascism from his approach, while using his critique of humanism as a basis for an amodern approach to ethics that does justice to nonhuman forms of morality and the ways humans have to deal with them.

### The Moral Significance of Obstetric Ultrasound

Elaborating a concrete case can make the ethical relevance of technology and the need for a nonhumanist ethical approach more clearly evident. The case I will elaborate here is obstetric ultrasound. I will analyze in what respects the roles played by this technology transcend the mere functionality of making visible an unborn child in the womb. Ultrasound might seem a rather innocuous medical technology. Expecting couples generally like to have a sonogram done, because it is an exciting form of contact with the unborn child in the body of its mother. But even though it may be a "noninvasive" technology in a physical sense, ultrasound is far from noninvasive in a moral sense.

In the Netherlands, pregnant couples are offered two routine ultrasound scans, one around the twelfth weeks of pregnancy and a second at about twenty weeks. The aim of the first scan is to determine the age of the fetus and the duration of pregnancy—but also to calculate the risk that the child will suffer from Down syndrome. This risk is calculated on the basis of the degree of nuchal translucency, which indicates the thickness of the nape in the neck of the fetus, most often in combination with a blood test. The aim of the second scan is to carefully examine the whole body of the unborn child in order to detect possible defects. It is done at twenty weeks because at this time it can reveal more defects than the earlier scan and because abortion in the Netherlands is legal—under strict conditions—until the twenty-fourth week. The examination can reveal a variety of defects, ranging from specific heart conditions to a harelip.

Postphenomenologically speaking, ultrasound constitutes the unborn in a very specific way: it helps to shape how the unborn can be perceptually present, and how it can be interpreted on the basis of the specific ways it is represented. In Ihde's terms, a sonogram establishes a hermeneutic relation between the unborn and the people watching it. In hermeneutic relations, technologies produce a representation of reality, which needs to be interpreted by its "readers." Moreover, the technology itself embodies a "material interpretation" of reality, because it has to make a "translation" of what it "perceives" into a specific representation—in this case, the scanner has to make a relevant translation of reflected ultrasonic sound waves into a picture on a screen.

This implies that a sonogram does not provide a neutral "window to the womb"—though a well-known pro-life movie that makes intensive use of ultrasound imaging (cf. Boucher 2004) takes this phrase as its title—but actively mediates how the unborn is experienced. The particular mediation brought about by ultrasound imaging has a number of characteristics. Some of these are directly related to how the unborn is represented on the screen; others have to do with the organization of this visual contact with the unborn and the context in which the unborn can be made present. In all cases, the unborn is constituted in a specific way and so are its parents in their relation to it.

#### THE FETUS AS A PERSON

First of all, the image on the screen has a specific *size*, and even though this representation suggests a high degree of realism, its size does not coincide with the size of the unborn in the womb. A fetus eleven weeks old measures about 8.5 cm and weighs 30 grams, but its representation on the screen makes it appear to have the size of a newborn baby (cf. Boucher 2004, 12). A number of techniques are used to construct a realistic image of the unborn. Further, a sonogram depicts the unborn independently from the body of its mother. As Margarete Sandelowski (1994, 240) put it: "The fetal sonogram depicts the fetus as if it were floating free in space: as if it were already delivered from or outside its mother's body." Ultrasound isolates the unborn from her or his mother.

All of these technological mediations generate a new ontological status for the fetus. Ultrasound imaging constitutes the fetus as an *individual person*; it is made present as a separate living being rather than forming a unity with its mother, in whose body it is growing. Obstetric ultrasound thus contributes

to the coming about of what has been called “fetal personhood”: the unborn is increasingly approached as a person (Mitchell 2001, 118; Boucher 2004, 13) or even as an unborn “baby” (Sandelowski 1994, 231; Zechmeister 2001, 393–95). This experience of fetal personhood is enhanced by the possibility of seeing the gender of the unborn: by its ability to reveal the genitals, ultrasound genders the unborn. The expectant parents, as a result, can begin to call the unborn by its name. It is not surprising, then, that a print of the first sonogram is often included in the baby album as “baby’s first picture”—as in the title of Lisa Mitchell’s 2001 book on obstetric ultrasound.

#### THE FETUS AS A PATIENT

Ultrasound constitutes the fetus not only as a person but also as a *patient*. An important goal of ultrasound screening is to detect abnormalities. In an early stage of pregnancy, ultrasound can be used for determining the risk of Down syndrome; in a later stage it can be used to detect a variety of other conditions. For these purposes, ultrasound scanners are equipped with sophisticated software that helps obstetricians to quantify the body of the unborn in various ways. These measurements help to determine the duration of pregnancy but also the risk of specific diseases. Ultrasound imaging lets the unborn be present in terms of medical variables and of the risks of suffering from specific diseases (cf. Landsman 1998).

In translating the unborn into a possible patient, ultrasound makes pregnancy into a medical condition that needs to be monitored and that requires professional health care. Moreover, ultrasound translates “congenital defects” into preventable forms of suffering. As a result, pregnancy becomes a process of choice: the choice to have tests like neck fold measurements done at all, and the choice of what to do if anything is “wrong.” The detection of a defect with the help of ultrasound translates “expecting a child” into “choosing a child”—or choosing to terminate the pregnancy.

In fact, the very possibility of having sonograms made, and therefore of detecting congenital defects before birth, irreversibly changes the character of what used to be called “expecting a child.” It inevitably becomes a matter of choice now: the choice not to have an ultrasound scan made is also a choice, a very deliberate one in a society in which the norm is to have these scans done, based on the predominant assumption that *not* scanning for diseases is irresponsible, because then you deliberately run the risk of having a disabled or sick child, causing suffering both for the child and for yourself and your family.

## RELATIONS BETWEEN UNBORN AND PARENTS

This isolation of the unborn from its mother creates a new relationship between them. On the one hand, the mother is now deprived of her special relation to the unborn, with the privilege of having knowledge about the unborn shifting to healthcare professionals (Sandelowski 1994, 231, 239). But on the other hand, these detaching effects have their counterpart in an increased bonding among mother, father, and unborn. Ultrasound can give expectant parents assurance of the baby's health and the feeling of being closer and more attached to the unborn (Zechmeister 2001, 389). This visual nearness to the unborn is used in pro-life campaigns to support the claim that abortion involves murdering a vulnerable person (Boucher 2004).

Another effect of this separation of mother and unborn is that the mother is increasingly seen as the *environment* in which the unborn is living, rather than forming a unity with it. And when the fetus is constituted as a vulnerable subject, its environment may potentially be harmful. This opens the way for using ultrasound screening as a form of surveillance, monitoring the lifestyle and habits of expecting women in order to enhance the safety of the unborn. Rather than an intimate place to grow, the womb now becomes a potentially hostile environment that needs to be guarded (Oaks 2000; Stormer 2000). The role of fathers in pregnancy, though, is often enhanced by ultrasound. Fathers often feel more involved once they have had such visual contact with their unborn. And because of the medical status of having a sonogram made, fathers are readily allowed to take a few hours off to attend the examination—while accompanying their partner to regular midwife or doctor visits is usually of greater concern to employers (Sandelowski 1994).

The most important mediating role of ultrasound imaging, however, is that it constitutes expectant parents as decision makers regarding the life of their unborn child. To be sure, the role of ultrasound is ambivalent here: on the one hand it may encourage abortion, making it possible to prevent suffering; on the other hand it may discourage abortion, enhancing emotional bonds between parents and the unborn by allowing the parents to visualize “fetal personhood.” But either way, ultrasound places expectant parents in the position of having to make a decision about the life of their unborn child. By constituting the unborn, the father, and the mother in very specific ways, it helps to organize a new relation between the three. What appears to be an innocent look into the womb may end up being the first step in a decision-making process that the expectant couple did not explicitly choose.

The use of obstetric ultrasound has important effects on practices of antenatal diagnostics and abortion. Nuchal fold measurement, for instance—

usually in combination with a blood test—does not provide certainty about the health condition of the unborn but only gives an indication of the risk that the unborn will suffer from Down syndrome. In order to get certainty, the mother must undergo an amniocentesis, which is an invasive examination, carrying a risk of miscarriage of about 1:250. Implicitly, for many parents excluding the risk of having a child with Down syndrome appears to be more important than running the risk of losing a healthy unborn child. Moreover, the week-twenty ultrasound examination offered in the Netherlands to all pregnant women appears to increase the number of abortions of fetuses with less severe defects like a harelip (*Trouw* 2006).

It appears hard to escape being technologically constituted as subjects who have to make a decision about the life of their unborn child. Even when people deliberately choose to use the ultrasound examination at twelve weeks only to determine the expected date of birth, the mere possibility that the radiologist might see the thickness of the nuchal fold will make it difficult not to try to interpret the expression on the face of the practitioner. Ultrasound inevitably and radically changes the experience of being pregnant and the interpretations of unborn life.

### Humanism in Ethics

This actively mediating role of ultrasound in moral interpretations, decisions, and practices is at odds with the humanist orientation of mainstream ethics. While ethics is commonly seen as an exclusively human activity, non-human entities like technological devices appear to have moral significance too. This state of affairs challenges both the flattened image of technology in many ethical approaches and the predominant view of the moral subject as an autonomous being. How are we to move beyond the humanist focus in ethics without giving up the undeniably crucial role of human beings in moral actions and decisions?

Humanism is surrounded by the same phenomenon as what Michel Foucault witnessed regarding the Enlightenment: there is a form of black-mail in it (Foucault 1997b). Whoever is not in favor of it is against it. While criticizing the Enlightenment often directly results in the suspicion that one is hostile toward the rationalist worldview and liberal democracy, criticizing humanism evokes the image of a barbarian form of misanthropy. Humanism embodies a number of values—like self-determination, integrity, pluriformity, and responsibility—that are fundamental to our culture in articulating human dignity and respect for human beings. Yet these humanist values do not need to be jettisoned when one is criticizing humanism as a *metaphysical*

position. The humanist metaphysics that lies behind contemporary ethics needs to be overcome if we are to include the moral dimension of objects and their mediation of the morality of subjects.

#### HUMANISM AND MODERNISM

Humanism is a very specific answer to the question of what it means to be a human being. As theorists like Latour and Heidegger have shown, modernity can be characterized by the strict separation it makes between subjects and objects, between humans and the reality in which they exist. Heidegger's work emphasizes how this modern separation of subject and object forms a radically new approach to reality. When humans understand themselves as subjects as opposed to objects, they detach themselves from the network of self-evident relations that arises from their everyday occupations. When one reads a book, is engaged in a conversation, or prepares a meal, just to mention a few examples, one does not direct oneself as a "subject" toward some "objects" but finds oneself in a web of relations in which humans and world are intertwined and give meaning to each other. To understand oneself as a subject facing objects, an explicit act of separation is needed. Humans are not self-evidently "in" their world anymore here but have a *relation* to it while being distanced from it.

Heidegger emphasizes that the word *subject* is derived from the Greek *hypokeimenon*, which he literally translates as "that which lies before" and "which, as ground, gathers everything onto itself" (Heidegger 1977a). The modernist subject becomes the reference point for reality; real is only what is visible to the detached and objectifying gaze of the subject. For such a subject, the world becomes a picture, a representation of objects in a world "out there," projected on the rear wall of the dark room of human consciousness. This is not to imply that the modernist metaphysics of subjects versus objects has no legitimacy. To the contrary, it is at the basis of modern science and has made possible a vast field of scientific research. But this modern "world picture" should not be made absolute as the only valid one. The subject-object separation is only one of the possible configurations in the relations between humans and reality—only one specific way to think this relation, which emerged at a particular moment.

In his book *We Have Never Been Modern*, Latour (1993) interprets modernity in a similar way as Heidegger did. For him, modernity is a process of purifying subjects and objects. Whereas the everyday reality in which we live consists of a complex blend of subjects and objects—or "humans" and "nonhumans," as Latour calls them, in his amodern vocabulary—modernity

proceeds as if subjects and objects had a separate existence. The modernist metaphysics divides reality into a realm of subjects, which form the domain of the social sciences, and a realm of objects, with which the natural sciences occupy themselves. As a result, the vast variety of hybrid mixings of humans and nonhumans among which we live remains invisible. The ozone hole, for instance, is not merely “objective” or “natural”: it owes its existence to the human beings who make it visible, who may have caused it, and who represent it in specific ways when discussing it. But it is not merely “subjective” or “social” either, because there does exist “something” that is represented and exerts influence on our daily lives. The only adequate way to understand it is in terms of its hybrid character; it cannot be reduced to either an object or a subject but needs to be understood in terms of their mutual relations. In Latour’s words, “One could just as well imagine a battle with the naked bodies of the warriors on the one side, and a heap of armor and weapons on the other” (Latour 1997, 77—translation mine).

Latour describes the rise of the modernist approach to reality as “the strange invention of an outside world” (Latour 1999, 3). Only when humans start to experience themselves as a consciousness separated from an outside world—as *res cogitans* versus *res extensa*, as Descartes articulated—can the question of the certainty of knowledge about the world become meaningful: “Descartes was asking for absolute certainty from a brain-in-a-vat, a certainty that was not needed when the brain (or the mind) was firmly attached to its body and the body thoroughly involved in its normal ecology. . . . Only a mind put in the strangest position, looking at a world *from the inside out* and linked to the outside by nothing but the tenuous connection of the gaze, will throb in the constant fear of losing reality” (Latour 1999, 4, emphasis his).

By making humans and reality absolute—in the literal sense of the Latin *absolvere*, which means “to untie” or “to loosen up”—modern thinking about the human can congeal into humanism and modern thinking about reality into realism. In the world in which we live, however, humans and nonhumans cannot be had separately. Our reality is a web of relations between human and nonhuman entities that form ever-new realities on the basis of ever-new connections. In order to understand this reality, we need a symmetrical approach to humans and nonhumans, according to Latour, in which no a priori separation is made between them. The metaphysical position of humanism is by definition at odds with this principle of symmetry. “The human, as we now understand, cannot be grasped and saved unless that other part of itself, the share of things, is restored to it. So long as humanism is constructed through contrast with the object . . . neither the human nor the nonhuman can be understood” (Latour 1993, 136).

## THE HUMANIST BASIS OF MODERN ETHICS

From their metaphysical and ontological analyses of modernity Heidegger and Latour only sporadically draw conclusions regarding ethics. Yet once reality has fallen apart in subjects with consciousness “within” on the one hand and mute objects in a world “out there” on the other, this has direct implications for ethics. After all, ethics now has to be located in one of the two domains. And almost automatically that domain is the one of the subject, which asks itself from a distance how to act in the world of objects. The core question of ethics then becomes “how should I act?.” Ethics is the exclusive affair of *res cogitans*, which judges and calculates to what extent its interventions in the outside world are morally right, without this world having any moral relevance in itself.

The development of modern ethics sharply reflects its modernist origins. Two principal approaches have developed, each centered on its own pole of the subject-object dichotomy. A *deontological* approach focuses on the subject as a source of ethics, while a *consequentialist* approach seeks to find its grip in objectivity. Put differently, while deontology directs itself to the “interior” of the subject, consequentialism emphasizes “outside” reality. Both options become possible on the basis of a metaphysics of subjects with consciousness “within” versus objects in a world “out there.”

The way Immanuel Kant formulated the principles of deontological ethics preeminently embodies the inward movement of the modern subject. Ethics here is centered on the question of how the will of the subject can be subordinated to a universally valid law while it is also kept “pure,” that is, free from the influence of accidental circumstances in the outside world. Because of this urge to purify the subject, only reason can provide something to go on, while any interference from the outside world must be rejected as polluting. “From what we have adduced it is clear that all moral concepts have their seat and origin fully *a priori* in reason . . . ; that these concepts cannot be abstracted from any empirical, and therefore mere contingent, cognition; that their dignity lies precisely in this purity of their origin, so that they serve us as supreme practical principles; that whatever one adds to them of the empirical, one withdraws that much from their genuine influence and from the unlimited worth of actions” (Kant [1785] 2002, 28).

In its striving for pure judgment the subject here isolates itself from reality and attempts to derive moral principles from the workings of its own thinking. With this approach, morality does not get its shape through human involvement with the reality in which they live but through a solitary inward

process of autonomous judgment that must not be disturbed by the outside world.

Consequentialist ethics, on the other hand, does not seek to find grip in the pure will of the subject but in determining and assessing as objectively as possible the consequences of human actions. To be sure, consequentialism does pay attention to the *ways in which* moral assessments can be made—for instance in the distinction between act-utilitarianism, which balances the desirable and undesirable consequences of an action against each other, and rule-utilitarianism, which seeks to find rules that result in a predominance of desirable consequences over undesirable ones. But the primacy is with determining the value of the consequences of actions. In order to make a moral assessment, one needs to make an inventory, as complete as possible, of all consequences of the action involved and of the value of these consequences.

Several variants of consequentialist ethics have developed, each of which attempts to assess the value of consequences of actions in a different way. They range from hedonist utilitarianism (which considers valuable what promotes happiness) and pluralist utilitarianism (which recognizes other intrinsic values beside happiness) to preferential utilitarianism (which does not seek intrinsic values but aims to meet the preferences of as many stakeholders as possible). All these variants share the ambition to determine which action in the world “out there” has the most desirable consequences for the people “out there.” They advocate putting effort into determining and assessing these consequences in order to make a substantiated decision.

Each of these approaches in modern ethics thus embodies one of the poles of the modernist subject-object dichotomy. Both poles represent a humanist ethical orientation in which humans are opposed as autonomous subjects to a world of mute objects. Both approaches take as their starting point a solitary human being that is focused either on the workings of its own subjective judgments or on the objective consequences of its actions.

This humanist orientation radically differs from its predecessor: classical and medieval virtue ethics. There what was central wasn't the question of the right action but the question of the good life. This question does not depart from a separation of subject and object but from the interwoven character of both. A good life, after all, is shaped not only on the basis of human decisions but also on the basis of the world in which it plays itself out (de Vries 1999). The way we live is determined not only by moral decision making but also by the manifold practices that connect us to the material world in which we live. This makes ethics not a matter of isolated subjects but, rather, of connections between humans and the world in which they live.

The example of obstetric ultrasound, or antenatal diagnostics in a broad sense, is illustrative here. As we saw, ultrasound and amniocentesis make it possible to determine during pregnancy whether the unborn suffers from spina bifida or Down syndrome. The very availability of such tests determines to a large extent which moral questions are relevant, and even which questions can be posed at all, in practices surrounding pregnancy. Moral questions regarding, for instance, aborting fetuses with congenital defects can arise only when these defects can be discovered and when abortion is an option at all, both from a technological and a cultural-ethical point of view.

The same can be said of larger-scale technologies, like weapons. Weapons of mass destruction, and especially nuclear weapons, have an important impact on decisions made by governments about how to deal with political conflicts and how to spend their money. And, as Günther Anders has argued, the use of such weapons does not match the capacity of our moral imagination: we can hardly imagine the meaning of and engage with several “megadeads” the way we can with the killing of one individual person (Anders 1988, 271–75; cf. Van Dijk 2000).

To a certain degree the moral charge of such technologies can be expressed in the vocabulary of humanist ethics. Questions like “is one allowed to abort a fetus with serious congenital defects?” and “is one allowed to give life to a child while knowing that it will suffer severely?” are phrased entirely in modern action-ethical terms, just like the more reflective question “is it morally right to delegate to parents the moral responsibility for deciding about the life of their unborn child on the basis of an estimation of risks?” A closer analysis of these moral questions, however, soon jams the modernist purification machine. For if ultrasound indeed helps to determine which moral decisions human beings make, this immediately breaks the autonomy of the subject and also the purity of its will and its moral considerations. Not only do we then appear to have failed in keeping the outside world “out there,” but this world also appears to consist of more than *res extensa*. Ultrasound imaging “does” something in this situation of choice; an ultrasound scanner is much more than a mute and passive object that is only used as an instrument to look into the womb.

Technologies appear to be able to “act” in the human world, albeit in a different way than humans do. By doing so, technologies painlessly cross the modernist border between subject and object. A humanist ethics, as Hans Harbers put it, is founded on a “human monopoly on agency” (Harbers 2005, 259). Because of this, such an ethics is not able to discern the moral dimension of artifacts, and this causes it to overlook an essential part of moral reality. In Latour’s words: “Modern humanists are reductionists because

they seek to attribute action to a small number of powers, leaving the rest of the world with nothing but simple mute forces" (Latour 1993, 138). This is not to say, to be sure, that Latour thinks artifacts are *moral* agents. In fact, he seldomly addresses ethics (except in Latour 2002). Moreover, he always approaches agency as part of a network of relations, which is why, on his view, artifacts can never "have" moral agency "in themselves." Yet this does not take away the fact that the "action" of artifacts that Latour thematizes can actually have moral relevance. Artifacts, after all, do help to shape human actions and decisions. Only a nonhumanist approach in ethics is able to address this moral relevance of nonhuman reality. But what might an ethical framework look like in which not only humans but also artifacts "act" and in which the actions of human beings are the results not only of moral considerations but also of technological mediations?

### Cultivating Humanity: Sloterdijk's Escape from Humanism

As a starting point for articulating a nonhumanist approach to ethics I will critically discuss Peter Sloterdijk's highly contested but also highly fascinating 1999 lecture "Regeln für den Menschenpark" (later translated as "Rules for the Human Zoo," Sloterdijk 2009). At the end of 1999 this text was the focus of a fierce and vicious debate in which Sloterdijk was accused of National Socialist and eugenic sympathies. Sloterdijk flirted with what can be seen as one of the biggest taboos in postwar Germany: the *Übermensch* or "Overman." His text is certainly not danger-free.

"Rules for the Human Zoo" is usually read as a text on biotechnology. But in fact it was written as a critique of humanism. Sloterdijk's lecture is a sparkling and contrarian answer to Heidegger's "Letter on Humanism" ([1947] 1976). In this text Heidegger distanced himself resolutely from the suggestion that his work could be seen, just like Sartre's existentialism, as a form of "humanism"—however convenient this would have been for the rehabilitation of both his work and his reputation after the Second World War. According to Heidegger, humanism entails a far too limited understanding of what it means to be human. Characteristic of humanism (also in its premodern manifestations), for Heidegger, is its approach to the human in terms of the animal: as *animal rationale* or *zoon logon echon*—an animal with speech and reason, or an animal with instincts that can and need to be controlled. Humanism, he says, "thinks the human from *animalitas* and does not think toward *humanitas*" (Heidegger [1947] 1976, 323, my trans.).

Heidegger therefore rejects humanism because it ultimately fixates humanity on its biological basis. A biological understanding of the human

ignores the radical distinction between human and animal, which for Heidegger exists in the ability to think the being of beings. Heidegger does not want to think *humanitas* from the animal, and even less from Sartre's "existence" which would precede "essence," like matter being molded into a form. Heidegger thinks humanity in terms of *ek-sistence*: "being open" to an always historically determined understanding of what it means to "be." Elaborating what Heidegger means by this would fall far beyond the scope of this chapter, but what matters here is Heidegger's rejection of an understanding of humans as animals-with-added-value, for it is precisely at this point that Sloterdijk turns his argumentation upside down. Sloterdijk shares Heidegger's resistance to humanism, but contrary to Heidegger, he does not elaborate his resistance into an *alternative* to the image of humans as "animals with reason" but into a *radicalization* of this image. As opposed to the emphasis Heidegger puts on the lingual aspect of being human ("Language is the house of being"—Heidegger [1947] 1976, 313), Sloterdijk emphasizes the *bodily* aspect of the human. What it means to be human, for him, gets its shape from language but also from corporality.

Sloterdijk shows that language has been the most important medium of humanism. Humanism has always made use of books, which he interprets as a kind of letters; they are written by people who are confident that their text will actually arrive somewhere and that people will actually be prepared to read it. For this reason Sloterdijk states that behind all forms of humanism there is the "communitarian phantasm" of a "literary society," a reading club (Sloterdijk 2009, 13). The literary character of our society, however, is rapidly decreasing—and therefore our society is also rapidly becoming posthumanist. To establish connections between people, letters will not do anymore. We need "new media of political-cultural telecommunication" because "the amiable model of literary societies" has become obsolete (Sloterdijk 2009, 14).

The literary epistles of the humanists aimed to cultivate humans. Behind humanism, therefore, for Sloterdijk, lies the conviction that humans are "animals under the influence" and that they need to be exposed to the right kinds of influences (Sloterdijk 1999, 17, trans. mine). But which media can take over the role of books? What would be appropriate to tame the human when humanism has failed? At this point Sloterdijk takes a path that gave some German intellectuals cause to connect his work with Nazism. Therefore this path needs to be trod carefully. I will briefly sketch the outlines of Sloterdijk's proposal, and after that I will make a counterproposal that makes his critique of Heidegger relevant for the ethics of technology in a broader sense than did Sloterdijk's own proposal.

Sloterdijk develops the thought that Heidegger's approach systematically overlooks the biological condition of humanity. He elaborates the thought that Heidegger's analysis of the *Lichtung*, the "open space" where "being" can manifest itself, ignores that this open space is no "ontological natural state" but a place that humans actually have to enter as physical beings. Being-in-the-world is possible only on the basis of coming-into-the-world, the biological and physical act of birth. This opens an entirely new space to understand what it means to be human and what shapes our humanity. Not only lingual forces that "tame" us are relevant, then, but also physical and material forces that help to "breed" us. Both aspects of shaping humanity are contained in the word *cultivation*. Human culture is both spiritual and material; it is the outcome of both "producing" and "refining," of "breeding" and "reading" (Sloterdijk 2009, 23). Not only the "lections" of the humanists help to shape *humanitas* but also the "se-lections" of the growers of humans that we have always been and that we will be ever more explicitly now that we have biotechnology (Sloterdijk 1999, 43). Because of the possibilities offered by new technologies, we cannot confine ourselves to disciplining humans. Inevitably the question will force itself upon us: which human beings will procreate, and which ones will not? This also lays bare a new social conflict: who are the breeders and who are the ones being bred? (Sloterdijk 2009, 23–24).

Friedrich Nietzsche already pointed out that Western culture has developed a smart combination of ethics and genetics, because of which it is no longer only the strongest that procreate but also those who are collectively weakened by an ethics of solidarity. Thus we already have an implicit ethics of breeding. The question that Sloterdijk raises for the future is, what will this ethics look like when it needs to be made explicit in the biotechnological revolution? Humanity is suddenly facing the need to make political decisions about the properties of its own species (Sloterdijk 2009, 24). When comparing society to a zoo—a metaphor that forces itself upon us when we think in biological rather than lingual terms about humanity—the issue is not only to determine the rules we need to follow for "keeping" ourselves in this park but also the rules for arranging procreation and population growth. The main question biotechnology raises is to what extent the humanist tradition will be able to guide us. Classical texts often abandon us in this context. They are on shelves in archives, "like posted letters no longer collected, sent to us by authors of whom we no longer know whether or not they could be our friends. . . . Letters that are not mailed cease to be missives for possible friends; they turn into archived things. . . . Everything suggests that archivists have become the successors of the humanists" (Sloterdijk 2009, 27).

Especially because of its explicit references to Plato's *Republic*, which I did not cite in this discussion, Sloterdijk's text has often been associated with the eugenic program of the Nazis. Against this interpretation, however, I propose to read Sloterdijk's text as an attempt to face the ultimate consequences of the biotechnological revolution (cf. Lemmens 2008). Appealing to the archives of the tradition allows philosophers to comfortably position themselves outside of reality and simply refuse to discuss the breeding of humans. But as soon as the technologies for such breeding become available and known, and therefore part of society, the discussion Sloterdijk has attempted to open becomes inevitable. Moreover, whoever sees with Nietzsche that the predominant humanist approach itself has genetic consequences has no argument to distance himself or herself from the posthumanist space opened by new technologies. Sloterdijk simply makes explicit the questions evoked by new technological possibilities, by placing them provocatively in front of us. He does not propose to design a specific transhuman entity or to breed a variant of the human being. He merely shows that the simple fact of our biological birth, added to our ability to alter our biological constitution, implies that the rules that have always organized our reproduction implicitly might have to be made explicit in the future, and might require a reorientation.

Within this book, however, I do not aim to contribute to the discussion about the biological future of *Homo sapiens*. My interest here is the ethics of technology and how to move beyond the humanist bias in ethics in order to make room for the moral relevance of technological artifacts. And for answering this question, the proposal to develop rules for the human zoo—however important it is—is the least interesting part of Sloterdijk's discussion with Heidegger. Much more interesting is Sloterdijk's ambition to think about ethics and technology beyond humanism. In his analysis it becomes clear how the biological and "material" aspect of the human has been neglected in the humanist tradition and how the media used by this tradition are losing their self-evident relevance. This "material" turn in approaching humanity creates applications for a nonhumanist ethics of technology. The "transhumanist" development toward an enhanced version of *Homo sapiens* is not central, then, but rather the "posthumanist" development beyond humanism as a predominant way of understanding what it means to be human.

The most important contribution of Sloterdijk's text to the ethics of technology therefore consists in opening a modern space to think about ethics. Precisely such a space is needed to escape from the humanist points of departure of contemporary ethics and to make room for the moral relevance of nonhuman entities. When we approach human beings not only in terms of their being-in-the-world but also in terms of their coming-in-the-world,

they do not only appear as “subjects” but also as “objects,” not only as the *res cogitans* of their consciousness but also as the *res extensa* of the bodies with which they experience and act in the world. Such a posthumanist approach to the human is as least as important for understanding the everyday life of the *Homo sapiens* we still are as it is for the transhuman forms of life on which Sloterdijk primarily focuses in this text.

### Humanities and Posthumanities: New Media for Cultivating Humanity

In order to elaborate the contours of a posthumanist ethics, we need to bracket Sloterdijk’s ideas about “breeding” human beings and focus on “taming” humanity. In *Rules for the Human Zoo*, Sloterdijk associates the activity of taming exclusively with the humanist tradition. Yet his observation that the lingual media of humanism are becoming ever more obsolete because of new technologies does not necessarily justify the conclusion that we also need to replace the humanist “taming” of humanity with a posthumanist “breeding.” A nonhumanist approach to humanity that does not separate the “objectivity” and “subjectivity” of human beings reveals applications of new forms of “taming” that remain undiscussed in Sloterdijk’s lecture.

In our technological culture, it has become clear that *humanitas* gains its shape not only by the influence of ideas on our thinking, or by physical interventions in our biological constitution, but also by material arrangements of the technological environment in which we live. Humanity and ethics do not spring exclusively from the cerebral activities of a consciousness housed in a bodily vessel but also from the practical activities in which human beings are involved as physical and conscious beings. By associating the “taming” of *res cogitans* only with texts and associating technology only with the “breeding” of *res extensa*, Sloterdijk ignores—at least in “Rules for the Human Zoo”<sup>2</sup>—how human beings, as *res extensa*, not only can be bred but are also being *tamed* by technology. If the lingual media of humanism have indeed become obsolete, as Sloterdijk observes, material media have taken their place. Beside the anthropotechnologies of writing and human engineering, there is a vast field of anthropotechnologies that need to be taken into account for understanding what it means to be human: the pile of technological artifacts that help to shape how we experience the world and live our lives, ranging from television sets and mobile phones to medical diagnostic devices and airplanes.

Again, the example of obstetric ultrasound is a good illustration here. The way this technology represents the unborn helps to shape a particular *practice* of dealing with uncertainties regarding the health of unborn children.

This new practice has important implications for the moral considerations of expecting parents. Because of the way ultrasound helps to shape how the unborn is experienced, new interpretations of pregnancy arise, along with new practices of dealing with the risk of congenital defects. After all, the very possibility of determining already before a child is born whether it suffers from a specific disease raises the question whether this pregnancy should be continued.<sup>3</sup>

This is not to say that ultrasound would only *stimulate* expectant parents to get an abortion when serious congenital defects are found. On the one hand, ultrasound imaging unmistakably has this effect, since an abortion can prevent suffering for both a seriously ill child and its parents. But on the other hand, ultrasound imaging establishes an intimate relation between parents and their unborn child, which enhances their bonding and makes abortion more difficult. In both cases, though, the very possibility of having an ultrasound examination done constitutes an entirely new ethical practice. Also *not* having such an examination done is a moral decision now, since this implies rejecting the option of sparing an unborn child an incurable disease and possibly dead-end suffering. An ultrasound scan of an unborn child is never a neutral peek into the womb. It helps to constitute the unborn as a possible patient and its parents as decision makers about the life of their unborn.

Ultrasound, therefore, is a nonlingual medium of morality; it “tames” human beings in a material way. Ironically, in this example the “taming” of humanity is also directly relevant to practices of “breeding.” This immediately makes clear that Sloterdijk’s work is relevant not only for analyzing wild scenarios of a transhuman future but also for making visible how the current everyday breeding practices of *Homo sapiens* are thoroughly posthumanist in character. Moral decisions about pregnancy and abortion in many cases are shaped in interaction with the ways in which ultrasound imaging makes visible the unborn child. Apparently, moral action cannot be understood here in terms of a radical separation of a human moral agent, on the one hand, acting in a world of mute material objects on the other. Ultrasound imaging actively contributes to the coming about of moral actions and the moral considerations behind these actions. This example therefore shows that moral agency should not be seen as an exclusively human property; it is distributed among human beings and nonhuman entities. Moral action is a *practice* in which humans and nonhumans are integrally connected, generate moral questions, and help to answer them.

In these connections, not only is *res extensa* more active than the modernist approach makes visible, but also is *res cogitans* less autonomous. From

a modernist orientation, it is impossible to classify an action induced by behavior-influencing technology as moral action. Everyone, for instance, who slows down near a school because there is a speed bump on the road shows steered behavior rather than moral and responsible action. The ultrasound example, however, shows that morality has a broader domain. Here, technology does not *impede* morality, but rather *constitutes* it. Ultrasound imaging organizes a situation of moral decision making while also helping to shape the frameworks of interpretation on the basis of which decisions can be made. As soon as we see that morality is not an exclusively human affair, material “interventions” in moral judgments of the subject are not taken to be pollutions of a “pure will” but media of morality. To paraphrase Kant: ethics without subjects is blind, but ethics without objects is empty. In the pure space of subjectivity the subject cannot encounter a world in which to find a moral relation; as soon as this world is there, practices come into being that help to shape the moral space of the subject. Mediated action is not amoral but is rather the preeminent place where morality finds itself in our technological culture.

Sloterdijk’s conclusion that the influence of the media of humanism is declining, therefore, does not need to imply that the “taming” of humanity is about to be replaced by “breeding.” Many more media appear to tame us than only the texts of humanism, and these new media especially need to be scrutinized: the technological artifacts that help to shape our daily lives. After all, the cohesion of the literary society in which humanity attempts to tame itself might be diminishing, but the human zoo in which humanity attempts to breed itself in sophisticated ways is far from being attractive enough to render the literary society completely obsolete. Instead, the posthumanist and amodern space opened by Sloterdijk shows that this literary society has never been as “literary” as it thought. The texts that were written, read, interpreted, and handed down have always been products of the concrete practices in which they were considered relevant, and the humanity of humans was always shaped not only on the basis of self-written texts but also of their self-created material environment in which their practices were formed. The detached autonomous human of modernist humanism has never existed.

### Conclusion: Toward a Nonhumanist Approach

How might we augment the ethics of technology in such a way as to include this posthumanist and amodern perspective? The most important prerequisite for such an expanded ethical perspective is the enlargement of the moral community to include nonhuman entities and their connections to human

beings. Only in this way can justice be done to the observation that the medium of ethics is not only the language of subjects but also the materiality of objects. This implies a shift of ethics. In addition to developing lingual frameworks for moral judgment, ethics consists in designing material infrastructures for morality. When matter is morally charged, after all, designing is the moral activity *par excellence*, albeit “by other means.” Designers materialize morality. Ethics is no longer a matter of only ethereal reflection but also of practical experiment, in which the subjective and the objective, the human and the nonhuman have become interwoven.

From this interwoven character two important lines of thought can be discerned in a posthumanist ethics: *designing* morally mediating technology (designing the human into the nonhuman) and *using* morally mediating technology in deliberate ways (coshaping the roles of the nonhuman in the human). These two lines might seem to reflect the modernist distinction between an actively reflecting subject and a passively designed world. But rather than reinforcing this distinction, a posthumanist ethics aims to think both poles together by focusing on their connections and interrelations.

Before addressing these lines in the ethics of technology, however, I will explore the implications of introducing the moral significance of technology into ethical theory. In chapter 3 I will articulate what the phenomenon of technological mediation implies for the role of the object in ethical theory; in chapter 4 I will investigate how the mediated character of moral actions and decisions calls for a reconceptualization of the role of the subject in ethical theory.

## Do Artifacts Have Morality?

### Introduction

How do we come to understand the moral dimension of technology?<sup>1</sup> Now that we have seen that technologies have moral relevance, and that ethics needs to expand its “humanist focus” to take this into account, the question rises how to conceptualize the morality of technology. What could it imply to say that technologies have a moral dimension? Do the examples that we have seen so far—ultrasound, speed bumps, cell phones—urge us to consider technologies to be moral entities, even moral agents? Or are there other ways to conceptualize the morality of technological artifacts?

Approaching things in moral terms is not a self-evident enterprise. It goes against the grain of the most basic assumptions in ethical theory. After all, it would be foolish to blame a technology when something immoral happens. It does not make sense to condemn the behaviour of a gun when somebody has been shot; not the gun but the person who fired it needs to be blamed. Tsjalling Swierstra is a good representative of such hesitations regarding “moralizing things.” He discusses how the moral community has been expanded many times since classical antiquity. “Women, slaves, and strangers were largely or entirely devoid of moral rights,” but “over time all these groups have been admitted” (Swierstra 1999, 317).<sup>2</sup> The current inclination to also grant things access to the moral community, however, goes too far, he argues from the two predominant ethical positions: deontology and consequentialism.

Consequentialist ethics evaluates actions in terms of the value of their outcomes. When the positive consequences outweigh the negative ones, an action can be called morally correct. From this perspective, Swierstra says, things can indeed be part of a moral practice, since they can incite human beings to behave morally—and from a consequentialist perspective it is only the result that counts. But things can do this only when human beings use

them for this purpose. Things themselves are not able to balance the positive and negative aspects of their influence on human actions against each other. They can only serve as instruments, not as fully fledged moral agents that are able to render account for their actions.

Deontological ethics is directed not at the consequences of actions but at the moral value of the actions themselves. From a Kantian perspective, for instance, the morality of an action depends on whether the agent has intended to act in accord with rationally insightful criteria. Artifacts, of course, are not capable of taking up such considerations. Moreover, if they incite human beings to act in ways that are morally right from a deontological point of view, these actions are not results of a rationally insightful moral obligation but simply a form of steered behavior.

This means that both from a deontological and a consequentialist perspective, artifacts can only be *causally* responsible for a given action, not *morally*. Artifacts do not possess intentions, and therefore they cannot be held responsible for what they “do.” In Swierstra’s words: “Compelling artifacts, therefore, are not moral actors themselves, nor can they make humans act truly morally. Therefore . . . there is no reason to grant artifacts access to the moral community.” (Swierstra 1999).

I share Swierstra’s hesitations regarding a too radically symmetrical approach to humans and things (cf. Verbeek 2005b, 214–17). Yet the argument that things do not possess intentionality and cannot be held responsible for their “actions” does not justify the conclusion that things cannot be part of the moral community. For even though they don’t do this intentionally, things do mediate the moral actions and decisions of human beings, and as such they provide “material answers” to the moral question of how to act. Excluding things from the moral community would require ignoring their role in answering moral questions—however different the medium and origins of their answers may be from those provided by human beings. The fact that we cannot call technologies to account for the answers they help us to give does not alter the fact that they do play an actively moral role. Take technology away from our moral actions and decisions and the situation changes dramatically. Things can be seen as part of the moral community in the sense that they help to shape morality.

But how to account for this moral role of technology in ethical theory? As stated in chapter 1, to qualify as a moral agent in mainstream ethical theory requires at least the possession of *intentionality* and some degree of *freedom*. Both requirements seem problematic with respect to artifacts—at least at first sight. Artifacts do not seem to be able to form intentions, and neither do

they possess any form of autonomy. Yet both requirements for moral agency deserve further analysis. From the amodern approach set out in chapter 2, the concept of agency—including its aspects of intentionality and freedom—can be reinterpreted in a direction that makes it possible to investigate the moral relevance of technological artifacts in ethical theory. This will be the main objective of this chapter. First, I will discuss the most prominent existing accounts of the moral significance of technological artifacts. After that, I will develop a new account in which I expand the concept of moral agency in such a way that it can do justice to the active role of technologies in moral actions and decisions.

### The Moral Significance of Technological Artifacts

The question of the moral significance of technological artifacts has popped up every now and then during recent decades. Several accounts have been developed, all of which approach the morality of technology in different ways. I will discuss the most prominent positions as a starting point for developing a philosophical account of the morality of technological artifacts.

#### LANDON WINNER: THE POLITICS OF ARTIFACTS

In 1980 Langdon Winner published his influential article “Do Artifacts Have Politics?” In this text, which was later reprinted in his 1986 book *The Whale and the Reactor*, Winner analyzed a number of “politically charged” technologies. The most well-known example he elaborated concerns a number of “racist” overpasses in New York, over the parkways to Jones Beach on Long Island. These overpasses, designed by architect Robert Moses, were deliberately built so low that only cars could pass beneath them, not buses. This prevented the African American population, at that time largely unable to afford cars, from accessing Jones Beach. Moses apparently had found a material way to bring forth his political convictions. His bridges are *political* entities. The technical arrangements involved *preceded* the use of the bridges. Prior to functioning as instruments to allow cars to cross the parkways, these bridges already “encompass[ed] purposes far beyond their immediate use” (Winner 1986).

Winner’s analysis obtained the status of a “classic” in philosophy of technology and in science and technology studies—even though it became the focus of controversy in 1999, when Bernward Joerges published the article “Do Politics Have Artefacts?” (Joerges 1999). In this article he showed that

Jones Beach can also be reached via alternative routes and that Moses was not necessarily more racist than most of his contemporaries. The controversy, however, did not take away the force of Winner's argument. Even as a thought experiment, the example shows how material artifacts can have a political impact—and in this case, a political impact with a clearly moral dimension (see Woolgar and Cooper 1999; Joerges 1999).

The low-hanging overpasses are not the only example Winner elaborated. For Winner, the political dimension of artifacts reaches further than examples like this, in which technologies actually embody human intentions in a material way. Technologies can also have political impact without having been designed them to do so. Many physically handicapped people can testify to this—unintentionally the material world quite often challenges their ability to move about and to participate fully in society.

To elaborate the nonintentional political dimensions of technological artifacts, Winner discusses the example of mechanical tomato harvesters. These machines have had an important impact on tomato-growing practices. Because of their high cost, they require a concentrated form of tomato growing, which means that once they are in use small farms have to close down. Moreover, new varieties of tomatoes need to be bred that are less tasty but can cope with the rough treatment the machines give them. There was never an explicit intention to make tomatoes less tasty and to cause small farms to shut down—but still these were the political consequences of the mechanical tomato harvester.

The example of Moses's bridges shows that technologies can have an impact that can be morally evaluated—the first kind of moral relevance of technologies. Moreover, the example of the tomato harvester shows that such impacts can occur without human beings explicitly intending them—they are in a sense “emergent,” which suggests a form of “autonomy” of technology, albeit without a form of consciousness or intentionality behind it. Technologies, according to Winner, are “ways of building order in our world.” Some technologies bring about this order at the intentional initiative of human beings, serving as “moral instruments” like Moses's bridges, and other technologies give rise to unexpected political impacts.

Winner's account is highly illuminating, yet in the context of this study his analysis leaves many knots untied. Showing that technologies can have a *politically* relevant impact on society, even when this impact was not intended by their designers, does not yet reveal how technologies can also have a *moral* impact. Moreover, we are still in the dark about the *ways in which* this impact comes, and an understanding of this is needed if we are to link me-

diation theory to ethical theory. Winner paved the way, but we need a more detailed account of the roles of technologies in moral actions and decisions if we are to grasp their moral significance.

BRUNO LATOUR:  
THE MISSING MASSES OF MORALITY

A second prominent voice in the discussion about the moral significance of technological artifacts is the French philosopher and anthropologist Bruno Latour. In 1992 he published an influential article titled "Where Are the Missing Masses? The Sociology of a Few Mundane Artifacts." In this text he elaborates the idea that morality should not be considered a solely human affair. Everyone complaining about the alleged loss of morality in our culture should open their eyes. Rather than looking only among people, they should direct their attention toward material things too. The moral decision about how fast one drives, for example, is often delegated to speed bumps in the road, which tell us to slow down. In some cars, blinking lights and irritating sounds remind us to fasten our seat belts. Automatic door closers help us to politely shut the door after entering a building. The "missing masses" of morality are not to be found among people but in things.

By attributing morality to material artifacts, Latour deliberately crosses the boundary between human and nonhuman reality. For Latour, this boundary is a misleading product of the Enlightenment. The radical separation of subject and object that is one of the cornerstones of Enlightenment thinking prevents us from seeing how human and nonhuman entities are always intertwined. Latour understands reality in terms of networks of agents that interact in manifold ways, continually translating each other. These agents can be both human and nonhuman. Nonhumans can act too; they can form "scripts" that prescribe that their users act in specific ways, just as the script of a movie tells the actors what to do and say at what place and time. Neither the intentions of the driver nor the script of the speed bumps in the road exclusively determines the speed at which we drive near a school. It is the network of agents in which a driver is involved which determines his or her speed.

Ten years later, Latour augmented his analysis in an article titled "Morality and Technology: The End of the Means." Here he shows how inadequate it is to think that technologies belong in the realm of means while human beings inhabit the realm of ends. This view results in what he calls an "archaic split between moralists in charge of the ends and technologists controlling

the means” (Latour 2002). Latour proposes instead to understand technology in terms of the notion of *fold*. In technical action, time, space, and the type of “actants” are folded together. Technologies cross space and time. A hammer, for instance, “keeps folded heterogeneous temporalities, one of which has the antiquity of the planet, because of the mineral from which it has been moulded, while another has the age of the oak which provided the handle, while still another has the age of the 10 years since it came out of the German factory which produced it for the market” (ibid., 249). The same holds true for *space* here: “the humble hammer holds in place . . . the forests of the Ardennes, the mines of the Ruhr, the German factory, the tool van which offers discounts on every Wednesday on Bourbonnais streets,” et cetera.

By “the type of actants,” the third element that is folded into technical action, Latour means that both human and nonhuman agents are involved and help to shape each other. Technologies should not be understood merely in terms of functionality, for this would limit us to seeing only how human intentions can be realized with the help of nonhuman functionalities serving only as means of extension. Technologies are not simply *used* by humans—they help to *constitute* humans. A hammer “provides for my fist a force, a direction and a disposition that a clumsy arm did not know it had” (ibid., 249). In the same way, speed bumps are not simply neutral instruments that fulfill the function of slowing down drivers. “What they exactly do, what they suggest, no one knows, and that is why their introduction in the countryside or in towns, initiated for the innocent sake of function, always ends up inaugurating a complicated history, overflowing with disputes, to the point of ending up either at the State Council or at the hospital” (ibid., 250). Technologies are not intermediaries, helping human intentions to be realized in the material world; they are *mediators* that actively help to shape realities. Technologies do not merely provide means but also help to form new ends; they do not provide functions but make possible detours. “Without technologies, humans would be contemporaneous with their actions, limited solely to proximal interactions. . . . Without technological detours, the properly human cannot exist” (ibid., 252).

The moral significance of technologies, for Latour, is part of this phenomenon of folding. Morality is a “regime of mediation” as well (ibid., 254). We usually recognize morality in the form of *obligation*, but this is not the only form it can take, since it “derives just as much from contract, from religious events, . . . from chains of references, from the law,” et cetera (ibid., 254). Rather than being a merely human affair, morality is to be found in nonhuman entities as well. “Of course, the moral law is in our hearts, but it is also in our apparatuses. To the *super-ego* of tradition we may well add the

*under-ego* of technologies in order to account for the correctness, the trustworthiness, the continuity of our actions" (ibid., 253–54). This "under-ego" is present in the speed bumps that tell us how fast to drive, or the coin locks on supermarket carts, demanding that we put the carts back in their rack rather than leaving them beside our parking place.

This does not imply, to be sure, that we need to understand technologies as moral agents *in themselves*. "In themselves" entities are quite meaningless anyway—they are given a character in the relations in which they function. In Latour's words, "Nothing, not even the human, is for itself or by itself, but always *by other things* and *for other things*" (ibid., 256; emphasis in original). Both morality and technology are "ontological categories" for Latour: "the human comes out of these modes, it is not at their origin" (ibid., 256). Technologies help to constitute humans in specific configurations—including the moral character of our actions and decisions.

#### ALBERT BORGMANN:

##### TECHNOLOGY AND THE GOOD LIFE

North American philosopher of technology Albert Borgmann has proposed a third position to describe the moral significance of technology. He has developed a neo-Heideggerian theory of the social and cultural role of technology. In this theory, he elaborates how our culture is ruled by what he calls the "device paradigm." According to Borgmann, the technological devices that we use call for a quite different way of taking up with reality than did pretechnological "things." While "things"—like water wells, fireplaces, musical instruments—evoke practices in which human beings are engaged with reality and with other people, devices primarily evoke disengaged consumption.

Borgmann understands devices as material *machineries* that deliver consumable *commodities*—for example, the boiler and radiators of a heating installation form a machinery that delivers warmth as a commodity. Devices ask for as little involvement as possible; they create the *availability* of commodities by keeping their machinery in the background as much as they can and putting their commodities in the foreground. Against this, "things" do not separate machinery from commodity. Rather, they *engage* people. Using a fireplace, for instance, requires people to collect and chop wood, to clean the hearth regularly, to gather around the fireplace to enjoy the warmth it gives, and so on.

In his article "The Moral Significance of Material Culture," Borgmann explains how his theory of the device paradigm makes visible a moral dimension in material objects. He focuses on the role of material culture in human practice, and shows how "material culture constrains and details practice

decisively” (Borgmann 1995, 85). In line with his device paradigm, he makes a distinction between two kinds of reality: one commanding, the other disposable. While a traditional musical instrument is a commanding thing, one that requires a lot of effort and skill and needs to be “conquered,” a stereo literally puts music at our disposal. The quality of sound can be even better than that of a live performance, but the music’s presence is not commanding like that of the music performed live by a musician. According to Borgmann, the device paradigm increasingly replaces commanding reality with disposable reality. In his book *Real American Ethics*, Borgmann elaborates the concept of *moral commodification* to analyze this phenomenon: “*a thing or practice gets morally commodified when it is detached from its context of engagement with a time, a place, and a community* and it becomes a free-floating object” (Borgmann 2006, 152; italics in original).

We find the moral significance of the material culture in its role in shaping human practices. While commanding reality “calls forth a life of engagement that is oriented within the physical and social world,” disposable reality “induces a life of distraction that is isolated from the environment and from other people” (ibid., 92). Human practices take place not in an empty space but in a material environment—and this environment helps to shape the quality of these practices. “If we let virtue ethics with its various traditional and feminist variants stand in for practical ethics, we must recognize that virtue, thought of as a kind of skilled practice, cannot be neutral regarding its real setting. Just as the skill of reading animal tracks will not flourish in a metropolitan setting, so calls for the virtues of courage and care will remain inconsequential in a material culture designed to produce a comfortable and individualist life” (Borgmann 1995, 92).

Even if we do not entirely follow Borgmann in his rather gloomy approach to technology—I think there is engaging technology as well: see Verbeek 2005b—his position highlights a significant form of the moral relevance of technology. Material objects, to summarize his position, help to shape human practices. And because the quality of these practices is ultimately a moral affair, material objects have direct moral relevance. Technological devices and nontechnological “things” help to shape the ways we live our lives—and the question of “the good life” is one of the central questions in ethics. Human actions and human life do not take place in a vacuum but in a real world of people and things that help to shape our actions and the ways we live our lives. And therefore, the good life is not formed only on the basis of human intentions and ideas but also on the basis of material artifacts and arrangements. Technologies provide a setting for the good life.

LUCIANO FLORIDI AND J. W. SANDERS:  
ARTIFICIAL MORAL AGENCY

A radically different but equally interesting approach was elaborated in 2004 by Luciano Floridi and J. W. Sanders in their influential publication “On the Morality of Artificial Agents.” Their article deals with the question to what extent artificial agents can be moral agents. Rather than focusing on the moral significance of technologies in general, they focus on *intelligent* technologies that could actually qualify as “agents.” Examples of such artificial agents are expert systems that assist people in making decisions, driving assistants that help people to drive their cars, and automatic thermostats in houses.

The approach Floridi and Sanders develop is so interesting because they give an account of artificial moral agency in which moral agents do not necessarily possess free will or moral responsibility. This way, they take away the obvious objection that technologies, lacking consciousness, can never be moral agents as human beings are. It is crucial to Floridi and Sanders’s analysis that they explicitly choose an adequate “level of abstraction” at which it becomes possible and meaningful to attribute morality to artificial agents—such an abstraction is needed in order to avoid the obvious objection that artifacts cannot have agency as humans do. As criteria for *agenthood*, therefore, Floridi and Sanders use “interactivity (response to stimulus by change of state), autonomy (ability to change state without stimulus) and adaptability (ability to change the ‘transition rules’ by which state is changed).” This implies that a system that interacts with its environment but is also able to act without responding to a stimulus and has the ability to learn how to “behave” in different environments could qualify as an agent. They use the ability to cause good or evil as the criterion for *morality*: “An action is said to be morally qualifiable if and only if it can cause moral good or evil. An agent is said to be a moral agent if and only if it is capable of morally qualifiable action” (Floridi and Sanders 2004, 12).

Their approach reveals what Floridi and Sanders call “aresponsible morality” (ibid., 13). They consider intentions—“intentional states,” in the vocabulary of the analytic tradition from which they work—as a “nice but unnecessary condition” for moral agency. The only thing that matters for them is whether the agent’s actions are “morally qualifiable”—that is, whether they can cause moral good or evil. However, Floridi and Sanders do not aim to declare the concept of responsibility obsolete. Rather, they separate it from moral agency as such, which opens for them the space needed to clarify the role responsibility actually plays in morality (ibid., 20).

It is an important contribution to understanding the moral significance of technology to reveal how normative action is possible even when there is no moral responsibility involved. The approach of Floridi and Sanders offers an answer to an obvious objection against attributing morality to technologies, that technologies do not have consciousness and therefore cannot “act” morally. If moral agency can be adequately understood in terms of showing “morally qualifiable” action, greater justice can be done to the moral relevance of technological artifacts than mainstream ethical theory allows. The problem remains, however, how to deal with forms of artifact morality that cannot be considered results of artificial agency. How should we deal with ultrasound imaging, for instance, in terms of this framework? And with Winner’s example of Moses’s bridges? These examples do not meet Floridi and Sanders’s criteria for agency—but they do actively contribute to moral actions and have impacts that can be assessed in moral terms. However illuminating Floridi and Sanders’s position is, we need more if we are to understand the moral relevance of technology. Artificial moral agency constitutes only a part of the moral relevance of technology; we need a broader understanding of “artifactual morality.”

### Moral Mediation

In the positions discussed above, various approaches to the morality of technology played a role. All authors agree that technologies are morally significant because they have a morally relevant impact in society. Technologies help to shape actions, inform decisions, and even make their own decisions, as some information technologies do; in all cases, they have an impact that can be assessed in moral terms. Yet there appear to be many ways to understand this “morally relevant impact.”

#### MORAL INSTRUMENTALISM

The first and minimum option is to approach technologies as moral instruments. Winner’s bridges, Latour’s speed bumps and door closers, and Hans Achterhuis’s turnstiles are examples of technologies that bring about a moral effect that humans seek to achieve through them. From the approach of technological instrumentalism, artifacts like these provide human beings with means to realize their moral ends: racial segregation, safety on the road, neatly closed doors, paying passengers on trains.

However, this approach is far too shallow to do justice to the complex moral roles of technologies. And to be sure, none of these authors actually

think that technologies are merely neutral means to realize human moral intentions. Winner's example of the tomato harvester, for instance, shows that technologies can have *unintended* consequences. Latour would readily acknowledge that speed bumps can invite local skaters to engage in behavior that actually diminishes rather than enhances traffic safety, and that automatic door closers might also embody forms of impoliteness by slamming doors in people's faces and making it difficult for elderly people to open them. Even though technologies can certainly function as moral instruments that enable human beings to generate specific moral effects, they always do more than this.

The behavior of technologies is never fully predictable—a thought that is vividly illustrated in Edward Tenner's book *Why Things Bite Back* (1996). Moral instrumentalism is too poor a position to account for the moral relevance of technology. Technologies inevitably enter into unforeseeable relations with human beings in which they can develop unexpected morally relevant impacts. Obstetric ultrasound is a good example again: this technology was not designed to organize new moral practices, and yet it plays an active role in raising moral questions and setting the framework for answering them.

#### TECHNOLOGIES AS MORAL AGENTS

Does this imply that we should take the opposite direction and approach technologies as moral agents? Should we simply start to acknowledge the fact that technologies can act morally? This is the position Floridi and Sanders defend. From the level of abstraction they elaborate, an entity is a moral agent when it is able to cause good or evil. This approach allows them to conclude that artificial agents can qualify as moral agents because they can "do" evil or good by producing effects that can be assessed morally. This approach is highly interesting and relevant, but unfortunately it applies to only a limited set of technologies. Not all morally significant technologies could qualify as agents based on Floridi and Sanders's criteria of interactivity, autonomy, and adaptability. Ultrasound imaging, for instance, would fail the criterion of autonomy, yet it has a moral impact beyond what human beings designed into it.

The position of Bruno Latour also attributes agency to technologies, but in a radically different way. While Floridi and Sanders focus on *artificial* agency, one could say that Latour focuses more broadly on *artifactual* agency. In his symmetrical approach, both humans and nonhumans can be agents, and nonhuman agents can also embody morality by helping to shape moral action. Yet, as indicated above, from a Latourian point of view it would not

be adequate to attribute moral agency to technologies “themselves”—as if “agency” were some intrinsic property of technology. Latour’s claims that nonhumans can be agents as well and that there is morality in technology need to be read in the context of his actor-network theory, in which all entities are understood relationally. From this perspective, technologies do not have moral agency *in themselves*; rather, when humans use technologies, the resulting moral agency is not exclusively human but incorporates nonhuman elements as well. Contrary to the position of Floridi and Sanders, for Latour technologies only “have” agency and morality in the context of their relations with other agents.

#### MORAL MEDIATION

Actually, Latour’s approach occupies a third position with respect to the moral relevance of technology. Rather than moral instruments or moral agents, Latour’s work makes it possible to see technologies as moral *mediators*. This position does justice to the active moral role of technologies in moral actions and decisions, without reducing this role entirely to human intentions. At the same time, it avoids characterizing morality as an intrinsic property of the technologies themselves. By mediating human experiences and practices—as elaborated in chapter 1—technologies mediate moral decisions and actions. Technologies help us to phrase moral questions and find answers to them, and they guide our actions in certain directions.

The notion of “mediator” expresses both the active moral role of technologies and the relational character of this moral role: they mediate, rather than being some kind of neutral “intermediary,” but mediators can function only in the context of an environment for and in which they mediate. The moral significance of Latour’s speed bumps and Winner’s overpasses can be understood best in terms of moral mediation. Understanding them as moral instruments for realizing the racist intentions or safety ambitions of city planners appeared to fall short, because this does not recognize the *unintended* roles these artifacts can play. Understanding them as moral agents would go too far, at least in the sense of being moral agents “in themselves,” capable of moral action. Only in the context of the practices in which they function do their moral roles emerge. Sometimes these roles coincide with the intentions of their designers, sometimes they don’t. In all cases, the moral roles of technologies come about in the context of their relations with their users and the environment in which they function.

Borgmann’s approach to the moral significance of technology is an inter-

esting supplement to the notion of moral mediation. He broadens the discussion from action-oriented ethics to the classical ethical question of the good life by focusing on technologies as providing a material setting for the good life. In Borgmann's approach, the moral role of technologies is not to be found in the ways technologies help to shape human actions but in how they help to answer the classical question of "how to live." Borgmann's example of the difference between a stereo set and a musical instrument does not revolve around the different actions involved in operating the two but around their roles in shaping a way of life.

By conceptualizing technologies as moral mediators, we can bring the postphenomenological approach to technological mediation into the realm of ethics. As we saw in the example of obstetric ultrasound, technologies-in-use establish a relation between their users and their world. Ultrasound imaging organizes a specific form of contact between expectant parents and unborn child, in which the parents and the child are constituted in specific ways with specific moral roles, responsibilities, and relevance. Along the same lines, larger-scale technologies mediate moral actions and decisions; energy production systems, for instance, help to organize a way of living in which it becomes ever more normal and necessary to use large quantities of energy, and in doing so they help to shape moral decisions regarding how we deal with environmental issues.

To be sure, approaching technologies as moral mediators does not imply that we need to reject Latour's ideas about nonhuman agency. Indeed the notion of moral mediation implies a form of technological agency. Moral mediation always involves an intricate relation between humans and nonhumans, and the "mediated agency" that results from this relation therefore always has a hybrid rather than a "purely nonhuman" character. When technologies are used, moral decisions are not made autonomously by human beings, nor are persons forced by technologies to make specific decisions. Rather, moral agency is distributed among humans and nonhumans; moral actions and decisions are the products of human-technology associations.

The way I use the notion of moral mediation is different from the way Lorenzo Magnani uses it in his book *Morality in a Technological World* (2007). Magnani lays out an approach to morality and technology that is congenial to the approach set out in this book but reaches different conclusions. Because his approach departs from the perspective of cognitive science rather than phenomenology, it cannot take into account the hermeneutic and pragmatic dimensions of technological mediation that are so central to the account developed here. For Magnani, moral mediators mediate moral *ideas*.

In his definition, “moral mediators . . . are living and nonliving entities and processes—already endowed with intrinsic moral value—that ascribe new value to human beings, nonhuman things, and even to ‘non-things’ like future people and animals.” Even though he discusses Latour’s work assentingly (ibid., 25–26), he does not acknowledge that Latour’s actor-network theory radically differs from his cognitive approach. Magnani’s strong focus on *knowledge* as the primordial variable in ethics and in moral mediation is rather remote from Latour’s focuses on practices, interactions, and materiality.

For Latour, and for the postphenomenological approach that uses his work, the cognitive approach makes too sharp a distinction between (subjective) minds that have knowledge and the (objective) world that this knowledge is about. In the approach I follow in this book, morality should not be understood in terms of cognitive “templates of moral doing” (Magnani 2007, 187–93) but in terms of ways of being-in-the-world which have both cognitive and noncognitive aspects and which are technologically mediated in more-than-cognitive ways. In my postphenomenological approach, technological mediation concerns action and perception rather than cognition; and moral mediation is not only about the mediated character of moral ideas but mostly about the technological mediation of actions, and of perceptions and interpretations on the basis of which we make moral decisions.

The concept of moral mediation has important implications for understanding the status of objects in ethical theory. As indicated in my introduction, in mainstream ethical theory “objects” have no place apart from being mute and neutral instruments that facilitate human action. Now that we have seen that technologies actively help to shape moral actions and decisions, we need to expand this overly simplistic approach. The mediating role of technologies can be seen as a form of moral agency—or better, as an element of the distributed character of moral agency.

I will rethink the status of the object in ethical theory in two ways. First, I will offer a “nonhumanist” analysis of two criteria that are usually seen as conditions *sine qua non* for moral agency. An entity can be called a moral agent if it can be morally responsible for its actions, and to be morally responsible, it needs at least (1) *intentionality*—the ability to form intentions—and (2) the *freedom* to realize its intentions. I will show that these two criteria can be reinterpreted along postphenomenological lines in such a way that they also pertain to nonhuman entities. Second, I will investigate the possibility that the predominant ethical approaches propose to take seriously the moral dimension of technologies. By elaborating what role objects could play in

deontological ethics, consequentialism, and virtue ethics, I will create the space needed to take the moral significance of technologies seriously.

### Technological Intentionality

The first criterion for moral agency—the possession of intentionality—directly raises a serious problem for anyone who intends to defend some form of moral agency for technology. While agency is not thinkable without intentionality, it also seems absurd to claim that artifacts can have intentions. Yet a closer inspection of what the concept of intentionality can mean in relation to what artifacts actually “do” makes it possible to articulate a form of “technological intentionality.”

The concept of intentionality actually has a double meaning in philosophy. In ethical theory, it primarily expresses the ability to form intentions. In phenomenology, though, the concept of intentionality indicates the directedness of human beings toward reality. Intentionality is the core concept in the phenomenological tradition for understanding the relation between humans and their world. Rather than separating humans and world, the concept makes visible the inextricable connections between them. Because of the intentional structure of human experience, human beings can never be understood in isolation from the reality in which they live. They cannot simply “think” but always think *something*; they cannot simply “see” but always see *something*; they cannot simply “feel” but always feel *something*. As experiencing beings, humans cannot but be directed at the entities that constitute their world. Conversely, it does not make much sense to speak of “the world in itself.” Just as human beings can be understood only through their relation with reality, reality can be understood only through the relation human beings have with it. “The world in itself” is inaccessible by definition, since every attempt to grasp it makes it a “world for us,” as disclosed in terms of our particular ways of understanding and encountering it.

In the context of this discussion of the possibility of “artifactual moral agency,” these two meanings of the concept of intentionality augment each other. The ability to form intentions to act in a specific way, after all, cannot exist without being directed at reality and interpreting it in order to act in it. Actually, the two meanings of intentionality have a relation to each other similar to the relation between the two dimensions of technological mediation that I discerned in chapter 1. The “praxical” dimension, concerning human actions and practices, cannot exist without the “hermeneutical” dimension, concerning human perceptions and interpretations—and vice

versa. Forming intentions for action requires having experiences and interpretations of the world in which one acts.

From the perspective of technological mediation, both forms of intentionality are not as alien to technological artifacts as at first they might seem. As for the phenomenological interpretation of the concept: the work of Ihde shows that the human-world relations that are central in the phenomenological tradition often have a technological character. Many of the relations we have with the world take place “through” technologies or have technologies as a background—ranging from looking through a pair of glasses to reading temperature on a thermometer, from driving a car to having a telephone conversation, from hearing the sound of the air conditioner to having an MRI scan made. Ihde shows that intentionality can work *through* technological artifacts, it can be directed *at* artifacts, and it can even take place against the *background* of them.

In most of these cases—with an exception for human relations that are directed *at* artifacts—human intentionality is *mediated* by technological devices. Humans do not experience the world directly here but via a mediating technology that helps to shape a relation between humans and world. Binoculars, thermometers, and air conditioners help to shape new experiences, either by procuring new ways of accessing reality or by creating new contexts for experience. These mediated experiences are not entirely “human.” Human beings simply could not have such experiences without these mediating devices. This implies that a form of intentionality is at work here—one in which both humans and technologies have a share. And this, in turn, implies that in the context of such “hybrid” forms of intentionality, technologies do indeed “have” intentionality—intentionality is “distributed” among human and nonhuman entities, and technologies “have” the nonhuman part. In such “hybrid intentionalities,” the technologies involved and the human beings who use the technologies share equally in intentionality.

The ethical implications of the second meaning of the concept of intentionality are closely related to those of the first. Intentions to act in a certain way, after all, are always informed by the relations between an agent and reality. These relations, again, have two directions; one pragmatic, the other hermeneutic. Technologies help to shape actions because their scripts evoke given behaviors and because they contribute to perceptions and interpretations of reality that form the basis for decisions to act. In the Netherlands, to give an example in the pragmatic direction, experiments are done with crossings that deliberately include no major road. The script of such crossings contributes to the intention of drivers to navigate extra carefully in order to be able to give priority to traffic from the right (Fryslân Province, 2005).

Genetic diagnostic tests for hereditary breast cancer, as mentioned in chapter 1, are a good example in the hermeneutic direction. Such tests, which can predict the probability that people will develop this form of cancer, transform healthy people into potential patients and translate a congenital defect into a preventable defect: by choosing to have a double mastectomy now, you can prevent breast cancer from developing in the future. Here technologies help to interpret the human body; it organizes a situation of choice and also suggests ways to deal with this choice.

In all of these examples, technologies are morally *active*. They help to shape human actions, interpretations, and decisions that would have been different without these technologies. To be sure, artifacts do not have intentions as human beings do, because they cannot *deliberately* do something. But their lack of consciousness does not take away the fact that artifacts can “have” intentionality in the literal sense of the Latin word *intendere*, which means “to direct,” “to direct one’s course,” “to direct one’s mind.” The intentionality of artifacts is to be found in their directing role in the actions and experiences of human beings. Technological mediation therefore can be seen as a distinctive, material form of intentionality.

There is another element that is usually associated with intentionality, though, and it is one that technologies seem to miss: the ability to form intentions that can be considered *original* or *spontaneous*, in the literal sense of “springing from” or “being originated by” the agent possessing intentionality. Yet the argument above can be applied here as well. For even though because of their lack of consciousness artifacts evidently cannot form intentions entirely on their own, their mediating roles cannot be entirely reduced to the intentions of their designers and users. If they could be, the intentionalities of artifacts would merely be a variant of what John Searle called “derived intentionality” (Searle 1983), entirely reducible to human intentionalities. Quite often, though, as pointed out already, technologies mediate human actions and experiences in ways that were never foreseen or desired by human beings.

Some technologies are used in different ways from those their designers envisaged. The first cars, which could go only 15 km/h, were used primarily for sport and for medical purposes; driving at a speed of 15 km/h was thought to create an environment of “thin air,” which was supposed to be healthy for people with lung diseases. Only after cars were interpreted as a means of long-distance transport did the car come to play its current role in the division between labor and leisure (Baudet 1986). In this case, unexpected mediations come about in specific use contexts. Unforeseen mediations can also emerge when technologies *are* used as intended. The introduction of mobile phones

has led to a different way of dealing with appointments, especially for young people—making plans far in advance for a night out does not make much sense when everyone can call each other anytime to make an ad hoc plan. This change in behavior was not intended by the designers of the cell phone, even though the phone is being used in precisely the context the designers had envisaged. And nobody foresaw that the introduction of the energy-saving lightbulb would actually cause people to use more rather than less energy. Apparently such bulbs are often used in places previously left unlit, such as in a garden or on the front of a building, thereby canceling out their economizing effect (Steg 1999; Weegink 1996).

It seems plausible, then, to attribute a form of intentionality to artifacts—albeit a form that is radically different from human intentionality. The intentional “dimension” of artifacts cannot exist without human intentionalities supporting it; only within the relations between human beings and reality can artifacts play the mediating roles in which their “intending” activities are to be found. For example, when expectant parents face a decision about abortion on the basis of technologically mediated knowledge of the chances that the child will suffer from a serious disease, this decision is not “purely” human, but neither is it entirely induced by technology. The very situation of having to make this decision and the very ways in which the decision is made are coshaped by technological artifacts. Without these technologies, either there would not be a situation of choice or the decision would be made on the basis of a different relation to the situation. Yet the technologies involved do not *determine* human decisions. Moral decision making is a joint effort of human beings and technological artifacts. Technological intentionalities are one component of the eventually resulting intentionality of the “composite agent,” a hybrid of human and technological elements.

Strictly speaking, then, there is no such thing as “technological intentionality”; intentionality is always a hybrid affair involving both human and nonhuman intentions, or, better, “composite intentions” with intentionality distributed among the human and the nonhuman elements in human-technology-world relationships. Rather than being “derived” from human agents, this intentionality comes about in associations between humans and nonhumans. For that reason it could best be called “hybrid intentionality” or “composite intentionality.”

### Technology and Freedom

A second requirement that is often connected to moral agency is the possession of freedom. If moral agency entails that an agent can be held mor-

ally responsible for his or her actions, this requires not only that the agent needs to have the intention to act in a particular way but also that he or she has the freedom to realize this intention. Now that we have concluded that artifacts may have some form of intentionality, can we also say that they have *freedom*?

The answer obviously seems to be no. Again, freedom requires the possession of a mind, which artifacts do not have. Technologies cannot be free agents as human beings are. The only degree of freedom that could be ascribed to them is their “ability” to have unintended and unexpected effects, like the increase in energy use brought on by the energy-saving lightbulb. But this is not freedom, of course, in the sense of the ability to choose and to have a relation to oneself and one’s inclinations, needs, and desires. Still, there are good arguments not to exclude artifacts entirely from the realm of freedom.

First of all, even though freedom is obviously required if one is to be accountable for one’s actions, the thoroughly technologically mediated character of our daily lives makes it difficult to make freedom an absolute criterion for moral agency. This criterion might exist in a radical version of Kantian ethical theory, where freedom is understood in terms of autonomy and where the moral subject needs to be kept pure of polluting external influences. But many other ethical theories take into account the situated and mediated character of moral agency. People do not make moral decisions in a vacuum, after all, but in a real world, which inevitably influences them and helps to make them the persons they are. The phenomenon of technological mediation is part of this. Technologies play an important role in virtually every moral decision we make. The decision how fast to drive and therefore how much risk to run of harming other people is always mediated by such things the layout of the road, the power of the car’s engine, the presence or absence of speed bumps and speed cameras. The decision to have surgery or not is most often mediated by all kinds of imaging technologies and blood tests, which help to constitute the body in specific ways and organize specific situations of choice.

Moral agency, therefore, does not require complete autonomy. *Some* degree of freedom can be enough for one to be held morally accountable for an action. And not all freedom is taken away by technological mediations, as the examples of abortion and driving speed make clear. In these examples, human behavior is not determined by technology but rather coshaped by it, with humans still being able to reflect on their behavior and make decisions about it. Nevertheless, we can in no way escape these mediations in our moral decision making. The moral dilemmas of whether to have an abortion and of how fast to drive would not exist in the same way without the technologies

involved in these practices. Such dilemmas are rather *shaped* by technologies. Technologies cannot be defined away from our daily lives. In this respect, technologically mediated moral decisions are never completely “free.” The concept of freedom presupposes a form of sovereignty with respect to technology that human beings simply do not possess.

This conclusion can be read in two distinct ways. The first is that mediation has nothing to do with morality at all. If moral agency requires freedom and technological mediation limits or even annihilates human freedom, only non-technologically mediated situations leave room for morality. Technology-induced human behavior then has a nonmoral character. Actions that are not products of our free will but induced by technology cannot be described as “moral.” This position does not help us much further, though. Denying that technologically mediated decisions can have a moral character throws out the baby with the bathwater, for it prevents us from conceptualizing the undeniably moral dimension of making decisions about unborn life on the basis of ultrasound imaging.

Therefore, an alternative solution to the apparent tension between technological mediation and ethics is needed. Rather than taking freedom from (technological) influences as a *prerequisite* for moral agency, we need to reinterpret freedom as an agent’s ability to *relate* to what determines him or her. Human actions always take place in a stubborn reality, and for this reason, absolute freedom can be attained only if we ignore reality and thus give up the ability to act at all. Freedom is not a lack of forces and constraints; rather, it is the existential space human beings have within which they can realize their existence. Humans have a relation to their own existence and to the ways it is coshaped by the material culture in which it takes place. The materially situated character of human existence *creates* forms of freedom rather than impeding them. Freedom exists in the possibilities that are opened up for human beings so that they might have a relationship with the environment in which they live and to which they are bound.

This redefinition of freedom, to be sure, does not imply that we need to actually attribute freedom to technological artifacts. Yet it does make it possible to take artifacts back into the realm of freedom, rather than excluding them from it altogether. Just as intentionality appeared to be distributed among the human and nonhuman elements in human-technology associations, so is freedom. Technologies “in themselves” cannot be free, but neither can human beings. Freedom is a characteristic of human-technology associations. On the one hand, technologies help to *constitute* freedom by providing the material environment in which human existence takes place and takes its form. And on the other hand, technologies can form associations with hu-

man beings, which become the places where freedom is to be located. Technological mediations create the space for moral decision making. Just like intentionality, freedom is a hybrid affair, most often located in associations of humans and artifacts. In chapter 4, which deals with the role of the technologically mediated subject in ethical theory, I will give a more extensive reinterpretation of the concept of freedom in relation to moral agency and technological mediation.

### Material Morality and Ethical Theory

By rethinking the concepts of intentionality and freedom in view of the morally mediating roles of technology, I have dispatched the major obstacles to including technological artifacts in the domain of moral agency. But how does this redefined notion of moral agency relate to mainstream ethical theory? Can it withstand the obvious deontological and consequentialist objections presented by Swierstra? And how does it relate to virtue-ethical approaches?

Let me start by discussing the deontological approach. The deontological argument against attributing moral agency to nonhumans revolves around the fact that objects lack rationality. Applying Kant's categorical imperative—the most prominent icon of deontological ethics—to things immediately makes this clear: “Act only in accordance with that maxim through which you can at the same time will that it become a universal law” (Kant [1785] 2002, 37). Technologies are obviously not able to follow this imperative—unless maybe they embody an advanced form of artificial intelligence. Yet that does not necessarily imply that there is no room for nonhuman moral agency in deontological ethics at all. It implies only that technologies cannot have moral agency *in themselves*. The position I have laid out in this chapter is based on the idea that the moral significance of technology is to be found not in some form of independent agency but in the technological *mediation* of moral actions and decisions—which needs to be seen as a form of agency itself.

Technologically mediated moral agency is not at odds with the categorical imperative at all. After all, technological mediation does not take away the rational character of mediated actions and decisions. A moral decision about abortion after having had an ultrasound scan can still be based on the rational application of moral norms and principles—and even on the Kantian question whether the maxim used could become a universal law. However, the rational considerations that play a role in the decision may be thoroughly technologically mediated. As we saw, the ways in which ultrasound constitutes the fetus and its parents help to shape the moral questions that are relevant and also the answers to those questions. The moral decision to have an

abortion or not is still made by a rational agent—but it cannot be seen as an *autonomous* decision. Human beings cannot alter the fact that they have to make moral decisions in interaction with their material environment.

Latour made an attempt to expand Kant's moral framework to the realm of nonhumans by providing a "symmetrical" complement to the categorical imperative. In *Groundwork for the Metaphysics of Morals* Kant actually gave several formulations of his categorical imperative. While the formulation given above is the so-called first formulation, Latour focused on the second, which reads "Act so that you use humanity, as much in your own person as in the person of every other, always at the same time as end and never merely as means" (Kant [1785] 2002, 46–47). In his book *Politics of Nature* Latour augmented this formulation with the imperative to act in such a way that you use *nonhumans* always at the same time as ends and never merely as means (Latour 2004, 155–56). In this way he tried to make room for ecological issues in ethical thinking; such issues by definition require us to bring nonhuman reality into the heart of ethical reflection.

This reformulation of the categorical imperative, though, approaches nonhumans primarily as moral *patients*, while the approach I develop here is primarily interested in nonhumans as moral *agents*—or, better, as active moral mediators. But Latour's reformulation leaves room for this other interpretation as well. "Using nonhumans at the same time as means and as ends," after all, can imply that using a technological artifact brings in not only means but also "ends"—the ends that are implied in the means of technology. Because of their mediating capacities, after all, technologies belong not only to the realm of means but also to the realm of ends (cf. Latour 1992b). And this makes possible a paraphrase of yet another formulation of the categorical imperative. Kant's third formulation reads "Every rational being must act as if it were through its maxims always a legislative member in a universal realm of ends"—but the approach of technological mediation makes clear that not only "rational beings" but technologies as well are "members in the universal realm of ends."

With regard to *consequentialist* ethics, the same line of argument applies. Utilitarianism, as the predominant variant of consequentialism, seeks to assess the moral value of actions in terms of their utility. This utility can be located in various things: the promotion of happiness (Jeremy Bentham's "greatest happiness for the greatest number of people"), the promotion of a plurality of intrinsically valuable things, or the fulfillment of as many preferences as possible. Obviously, technological artifacts are generally not able to perform an assessment like this—with the possible exception of artificially

intelligent devices. Yet such assessments are not products of autonomous human beings either. In our technological culture, the experience of happiness, the nature of intrinsically valuable things (like love, friendship, and wisdom), and the specific preferences people have are all technologically mediated.

Making a utilitarian decision about abortion, to return again to this example, clearly illustrates this. A hedonistic-utilitarian argument in terms of happiness, for instance, inevitably incorporates a thoroughly technologically mediated account of happiness. The medical norms in terms of which the fetus is represented, and the fact that ultrasound makes expectant parents responsible for the health of the unborn child, changes how abortion is connected to the happiness of the people involved here. Similarly, a preference-utilitarian argument will rest upon preferences that are highly informed by the technology involved. Preferences to have a healthy child, to avoid feelings of guilt if a child is born with a serious disease, or to prevent a seriously ill child from threatening the happiness of other children in the family—to mention just a few preferences that are likely to play a role in this case—could not exist without the whole technological infrastructure of antenatal diagnosis and abortion clinics.

From a *virtue-ethical* position it is much easier to incorporate the moral roles of technologies. As Gerard de Vries has noted (de Vries 1999), this pre-modern form of ethics does not focus on the question of “how should I act” but on the question of “how to live.” It does not take as its point of departure a subject that asks itself how to behave in the outside world of objects and other subjects. It rather focuses on “life”—human existence, which inevitably plays itself out in a material world. From this point of view, it is only a small step to recognize with de Vries that in our technological culture, not only ethicists and theologians answer this question of the good life but also all kinds of technological devices tell us “how to live” (ibid.). The next chapter, in which I will discuss the technologically mediated moral subject, will give a more extensive elaboration of the importance of classical virtue-ethical conceptions for understanding the moral significance of technologies.

### Conclusion: Materiality and Moral Agency

Technologies appear to be thoroughly moral entities—yet it is very counterintuitive to attribute morality to inanimate objects. In this chapter I have developed a way to conceptualize the moral significance of technological artifacts which aims to do justice to both of these observations by developing the concept of moral mediation in the context of ethical theory. This concept

makes it possible to address the moral significance of technologies without reverting to a form of animism that would treat them as full-blown moral agents. The example of the gun, used at the beginning of this chapter, can also serve as a conclusion. Now we can come to a more nuanced picture of the moral significance of a gun. Rather than simply stating that it would be ridiculous to blame a gun for a shooting and using this as an argument against the moral agency of technology, we can find our way to a more sophisticated understanding via the concept of moral mediation. After all, it would not be satisfactory either to completely deny the role of the gun in a shooting.

This related to an example explored by Latour: the debate between the National Rifle Association in the United States and its opponents. In this debate, those opposing the virtually unlimited availability of guns use the slogan “Guns Kill People,” while the NRA replies with the slogan “Guns don’t kill people; people kill people” (Latour 1999, 176). The NRA position seems to be most in line with mainstream thinking about ethics: if someone is shot, nobody would ever think of holding the gun responsible. Yet the antigun position also has a point: in a society without guns, fewer fights would result in murder. The problem in this discussion, however, is the separation of guns and people—of humans and nonhumans. Only on the basis of such a modernist approach does the question “can technologies have moral agency?” become a meaningful problem. From an amodern perspective, as I suggested in chapter 2, this question leads us astray. It seeks to find agency in technology *itself*, isolated from its relations with other entities, human and nonhuman.

A gun is not a mere instrument, a medium for the free will of human beings; it helps to define situations and agents because it offers specific possibilities for action. A gun constitutes the person holding the gun as a potential killer and his or her adversary as a potential lethal victim. Without denying the importance of human responsibility in any way, we can conclude that when a person is shot, agency should not be located exclusively in either the gun or the person shooting, but in the assembly of both. The English language even has a specific “amodern” word for this example: *gunman*, as a hybrid of human and nonhuman elements. The gun and the man form a new entity, and this entity does the shooting.

The example illustrates the main point of this chapter: in order to understand the moral significance of technology, we need to develop a new account of moral agency. The example does not suggest that artifacts can “have” intentionality and freedom, just as humans are supposed to have. Rather, it shows that (1) intentionality is hardly ever a purely human affair—most often it is a matter of human-technology associations; and (2) freedom should

not be understood as the absence of “external” influences on agents but as a practice of *dealing* with such influences or mediations. Chapter 4 will further explore this new understanding of moral agency—not from the perspective of the object but from the point of view of the technologically mediated *subject*.

## Technology and the Moral Subject

### Introduction

How should we understand the moral subject in our technological culture?<sup>1</sup> Once the ways that technologies help to shape moral practices and decisions have been examined, it makes sense to ask what this analysis implies for the conceptualization of the subject engaged in these practices and decisions. Is it possible to approach ethics not as a solely human affair but as a matter of associations between humans and technologies? Just like objects, mediated subjects do not automatically meet the common criteria for moral agency. How are we to conceptualize this mediated moral subject? How can we grasp its moral character when its actions and decisions do not result entirely from autonomous choice and proper intentions?

In this chapter I will try to answer these questions by engaging in a critical discussion of the work of Michel Foucault. Foucault's oeuvre embodies precisely the tension that needs to be dealt with if we are to understand the technologically mediated moral subject. Foucault's early work focuses on the forces and structures that determine the subject, or better, that produce specific subjects. And his analysis of the Panopticon prison design in *Surveiller et punir* (*Discipline and Punish*) shows that material artifacts can be reckoned among these forces and structures (Foucault 1975). Human intentions are not "authentic" but result from structures of power that can also be present materially; instead of being autonomous, human beings are heteronomous. The later Foucault, however, addressed the ways in which human beings can find a relation toward structures of power. He did so in the new ethical perspective that he developed in volumes 2 and 3 of his *History of Sexuality*. In this perspective—which is less known but is currently being rediscovered in such approaches as "life ethics" and "aesthetics of existence" (Schmid 1991; Schmid 1998)—he does not revoke his earlier analyses but investigates how,

amid these structures of power, human beings can constitute themselves as (moral) subjects. Humans are not only the objects of power here but also subjects that create their own existence against the background of and in confrontation with these powers.

This shift makes Foucault's work highly important for the ethics of technology. Not only was he one of the first to discern the moral significance of material artifacts and the constitutive role of objects in the coming about of subject definitions, but he also tried to articulate a redefinition of ethics beyond the concept of the autonomous moral agent. Especially interesting is the fact that he connects with ethical approaches from classical antiquity that focus on the question of *the good life*. Because of their premodern character, such approaches can be helpful for articulating an ethical approach beyond the autonomous subject. Foucault's work makes it possible to redefine the concept of freedom in a way that it is in line with the phenomenon of technological mediation.

In this chapter, I will investigate to what extent Foucault's analysis of subject-constitution and his association with classical Greek ethics could form the basis for an ethical framework that can do justice to the moral significance of technological artifacts and the mediated character of moral agency. First I will discuss Foucault's work on power from the perspective of the philosophy of technology and technological mediation. After that, I will elaborate the "ethical turn" in his work in order to develop an interpretation of the moral subject that incorporates the phenomenon of technological mediation rather than being at odds with it.

### The Power of Technology

Foucault is not generally considered a philosopher of technology. Only in the last phase of his work did he explicitly use the word *technology*, and there it was to indicate what he called "technologies of the self," which are primarily existential techniques and not technological artifacts. Yet his work contains highly relevant contributions to the philosophy of technology. Foucault's scholarship is generally divided in three periods, the first focusing primarily on knowledge, the second on power, and the third on ethics. Several authors have argued that his work on power is especially relevant to the philosophy of technology (Gerrie 2003; Sawicki 2003).

The American philosopher Jim Gerrie, for instance, shows that Foucault's analyses of power are very much in line with a particular strand in the philosophy of technology that approaches technology not in terms of specific material devices and appliances but as "a set of structured forms of action by

which we inevitably also exercise power over ourselves” (Gerrie 2003, 14). The philosophy of Martin Heidegger is a good representative of this approach. In Heidegger’s philosophy, which is still influential today, technology plays a role not as technological objects but as a way of “disclosing” reality. Technology, in Heidegger’s terms, is “a way of revealing”: it is an approach to reality in which entities derive their meaning from the ways they can be manipulated and used as raw material for the human will to power. In our technological era, according to Heidegger, we are immersed in a technological way of thinking and interpreting the world. This ruling way of disclosing reality colors and steers all of our actions and perceptions; it is the ground for our existence in this technological era. Heidegger thus developed a hermeneutic approach to technology: the technological way of interpreting reality is the central focus of his work, not the technological artifacts themselves (cf. Verbeek 2005b, 47–98).

In Foucault’s work, power plays a role that is comparable to yet also different from technology’s role in Heidegger’s work. For Foucault, power is what structures society and culture. The ways we live, think, and act are all shaped by structures of power—just as they are shaped by technology in Heidegger’s approach. But rather than monolithically analyzing power as a particular volition or a metaphysical relation to reality, as in the Heideggerian perspective, Foucault investigates how structures of power are at work in concrete practices, objects, and ideas. Human existence does not take place in a vacuum but in a world made of ideas, artifacts, institutions, organizations that all have impacts on human subjectivity. Vocabularies and scientific theories help to shape how we think, our material environment organizes our actions, and social institutions like schools, hospitals, armies, and prisons give shape to how we live our lives and deal with illness, criminality, and madness. Technology, as I will show, can be seen as one of these sources of power that help to shape the subject.

#### THE POWER OF TECHNOLOGY

In *Discipline and Punish*—probably his most relevant work for the philosophy of technology—Foucault elaborates how in modern society a new form of power has developed, which he calls *disciplinary power*. Many practices and institutions have come to regulate, discipline, and normalize the human subject. Prisons regulate the behavior of prisoners, schools extensively train and drill the behavior and even gestures of pupils, hospitals draw new boundaries between normal and abnormal, healthy and ill, sane and insane. Even though the modern subject, as a product of the Enlightenment, has of-

ten been approached as autonomous and transparent, Foucault shows in many ways how it is actually (literally) *subjected* to a multitude of powers.

These powers, to be sure, are not simply oppressive and alienating. Rather, they are *productive* in the sense that they produce the subject in ever new ways. And this production of subjects occurs in a very material and concrete fashion. Training practices for writing at schools, the architectural designs of prisons, drilling techniques to learn to operate weapons, observation and surveillance techniques—all these arrangements that operate directly on the body help a specific subject to come about. “Modern technologies do not control the body by conquering it (as did the techniques of torture and execution under sovereign power), but by simultaneously rendering it more useful and docile” (Sawicki 2003, 62). Subjects are produced by being “subjected.” Specific forms of power introduce and enforce specific forms of normalcy and abnormalcy, which generate a specific subject.

Such forms of power do not necessarily find their origin in powerful human individuals. Rather, power is simply something that is “at work” through everyday practices, ideas, and objects and that can be operative without the explicit initiation of human agents. One of the most important and well-known examples Foucault uses to illustrate the workings of disciplinary power beyond human agency is Jeremy Bentham’s “Panopticon.” The Panopticon is a prison design that ensures optimal observability of prisoners. It consists of a ring of cells that can be observed from a central tower. Prisoners cannot see whether they can be observed, but the very possibility of being observed effectively regulates their behavior. This design completely organizes its environment. Not only the prisoners but the guards, too, are part of the disciplinary machinery, because they need to observe the prisoners at least occasionally for the design to be functional. Thus the Panopticon for Foucault provides a forceful image of what disciplinary power is. In our disciplinary society “power takes the form of self-control and does not necessarily represent a system of rules only imposed from without, but a system of rules we also self-impose in order to create and maintain a functioning community, or society” (Gerrie 2003, 20).

This way that power operates beyond human intentions and actions links Foucault’s analysis to Heidegger’s work in yet another way. For Heidegger, after all, technology is not “a human activity” either. Just as power in Foucault’s work cannot be understood in terms of underlying human actions and intentions, in Heidegger’s approach technology is not the outcome of human actions. Rather, it is a way of taking up with reality in which we always already find ourselves. We cannot help interpreting reality in a technological way, because this framework is the only context within which reality can be

interpreted. And for Heidegger, this technological relation to reality also produces a particular subject. Actually, the very notion of a “subject” is the product of a modernist way of thinking in which human beings approach their world as a reality “out there,” represented “in” a knowing subject, as we saw in chapter 2 (cf. Heidegger 1977a). The technological way of thinking radicalizes this modern subject. Not only does it separate subject and object rather than starting from a notion of “being-in-the-world,” it also equips the subject with the power to dispose of the objects.

The Foucauldian subject, as the product of power, however, is a subject of flesh and blood. It does not only *think* but also has a *body*. It is trained, disciplined, and observed. Whereas Heidegger places technology primarily in the history of ideas, Foucault follows a much more empirical approach. Because of this, his analysis of power offers many points of application that connect it to technological mediation. The design of the Panopticon prison, for instance, can be analyzed entirely in terms of its mediating role in human actions and perceptions. Power can be “at work” through the material environment in the form of technological mediation. From this perspective, the subject in a technological world is a product not only of metaphysical thinking but also of material mediation.

Read in this way, Foucault’s work can be seen as an intermediate step between Heidegger’s philosophy on the one hand and the work of Latour and Ihde on the other. Just like technology for Heidegger, power does not have an “author” or a “prime mover”; it is simply at work and organizes human existence. And just like technologies for Latour and Ihde, power can work through material artifacts and not only through interpretive ways of “revealing.” Studying technological mediation necessarily involves studying power on a micro level. By mediating our actions and perceptions, technologies form a structure of power, disciplining, organizing and normalizing the subject.

#### RESISTANCE AND FREEDOM

Now that we have seen that Foucault’s work can help us analyze the social role of technology, how can it help us further in conceptualizing the technologically mediated moral subject? If the subject is so profoundly shaped by structures of power, after all, how much room remains for moral agency? As Ladelle McWorther puts it: “If it is the case that power is the source of conscience and self-knowledge, then it would appear that individual selves have no control over their own beliefs and hence their own actions; agency is an illusion” (McWorther 2003, 114). This conclusion would bring us back to

where we started: that it is very complicated to integrate technological mediation in the notion of the moral subject.

A more adequate conclusion can be drawn from Foucault's analysis, however. As McWorther rightly showed as well, power should not be seen as forces that operate on the subject "from the outside"; rather, what subjects are and do, and how they understand themselves, comes about in relations and networks of power. "Selves are not constrained by powers external and foreign to them. Relations and networks of power *are* selves, *are* subjects" (ibid., 114). But how does this help us to escape the conclusion that agency—especially *moral* agency—is merely an illusion? Even if the self and the subject are produced rather than oppressed by power, the subject does not seem to have a "prime mover" status in its actions. And can such a subject be a moral subject? Or does the subject need to find a way out of the realm of power and technology in order to regain morality?

For Heidegger, the answer to this question was clear. The only way out of the technological framework is an attitude of *releasement*. This attitude, for Heidegger, consists in "the will not to will." Every attempt to design a new relation to reality would only reconfirm the hegemony of the will to power. The only possibility we have is to wait in openness for a new way of engaging with reality to develop. Foucault's account of power, by contrast, leaves much more room for a human role in changing power structures. His historical analyses show the *contingency* of the structures of power that are at work in society. They could have been otherwise, and therefore human beings can change them. In Jana Sawicki's words: "Freedom lies not in the discovery of essential features of the human situation, in complete mastery of reality, or in releasement"; it rather lies in the relations people develop toward the "dominating powers of technology" (Sawicki 2003, 69).

Yet the way one approaches such relations toward power is crucial. Sawicki, quoting John Rajchman (1985, 62), describes the freedom of the subject in terms of "*rebell*ing against the ways in which we are already defined, categorized, and classified" (cited in Sawicki 2003, 69, emphasis mine). Where power is there is also the possibility of resistance—this seems to be the parallel of the poetic answer to technology Heidegger finds in Hölderlin: "But where danger is, grows / the saving power also" Heidegger 1977b, 28). In this view, the freedom of the subject is to be found in resistance and opposition.

This is a popular reading of Foucault's work. In the ethics of information technology, for example, analyses of privacy threats by new media are often inspired by the Foucauldian image of a panoptic "surveillance society" that we should oppose (cf. Lyon 2006). This, however, is not the approach I will take here. When power is what makes us the subjects we are, after all,

a merely subversive and rebellious attitude toward power does not offer a real alternative. I read Foucault's concept of power in a hermeneutic way rather than incorporating it in a Marxist dialectic of oppression and resistance. In a hermeneutic reading, human beings derive their subjectivity from their interplay with the structures of power in society—just as entities in general derive their meaning in interaction with the context in which they exist. Opposing these power structures would be nonsensical, since every attempt to escape them can be made only in terms of these powers themselves.

If the dialectical scheme is of any use here, it is only to show that any "antiposition" is necessarily formulated in terms of the position it aims to counter, and that a "synthesis," a "sublation," is needed if we are to move beyond this oppositional scheme. Such a reflexive step reveals that thesis and antithesis define each other and need to be seen as elements of a more encompassing reality—just as adolescence largely consists in opposing childhood, while adulthood starts with a positive discovery and development of one's own personhood.

What such a "sublation" can entail—to use a final dialectical image—is pregnantly described by Nietzsche's *Zarathustra* in his lecture about "the metamorphoses of the spirit." Zarathustra tells how the spirit initially was a camel, bearing much and kneeling down to be loaded with more. After that it became a lion, which opposes the camel and replaces the oppressive "Thou shalt" of the camel with a subversive "I will." The lion, however, embodies only a "sacred No"; it exists by opposing what it does not want to be, and is not able to create something *new*. For that, the spirit needs to take the shape of a child: a "sacred Yes," a "new beginning," a "first movement" (Nietzsche [1883] 1969). In somewhat dramatic terms, this parable illustrates how the subject can be articulated, not *over against* but *in relation to* the powers that shape it.

A more positive relation toward power needs to be articulated in order that the freedom of the subject can be made visible. The needed relation does not merely *oppose* power but recognizes that subjectivity is shaped in interaction with power—including the technological mediations that are central in this book—and does not let this happen passively but actively *engages* in it. Rather than only undermining structures of power, subjects can also take them as a starting point in order to contribute actively to the way they are constituted as subjects. Instead of being merely subversive and antithetic, the subject can be engaged, seeking to shape itself in the context of "the powers that be."

This relation to power opens up a different form of freedom. It is to be

found not in the absence of influences that constrain the subject but, rather, in *dealing with* these influences. Freedom becomes an *activity*, a practice of dealing with power, not a desirable final state of the subject in the absence of power. As Leslie Paul Thiele has argued, “Foucault insists that freedom is not something to be secured, like the individual rights and opportunities that Isaiah Berlin described as negative liberty. Freedom is an activity to be engaged” (Thiele 2003, 225–26). In Foucault’s own words: “The claim that ‘you see power everywhere, thus there is no room for freedom’ seems to me absolutely inadequate. The idea that power is a system of domination that controls everything and leaves no room for freedom cannot be attributed to me” (Foucault 1997a, 293). The freedom of the subject does not consist in being liberated from power but in interacting with it. One becomes a subject not by securing a place outside the reach of power but by shaping one’s subjectivity in a critical relation to it.

This approach is in line with a hermeneutical reading of Foucault’s concept of power. Power helps to shape human subjects by providing a context for their existence, but its ability to operate depends on how it is interpreted and constituted in the actual relations humans have with it. Moreover, this freedom-in-relation-to-power can form the basis for a specific way of *dealing with* power. Freedom does not need to take the form of subversion and looking for an escape. In the words of James Bernauer and Michael Mahon: “If one side to . . . resistance is to ‘refuse what we are,’ the other side is to invent, not to discover, who we are by promoting ‘new forms of subjectivity’” (Bernauer and Mahon 2005, 155). Dealing with power in practices of freedom opens the possibility of modifying its impact on human subjectivity.

It is this approach to freedom as a *practice of subject constitution* that connects Foucault’s work on power with his later work on ethics. The ethical work of Foucault focuses on the constitution of moral subjectivity. It is not rules, norms, and codes for behavior that form the center of this ethical approach but the ways in which human beings constitute themselves as moral subjects. And as Steven Dorrestijn has convincingly shown (Dorrestijn 2004), this ethical approach offers many interesting points of application to articulate a fruitful approach to technology. As I will show, inspired by Dorrestijn’s work, in our technological culture moral self-constitution inevitably takes place in relation to the structures of power in the lifeworld of the subject—including technological mediations. From this approach, technological mediation is not a threat to the moral subject but rather a starting point for it. Therefore ethics should not position itself in opposition to power but incorporate power into its approach to morality and moral agency.

### Morality and the Subject of Power

The late work of Michel Foucault opens a perspective on ethics that can serve as a way to do justice to the intricate relations between ethics and technology and to the technologically mediated character of moral action. In the last two volumes of his *History of Sexuality* he elaborates an ethical approach that differs radically from most predominant ethical frameworks (Foucault [1984] 1990a, [1984] 1990b). For Foucault, ethics is not primarily about which imperatives we need to follow and how we need to act, but about how human beings constitute themselves as “subjects” of a moral code. And rather than aiming to develop a new code himself, Foucault investigates what these codes “do” to people and how humans “subject” themselves to them.

Foucault argues that any moral system or approach consists of three elements: morality encompass not only a *moral code* people have to comply with and the *behavior* corresponding to this code but also the way human beings constitute themselves as *moral subjects* that follow this code (Foucault 1992, 25–32). A moral code of chastity, for instance, regulates the sexual behavior of human beings, and for doing to it requires moral subjects that organize their lives in such a way that they can subordinate their passions to the code. Ethics, for Foucault, primarily concerns this third element of morality: the ways in which human beings constitute themselves as moral subjects. The word *subject* aptly expresses that ethics is not only a matter of a person who is the “subject” of his or her actions—like the grammatical “subject” of a sentence—but of a person who also “subjects” himself or herself to a moral code, a vision of what constitutes a good life or good behavior. The moral subject is not an *autonomous* subject; rather, it is the outcome of active *subjection*.

The moral subject has already taken many forms, such as the Kantian subject that aims to keep its intentions pure, assessing them in terms of their potential to function as universal laws, or the utilitarian subject that aims to examine the consequences of its actions in order to attain a prevalence of positive outcomes over negative outcomes. These forms of “subjection” required by specific moral systems usually remain implicit. Ever since the Enlightenment it has been self-evident that one considers one’s intentions and one’s capacity to balance desirable and undesirable consequences as the domain of ethics. Foucault’s approach to ethics questions the naturalness of this assumption. He argues that any form of ethics requires a moral subject and is therefore necessarily based on a form of “subjection.” Every moral system defines not only a code of behavior but also a subject that is supposed to follow this code. Following the Kantian categorical imperative or the con-

sequentialist principle of utility requires a moral subject, after all, and this “subjects” a particular aspect of one’s person to particular criteria.

#### AUTOPOIÈSIS: ETHICS AND AESTHETICS

In classical antiquity, however, Foucault discovers an ethical approach that did *not* implicitly define a moral subject but explicitly directed itself at the constitution of one’s moral subjectivity. Foucault’s investigations of classical ethics were primarily directed at sexuality. He argues that in ancient Greece, sexuality was organized not primarily via a moral code of imperatives and prohibitions but rather in terms of styling one’s dealings with pleasure. Ethics consisted in finding a relationship to one’s sexual desires and drives such that these did not determine the self but became the object of active “design.”

This “design” or “styling” took the form of what Foucault called “self practices,” ways to experiment with and give shape to one’s way of dealing with pleasure. Foucault also termed them “technologies of the self” (Foucault 1997a, 223–52). Rather than simply following one’s passions and desires, self practices aimed at gaining a productive distance. From this distance, the subject itself, not pleasure, could have the central place in determining how it lives life. The purpose of such self practices was not to resign one’s passions to a code but to give shape to one’s dealings with pleasure. In a variety of ascetic and aesthetic practices, the subject was shaped in an explicit way, rather than being a product of obedience to an external code. Ethics was not primarily about showing morally right behavior; its main focus was not the question “how should I act?” but “what kind of subject do I want to be?” Ethics was a matter of “care of the self”: paying careful attention to one’s subjectivity and shaping one’s life in a desirable way.

This approach introduces an aesthetic dimension into the realm of ethics. Rather than approaching morality in terms of moral obligations, Foucault draws attention to “styles” in moral subjectivity. And rather than making it possible to produce normative judgments about actions, he focuses on “designing” one’s life in “self practices.” In classical ethical approaches, seen through Foucault’s eyes at least, the good life becomes a work of art, and ethics becomes a “technology of life”—*techné tou biou* (cf. O’Leary 2002, 53).

This aesthetic dimension, however, is not as alien to ethics as it might seem to be from our modernist frameworks. First of all, for the ancient Greeks there was a close connection between the ethical and the aesthetic. The classical Greek word *kaloskagathos* testifies to this; it indicates a unity of the good and the beautiful (cf. O’Leary 2002, 53 128). Second, this aesthetic dimension

should be understood in the context of the classical Greek conception of art, which is closer to “craft” and embodies a close relation between technique and aesthetics. Art was conceived as *technè*, and as such it was a form of *poièsis* (making), just like *physis* (nature) (cf. Heidegger 1977b). While *physis* pertained to that what “makes itself,” *technè* belonged to the realm of what comes into being with the help of a human being. *Technè*, therefore, comprises both works of art and useful objects—just as *craft* comprises both technical skills and aesthetic refinement. It is form-giving, designing, helping to come into being.

*Techné tou biou*, therefore, should be understood as the “art of living” in the broadest sense; it consists in shaping one’s subjectivity and giving it a style that can be good and beautiful at the same time. This aesthetic elaboration of ethics did not imply a relativist reduction of ethics to a choice of style—rather, giving style to one’s existence was an ethical activity. Returning to Foucault’s analysis of the ethics of sexuality: whereas Christian thought approached sexuality in terms of *renunciation*, the ethical choice of the ancient Greeks was *styling* one’s way of dealing with sexual pleasures, rather than losing oneself in their intensity (cf. May 2006, 110). This styling was a form of moderation—not because pleasures were sinful temptations but because the aim was to develop a free relation to them. It is ethical because it is explicitly directed at the question “how are we to live?” In O’Leary’s words: “For Foucault, the aesthetic is the realm in which we work to develop techniques which will allow us to give a form to our lives, our behavior and our relationships with others” (O’Leary 131–32).

Foucault’s turn toward the aesthetics of existence, after having analyzed the pervasive role of power structures in society, forms yet another link to the work of Heidegger. Both authors give art a prominent role in finding a relation to technology—even though their views are wide apart. For Heidegger, art forms an important answer to the predominating technological way of approaching reality. Art embodies a radically different approach. When experiencing a work of art, we experience a reality coming into being. The materials used by the artist—pigment, canvas, bronze, sounds—open up a new world. This way of coming into being is radically different from technological manipulation and “ordering” of reality. It makes visible the phenomenon of coming-into-being itself. As such, art has the potential to preserve a non-technological approach to reality that does not reduce reality to a product of the human will to power.

Foucault’s turn toward art is radically different from Heidegger’s. For Foucault, art does not embody a contemplative openness to a different way of disclosing reality. Rather, it addresses structures of power by actively engag-

ing with them, shaping one's subjectivity in a productive interaction. Regarding technology, as I will elaborate below, this results not in a Heideggerian attitude of "releasement" but in giving direction to the technological mediation of one's subjectivity. Not refraining from technology but recognizing its constitutive role in human existence will be the starting point for a Foucault-inspired ethics of technology. Before we examine this in more detail, though, we need to take a closer look at the various aspects of the constitution of the subject.

#### CONSTITUTION OF THE MORAL SUBJECT

How might we conceptualize the phenomenon of moral subject-constitution? How might we understand the ways in which human beings shape their subjectivity in relation the world in which they live? In chapter 3 of the introduction to *The Use of Pleasure* (Foucault 1992, 25–28), Foucault distinguishes four aspects of the constitution of moral subjectivity. First of all, there is the *ethical substance*: the part of oneself that is "subjected" to a moral code and that becomes the object of ethical work. Second, there is the *mode of subjection* that is applied: the specific ways in which people are invited to put themselves under obligation. Third, Foucault distinguishes the *self practices*: the self-forming activities that shape the ethical substance into an ethical subject. And fourth, there is the *teleology* of these practices, which consists in the way of existing we aim to realize by acting in a moral way.

The *ethical substance* concerns what people take as the "material" of ethics. It is, so to speak, the point of application for giving shape to one's moral subjectivity: the object of ethical self-work. In Foucault's own words, the ethical substance is "the way in which the individual has to constitute this or that part of himself as the prime material of his moral conduct" (Foucault 1992, 26). Various aspects of the self have served as such a point of application. As Todd May points out, "Many modern philosophers believe that behaviour is the ethical substance. For others, like Kant, the ethical substance is the will. It need not be either of these, however. It can be the soul, or desire, or the emotions or passions" (May 2006, 108; cf. Foucault 1997a, 253–80). Any form of ethics requires that a specific aspect of the self is "subjected," molded, brought under control. In classical antiquity, according to Foucault, the primary ethical substance of sexual ethics was the *aphrodisia*—the pleasures (cf. O'Leary 2002, 12).

The second aspect of moral subject constitution is what Foucault calls the *mode of subjection*. By this he means "the way in which people are invited or incited to recognize their moral obligations" (Foucault 1997a, 264). In other

words, the mode of subjection is what makes people subject themselves to a moral code. As Foucault points out, many such modes of subjection have come into being through the centuries. The “authority” that causes people to subject themselves to a given moral code has taken on many different shapes—for example, a divine law revealed in a book, a cosmological order of natural law, or a universal and rational rule (ibid.). All these modes of subjection represent answers to “the question of my relation to what is asked of me, and in that relation I establish myself as a particular kind of ethical subject” (May 2006, 108). In the time of the Greeks, the mode of subjection was the “free, personal choice . . . to give one’s life a certain noble, perfect, or beautiful form” (O’Leary 2002, 12).

The third element in moral self-constitution is the *self practices* or the *ethical work* in which the moral subject is actively shaped. Beside identifying the aspect of oneself that should be “subjected” and the mode of this subjection, moral subject constitution requires activities of “shaping oneself.” Foucault describes it as the work “that one performs on oneself, not only in order to bring one’s conduct into compliance with a given rule, but to attempt to transform oneself into the ethical subject of one’s behavior” (Foucault 1992, 27). He calls these self-forming activities forms of *ascesis* (ibid., 72–77). With this concept, to be sure, Foucault does not aim to defend a form of austerity. Ascesis does not necessarily consist in radically *abandoning* things such as comfort, sex, or rich foods, to mention some ascetic examples from the past. What is crucial in asceticism for Foucault is that the subject develop a *distance* from anything that otherwise remains self-evident, in order to find a productive relation to it. From an ascetic distance, the subject is not simply handed over to the powers that shape it but explicitly *takes a stance* toward these powers, actively accompanying and reshaping them. Rather than letting the pleasures become the central focus of one’s existence, the Greeks advocated developing a style in dealing with them. They did not renounce sex as a necessary but indecent activity, but sought to know “how and when to indulge. . . . There is no shame in sex; there is, however, a shame in overindulgence. One needs to engage moderately” (May 2006, 110).

*Teleology*, to conclude, takes up the question of what kind of beings we aspire to be when we behave morally. What do we aim at when we subject ourselves to a moral code? What kind of subjects do we want to be? Shaping one’s subjectivity is inevitably guided by some ideal of subjectivity and human existence. Every ethical system is implicitly or explicitly driven by a particular ideal of moral subjectivity. Foucault lists a number of examples from the past: “Do we want to become pure, or immortal, or free, or masters

over ourselves?” (1997a, 265). In classical antiquity the principal ideal was self-mastery—not being mastered by somebody or something else but gaining a free relation to the surrounding powers (cf. O’Leary 2002, 12).

Gilles Deleuze has argued that these four aspects of moral subject constitution mirror the four forms of causation that Aristotle elaborated in his *Physics* (Deleuze 1988, 104; cf. O’Leary 2002, 85). Causation, for Aristotle, it what helps an entity to come into being (*Physics* 2.3; Aristotle 1970, 129–31)—and for that reason it is no surprise that the classical Aristotelian approach to causation mirrors Foucault’s understanding of how human beings help their own subjectivity to come into being. For Aristotle, a cause is not a “prime mover” or a kind of “fountainhead” of reality. Rather, it is one of the things to which an entity owes its existence. Aristotle distinguishes four forms of causality: the material cause (*causa materialis*), the material from which an entity comes into existence; the formal cause (*causa formalis*), the form that this materiality should take; the efficient cause (*causa efficiens*), the “incentive” to help an entity come into being—this is currently the most common meaning of the word *cause*; and the final cause (*causa finalis*), or the goal or aim for the sake of which the entity comes into being.

Read in this way, Foucault’s moral substance has the role of the material cause of one’s subjectivity, the mode of subjection is its efficient cause, self-practices are its formal cause, and teleology, of course, the final cause. This Aristotelian framework also makes clear that subject constitution should not be understood as some kind of postmodern voluntaristic activity in which human beings can design any subjectivity for themselves that they want. The classical conception of causation, as Heidegger noted, needs to be understood in terms of “being indebted to” or “owing existence to,” rather than as indicating forces of power that create reality. Causation is cooperation, being involved in a process of helping to come into being. Similarly, ethics for Foucault is being engaged in a process of subject constitution, cooperating with the powers that be. And as I will show in the next section, technology is one of these powers.

#### FREEDOM AND THE MODERN SUBJECT

By articulating this Greek account of ethics, Foucault did not of course aim to return to the times of classical antiquity. He did not seek to reintroduce the ancient ethics of dealing with pleasure but was primarily interested in re-conceptualizing the moral subject. In view of his work on power, after all, the predominant view of the moral subject as a rational, autonomous subject is

highly problematic. A subject that is at least partly a product of the structures of power that surround it cannot be an autonomous subject. At the same time, however, Foucault's ethical approach shows that power relations do not by definition *dominate* the subject. His ethical approach seeks a middle ground between autonomy and domination, the *freedom* of the subject as it is formed in self-practices in relation to the powers that surround it. As he stated in an interview, in reaction to Jürgen Habermas's critical attitude toward power: "I do not think that a society can exist without power relations. . . . The problem, then, is not to try to dissolve [power relations] in the utopia of completely transparent communication, but to acquire the rules of law, the management techniques, and also the morality, the ethos, the practice of the self, that will allow us to play these games of power with as little domination as possible" (Foucault 1997b, 298).

Foucault's concept of freedom needs to be read in the context of his particular relation to modernity. In his 1978 lecture *What Is Enlightenment*, he developed a rebellious interpretation of what enlightenment can be, providing an alternative to the predominating view of the modern subject as an *autonomous* subject. Foucault goes on to argue that in the work of Kant "Enlightenment" does not play the role of the birth of an autonomous and rational subject; it rather indicates a "way out," an "exit"; it is "a process that releases us from the status of immaturity." And immaturity, for Kant, is "a certain state of our will which makes us accept someone else's authority to lead us in areas where the use of reason is called for" (Foucault 1997a, 305). Enlightenment is a form of liberation. Foucault consequently describes modernity not as "a set of features characteristic of an epoch" but as an *attitude*. Modernity is "a mode of relating to contemporary reality" (*ibid.*, 309). Foucault characterizes this attitude as a "limit attitude"; it actively looks for the limits of any situation and state of affairs. Its central question is "In what is given to us as universal, necessary, obligatory, what place is occupied by whatever is singular, contingent, and the product of arbitrary constraints?" (*ibid.*, 315).

Kant's "critique" can be seen as a variant of this limit attitude, indicating "the moment that humanity is going to put its own reason to use, without subjecting itself to any authority" (*ibid.*, 308). But the "limit attitude" can take other forms as well which do not necessarily lead to the autonomous and rational subject that is commonly seen as "the modern subject." By questioning the self-evidence of the autonomous subject, Foucault aims to challenge what he calls "the blackmail of the Enlightenment." This blackmail consists in the conviction that one can only be "for" or "against" the Enlightenment

and that whoever dares to criticize the Enlightenment is directly accused of “trying to escape its principles of rationality” (ibid., 312–13).

Foucault claims that his problematization of the self-evidence of “the constitution of the self as an autonomous subject” is rooted in the Enlightenment just as much as is Kant’s work, because both approaches embody the limit attitude of modernity (ibid., 312). Foucault’s reinterpretation of what it means to be modern thus allows him to articulate a nonautonomous yet modern subject. In his version of the modern subject, Foucault replaces the notion of autonomy with freedom. Against the splendid isolation of the pure, rational, and autonomous subject, Foucault develops a situated and historically contingent subject that has the freedom to develop a relation to its own contingency and to contribute actively to how it is constituted.

This approach is congenial to yet different from Latour’s approach to modernity. Foucault shares with Latour a critical approach toward modernity. Both approaches move beyond the autonomous subject and the radical separation of subjects and objects. Foucault would agree with Latour that we have never been “modern” in that sense; modernity is an *attitude* for which the autonomous subject is not a given but a historical phenomenon that is contingent rather than self-evident. Yet for Foucault it is crucial to realize that we are products of the Enlightenment nonetheless. While Latour claims that “we have never been modern,” Foucault rather seeks to work out a *different interpretation* of modernity. He does not claim that the autonomous modern subject has never existed, but he approaches it as a contingent historical entity: “We must try to proceed with the analysis of ourselves as beings who are historically determined, to a certain extent, by the Enlightenment” (ibid., 313). Foucault chooses not to dissolve the distinction between subject and object in a radical symmetry, as Latour does, but to articulate the subject in its relations to objects, including relations of power. In our technological age, these objects and power relations are primarily technological. For that reason, as I will show below, Foucault’s ethical approach can be a very fruitful framework for understanding the relations between technological mediation and the moral subject.

### Technological Mediation and Moral Subjectivity

Foucault’s analysis of the constitution of moral subjectivity offers many relevant points of application for understanding the technologically mediated moral subject. Foucault’s ethical perspective unites two elements that usually remain opposites in ethics: on the one hand the radically mediated character

of human actions, which causes the subject to lose the autonomy it was assumed to have ever since the Enlightenment, and on the other hand the ability of the subject to relate itself to the powers that help to shape it, which makes it possible to modify their impact. While Foucault directs his attention to the role of the pleasures in ancient Greece, in our technological culture technology is a preeminent example of the powers that help to shape the subject. Not only the religious frameworks, views of life, and philosophical systems handed down to us impose moral obligations and visions of the good life upon us, but so do technological artifacts. By helping to shape our actions and interpretations, technologies also help to shape what can be recognized as a moral obligation, what constitutes a good life, and what moral responsibilities we have. And by finding a relation to these mediations, incorporating them in our existence, human beings can further shape and stylize their moral subjectivity.

Just as the Greeks in classical antiquity did not have to deny or renounce the pleasures to be moral subjects, we need not deny the technologically mediated character of our existence to be moral subjects. If technology fundamentally mediates what kind of humans we are, this does not imply that “humanity” is mastered by “technology,” as those advocating some Heideggerian positions want us to believe. Neither does it imply that “the system” has entered “the lifeworld” and causes humans to be treated not as subjects but as objects. In such a Habermasian approach, technology is connected only to an instrumental form of rationality that threatens the lifeworld of the human subject (Habermas 1969), without taking into account how deeply technological objects have always been interwoven with the various forms of the lifeworld. From a Foucauldian perspective, the technologically mediated character of life today does not form a threat to the subject. As Steven Dorrestijn has elaborated in his study of the ethics of behavior-influencing technology, technology forms a way in which the subject is constituted, which can be the starting point for moral self-practices (Dorrestijn 2004, 89–104). Ethics, then, should not aim at protecting “humanity” from “technology” but should consist in carefully assessing and experimenting with technological mediations, in order to explicitly fashion the ways they help to shape subjects in our technological culture.

#### TECHNOLOGICALLY MEDIATED SUBJECT CONSTITUTION

The four aspects of moral subject constitution I discuss above shed a new light on the relations between technological mediation and moral subjectivity. In our technological culture, moral codes—as one of the three elements

of morality, beside behavior and ethics—are embodied in technological devices, appliances, and systems. These “material codes” help to shape our behavior and our moral subjectivity. Ultrasound imaging, to return to the same example, does have an impact on human behavior, but only because it also has an impact on moral subjectivity: it “subjects” a specific aspect of human beings. At the same time, though, as I will elaborate below, humans can develop a relation to its mediating roles and get involved actively in how they are constituted as moral subjects.

In terms of Foucault’s ethical “fourfold,” the preeminent *ethical substance* in our technological culture is our technologically mediated subjectivity. The focus of “self-care” should be the close interaction between humans and technologies, in which technologically mediated subjects come into being. Just like pleasures, intentions, and rational arguments, mediating technologies contribute to the actions and decisions of the subject. And therefore the impacts of technology on one’s subjectivity are what should be at stake in an ethics of self-constitution. Rather than being passive objects of mediating technologies, then, human beings develop an engaged relation to technological mediations and actively contribute to the ways their mediated subjectivity is formed.

The *mode of subjection* in the context of technological mediation subsequently exists in the specific shapes that technological mediations take. Technological artifacts, as I will explain more extensively in chapter 5, can have an impact on the moral subject in many different ways. First of all, there is a difference between explicit and implicit forms of mediation. While some technologies, like speed bumps and coin locks, were explicitly designed to influence the behavior of their users, other technologies, like ultrasound imaging, mediate human actions and decisions without having been designed to do so. These are entirely different forms of mediation, which organize different responses from the users of these technologies.

Second, mediating technologies can have different “modes” of impact. Technologies can actually *force* people to behave in specific ways, like a speed bump that makes it impossible to drive at high speed without damaging your car. But mediation can also take the form of *persuasion*: by giving feedback on people’s actions, technologies can convince people that they should behave differently. Experiments have been done, for instance, with washing machines that give feedback about the economically and environmental impacts of the ways they are used: whether they are filled enough, whether the water filter needs to be cleaned, and so on (McCalley and Midden 2006).

Third, technologies can also mediate by means of *seduction*—which can be seen as a noncognitive form of persuasion. Shop design is a good example

here: the specific ways in which various goods are displayed helps to seduce people to buy specific things.

In a technological context, *self practices* consist in using technology deliberately by anticipating and modifying its mediating role in our existence, realizing that each way of using it also helps to shape our subjectivity. Such self practices could be described as “techniques of using technology.” They require a form of ascesis, without implying, again, that one should refrain from technology or use it only reluctantly with a Heideggerian attitude of “releasement” (*Gelassenheit*). Technological ascesis consists in using technology resolutely but in a deliberate and responsible way, such that the subject that emerges from it—including its relations to other people—acquires a desirable shape. As I will elaborate in the next chapter, “techniques of technology use” require *experimentation*. The distance needed to gain a free relation to technology and to modify and shape its impact on our existence can be obtained only by deliberately allowing technologies to play their mediating roles in different settings. Human beings can coshape their technologically mediated subjectivity by styling the impacts of technological mediations.

Besides the *use* of technologies in deliberate ways in order to modify their mediating impact on human subjectivity, the *design* of technologies can be an important “self practice” in our technological culture. Any technological design is the starting point of an artifact that not merely is instrumental but also helps to shape the subjectivity of its users. As chapter 5 will show, designing technologies from this perspective can be both an enrichment of existing design methods and an important way to “care for the self.”

The central *teleological* question in a technological culture, to conclude, is “what kind of mediated subjects do we want to be?” Integrating Foucault’s analysis of moral subject constitution and the postphenomenological analysis of technological mediation, a teleological perspective in our technological culture should address the question of how to shape our selves in dealing with technology. Rather than separating the human domain from the domain of technology, we need to ask ourselves in what ways we want both domains to interface. Rather than trying to save humanity from technology, the aim of our moral self-constitution should be to let humanity and technology blend in desirable ways.

For answering the question of what kind of mediated subjects we want to be, to be sure, the ethical frameworks from classical virtue ethics and modern deontological and utilitarian systems can continue to play important roles. Foucault’s thesis that any ethical system eventually generates a particular form of subject constitution, after all, does not take away the fact that the frameworks that were handed down to us from the past may still prove to

be valuable for dealing with the technological mediation of our subjectivity and the question of what kind of subjects we want to be. Moral self practices in a technological culture, in which human beings attempt to give a desirable shape to the technological mediation of their subjectivity, offer plenty of space for the virtue-ethical pursuit of the good life, the deontological ambition to meet moral norms, and the utilitarian goal of reaching a preponderance of positive effects over negative effects.

As I stated above, the central concept in Foucault's ethical approach is *freedom*—which can be seen as a generalization of the Greek ideal of self-mastery. For Foucault, the *telos* of subject constitution is freedom. Again, this notion of freedom does not consist in an absence of power but in gaining a new relation to power. As Foucault said in an interview: "I am sometimes asked: 'But if power is everywhere, there is no freedom.' I answer that if there are relations of power in every social field, this is because there is freedom everywhere" (Foucault 1997b, 292). It is not power that should be rejected but domination, as the "perversion of power" that takes away the possibility of freedom (O'Leary 2002, 158).

Freedom in Foucault's work functions both as the condition for ethics and as its ultimate aim (O'Leary 2002, 154–70). On the one hand, there can be ethical behavior only when people are not completely dominated by power; on the other hand, ethics consists in developing "practices of freedom" in which people interact with power to constitute their subjectivity. Given this, the Foucauldian concept of freedom offers an interesting alternative to the criterion of autonomy that is often used in ethical theory. While the concept of autonomy stresses the importance of the absence of "external influences" in order to keep the moral subject as pure as possible, the concept of freedom recognizes that the subject is formed in *interaction* with these influences. The subject is not what remains when all powers and mediations are stripped from it; it is what results from an active designing and styling of the impact of these powers and mediations. The core of a Foucauldian ethics of technology is gaining a free relation to technology, which allows one to style the way one's technologically mediated subjectivity is shaped.

#### THE SUBJECT OF OBSTETRIC ULTRASOUND

Again, obstetric ultrasound can helpfully illustrate the implications of this ethical approach. As we have seen, ultrasound substantially contributes to the experience of expecting a child by framing pregnancy in medical terms and by confronting expectant parents with a dilemma if their unborn appears to have a significant risk of a serious disease. From a moral point of

view, this role of ultrasound imaging is at least as important as, for example, the possible health risk to the fetus of ultrasonic sound waves, which would be the natural focus of many ethical approaches to technology. This is especially true when taking into account that such dilemmas have a tragic dimension. As explained above, the risk estimation provided by ultrasound can be converted into certainty only by having an amniocentesis done, which has a risk of provoking a miscarriage—and in many cases this risk is higher than the risk of having a child with Down syndrome. From the Foucauldian perspective elaborated in this chapter, what is at stake here is the impact of ultrasound imaging on our moral subjectivity. What kind of subjectivity is implicitly organized by the mediating role of this technology? And how could human beings deal with this, “styling” this mediation in order to constitute a positive form of technologically mediated moral subjectivity?

Having antenatal ultrasound examinations done inevitably implies the choice of a kind of subjectivity in which humans are constituted as subjects that have to make decisions about the life of their unborn child, and in which obtaining certainty about the health of the unborn child is worth the price of losing a healthy unborn child as a result of the required test. The moral substance here consists of the moral actions and decisions of expectant parents mediated by ultrasound, while the mode of subjection is an implicit form of mediation that helps to shape how the fetus can be interpreted. When this moral impact of ultrasound becomes the subject of moral reflection, we gain the space to explicitly relate ourselves to it from a Foucauldian telos of freedom.

This relation to ultrasound imaging can take the form of self-practices in which the resulting mediated subjectivity is modified, changed, and refined—for instance, by using ultrasound only for determining an approximate date of birth, without gaining further information about nuchal translucency or neural tube defects. Or by using antenatal examinations only to estimate a risk, in order to be prepared for the possible birth of a child with health problems, without running the risks of having an amniocentesis done. Or by having all tests done, as an explicit choice rather than an unintended side effect of the normative workings that are hidden behind the provision of such diagnostic tests on a large scale. Or by refusing ultrasound examinations at all (cf. Rapp 1998). In all these cases there is a deliberate shaping of the ways humans are being constituted as moral subjects, based on the realization that technology plays a mediating role here too. Human beings are not fully autonomous in their subject constitution; they have to accept both the pregnancy and the possibility of having ultrasound screening done as given facts. But they do have the freedom to let themselves be constituted as

subjects—subjects that will have to decide about the life of its unborn child; subjects that orient themselves based on norms that exist separately from the situation in which they need to be applied; or subjects that want to use the availability of a technological form of contact with unborn life for a careful assessment of all possible consequences of letting a child be born with a serious disease.

### Conclusion: Moral Agents and Mediated Subjects

Foucault's approach to moral subjectivity makes it possible to articulate a notion of moral agency that includes the notion of technological mediation. While the technologically mediated character of human actions and decisions at first seemed to be incongruent with the freedom that is required to be a moral agent, the Foucauldian concept of freedom offers a fruitful way out of this tension. When we approach freedom not as the absence of limits and constraints but in terms of a *relation* to these, a rich understanding of technologically mediated moral agency becomes possible. In this approach, technological mediation is not the *end* but rather the *starting point* of moral agency. Except in cases of complete domination, where technological mediation makes room for force and compulsion, the mediated character of actions and decisions appears not to obstruct moral agency at all. Rather, it is a point of application for a sophisticated form of moral agency: the careful coshaping of one's moral subjectivity.

In earlier chapters I used a definition of moral agency that was derived from the concept of moral responsibility; in order to be held morally responsible for one's actions, an agent needs to have at least the *intention* to act in a specific way and the *freedom* to realize that intention. The Foucauldian understanding of freedom examined above makes possible an expanded understanding of moral responsibility. Rather than being a mere product of technological mediations, the mediated subject becomes responsible for the form its mediated subjectivity takes. Its actions are not simply results of technological determination but spring from an active appropriation of technological mediations. By engaging with the ways that technologies help to shape one's actions in and interpretations of the world, a moral subject actually takes on a double moral responsibility: one for its moral subjectivity and another for its actions and the way it lives its life. A free subject acts but also "cares" for its moral subjectivity, which forms the basis for moral actions and decisions. Seen from the perspective of predominant views of moral agency, self practices can be seen as a form of meta-agency—agency directed at shaping one's agency.

This Foucauldian reinterpretation of moral agency overcomes the separation between humans and technology that characterizes so many ethical approaches. In these approaches, as we saw, humans are placed on one side of the line, technologies on the other side, and humans must see to it that technologies do not cross the line and begin to interfere in the human world in undesirable ways. This separation underlies precautionary approaches that aim to pull the emergency brake when a certain technological development might be a threat to society. It also underlies approaches that aim to find the most prudent and just way to deal with the risks connected with the introduction of a new technology. Positions like these show perfectly that very close relations can exist between humans and technologies—contrary to the at least equally influential position of *instrumentalism*, which (wrongly) holds that technology is primarily an instrument that can be used for good and bad purposes. Rather than taking the interwoven character of the human and the technological as a point of departure for ethical reflection, this approach treats the technological as a threat that needs to be kept away from the human with the help of ethics.

Such approaches, as we saw in the previous chapters, fail to take into account how moral actions and decisions are thoroughly technologically mediated. Accepting this interwoven character of subjects and objects has large implications, though. Not only does it imply that technological artifacts become morally relevant and that moral subjectivity becomes technologically mediated, but it also requires that we recognize that morality develops alongside technology. Because of their interwoven character, morality cannot claim a “pure” and isolated position outside the realm of the technology at which it is directed. A good example of this interwoven character was elaborated by Gerard de Vries, who showed how the moral evaluation of anesthesia has changed drastically over time (de Vries 1993). While the application of anesthesia was initially severely condemned on various moral and theological grounds, nowadays it would be considered highly immoral to perform surgery *without* anesthesia. From a modernist perspective, the critics of the past could interpret this development only as the inevitable outcome of entering a slippery slope, but from an amodern perspective it illustrates that ethics is a dynamic phenomenon that develops in interaction with technology.

From an amodern ethical approach, as we have seen, ethics is a thoroughly hybrid affair. Not only do objects have moral significance, but subjects are also technologically mediated; technologies have moral qualities, and ethics is formed in interaction with technology. By expanding Foucault’s ethical work to the realm of technology, this chapter sketched the outlines of a amodern conceptualization of the moral subject. This subject makes moral decisions

and acts morally on the basis of its interweaving with the technologies it uses. By helping to shape the practices and experiences of human beings, these technologies also help to shape the actions and decisions of moral subjects. And conversely, by finding a relation to these technological mediations, the technologically mediated moral subject can “care for itself” by actively “designing” and “styling” the way it is formed in interaction with technology. Moral agency in a technological culture comprises acting morally in a world of technological objects and taking responsibility for one’s mediated moral subjectivity.

Self practices can entail more than merely styling the impact of technological mediations on one’s moral subjectivity, however. In addition to shaping our existence in relation to the mediating role of technologies, we can intervene in the *design* of technologies. The impact of technological mediations, after all, results not only from the roles human beings allow technologies to play in their lives but also from the characteristics of technologies that help to shape their mediating roles. This moral role of technology design will be explored further in chapter 5.

## Morality in Design

### Introduction

The analysis of the moral significance of technological artifacts I developed in the previous chapters has important implications for the ethics of technology and of technology design.<sup>1</sup> Even when designers do not explicitly reflect morally on their work, the artifacts they design will inevitably play mediating roles in people's actions and experience, helping to shape moral actions and decisions and the quality of people's lives. Technology design, therefore, appears to be an inherently moral activity. Designers cannot but help to shape moral decisions and practices. Designing is "materializing morality." For this reason, moral issues regarding technology development involve more than minimizing technological risks and disaster prevention, however important these activities are. All technologies-in-design will eventually mediate human actions and experiences, thus helping to form our moral decisions and the quality of our lives. So the ethics of technology design should take up these future mediating roles.

The phenomenon of technological mediation therefore burdens designers with a specific responsibility. Ethical questions regarding the design of technologies are not limited to questions about the goals for which technologies are designed and applied or the quality of their functioning. Since technologies are inherently moral entities, designers have a seminal role in the eventual technological mediation of moral actions and decisions. Designers are in fact practical ethicists, using matter rather than ideas as a medium of morality. Usually this "material ethics of design" remains implicit: designers shape a new technology with certain functionalities in mind, without explicitly aiming to influence the actions and behavior of users. The question, then, is how considerations regarding the mediating role that the

technology-in-design will eventually play in society could be explicitly integrated in the design process.

There are two possible ways to take technological mediation into account in design activities. A first, minimal option is that designers try to assess whether the product they are designing might have undesirable mediating capacities. A second goes much further: designers could explicitly try to “build in” forms of mediation that are considered desirable. Morality then, in a sense, becomes part of the intended “functionality” of the product. This second option immediately raises many questions, however. For how could a designer predict the eventual impact of the technology-in-design? Technologies can be used in unforeseen ways and can give rise to unforeseen mediations. Also, deliberately influencing human behavior through technology design is likely raise moral objections, because it might limit human freedom, and because of fears of a technocracy in which not humans but technologies are in control.

In this chapter I will analyze various implications of the moral significance of technologies for the ethics of design. After discussing the implicit and explicit design of technological mediations, I will consider how designers could anticipate and help to shape the moral mediations of their products. Also, I will extensively discuss the moral issues raised by the active “moralization of technology.” Moreover, I will propose some methods designers could use here and point to several examples. In this way I aim to translate the nonhumanist ethical approach set out in previous chapters into design practices. Rather than proving designers with instrumental tools and a list of issues to deal with when designing a technology, this chapter should be read as philosophy-in-practice, exploring the ways moral questions and issues are articulated in design processes and how designers could deal with the moral significance of the technologies they are designing.

### Designing Mediations

The theory of technological mediation reveals an inherently moral dimension in technology design. The fact that technologies always help to shape human actions and interpretations on the basis of which (moral) decisions are made has important implications for our understanding of the ethical roles of both technological artifacts and their designers. If ethics is about how to act and designers help to shape how technologies mediate action, designing should be considered a material form of doing ethics. Every technological artifact that is used will mediate human actions, and every act of design therefore helps to constitute moral practices.

## EXAMPLE: DESIGNING SUSTAINABLE TECHNOLOGY

A good first example of the need for a design approach that takes technological mediation seriously is the design of sustainable technology. In environmental policy, activities are usually focused either on the development and promotion of clean technologies—that is, technologies that have the smallest environmental impact—or on stimulating environmentally friendly behavior, mostly with the help of information campaigns directed toward changing attitudes and the behavior of the public. However, this two-track approach has proved to be highly unproductive (cf. Slob and Verbeek 2006).

An exclusive focus on developing clean technologies would risk overlooking the role of user behavior, which sometimes generates unexpected and unintended side effects. Such side effects can cancel out the desired effects of the technology. The increasing availability of safety devices in cars and the strongly reduced noise car engines produce, for instance, have created a safe-feeling and comfortable environment that invites people to drive faster, thus threatening the effect of the technological safety measures. However, an exclusive focus on influencing people's behavior with the help of information campaigns has severe limitations as well. This approach overlooks the important behavior-influencing roles of technology, and therefore it misses important opportunities. In some cases, technological devices can be more effective for changing human behavior than persuasive communication campaigns. Speed bumps, for instance, succeed much better in making people drive more slowly than information campaigns about the risks of driving too fast.

An interesting example here is the introduction of the so-called green bin (*groenbak*) in the Netherlands, in the context of Dutch environmental policy. The green bin is a separate bin, placed outside the house, in which only food- and garden-related waste is collected, with the help of a small bin inside the house. The introduction of the green bin was accompanied by an information campaign in which Dutch households were encouraged to separate organic from nonorganic waste. But one of the main problems of the small indoor bin proved to be the high speed of the decaying process during the summer months, causing a terribly bad smell and making the emptying of the bin a quite distasteful job. This often discouraged people from separating their waste—until a new product entered the practice of waste separation, that is; a small paper bag (nowadays made from biodegradable plastic) that can be placed inside the indoor bin, making it much easier to empty and clean. A material artifact appeared to be able to effect the change of behavior for which the information campaign appeared not to be strong enough.

A second example in the context of Dutch environmental policy concerns the emission of carbon dioxide (cf. Slob and Verbeek 2006). Curiously enough, the development of many new energy-saving appliances has led to an increase rather than decrease in energy consumption. Precisely because they are so cheap to use, people appear use them more intensively. This phenomenon is often called the “rebound effect” (Tenner 1996); the introduction of technologies to solve certain problems then appears to be counterproductive, resulting in precisely the opposite of what was intended. A technological device that has become significantly more energy efficient in a technological sense but that stimulated more energy-consuming behavior is the washing machine. Washing machines use ever less energy, but people increasingly use their machines for small quantities of laundry (Slob et al. 1996). Such rebound effects clearly show the need to bridge the gap between a technological approach and a behavioral approach; it is the interface between them that deserves the attention of policy makers and designers.

Several types of rebound effects can be distinguished. Unexpected effects of introducing a new technology might include users’ bypassing the technology or not using it at all, or using the technology in a way that differs radically from what designers intended. A “bypassing” rebound effect occurs frequently in the use of automatic control systems like motion detectors that switch on lights or a heating installation. When consumers prefer to be in control themselves, they devise ways to escape the control exerted by the system (Van Kesteren, Meertens, and Franssen 2006). A third type of rebound effect consists in a mismatch between the product’s design and the expectations and routines of users. Jaap Jelsma (2006), for example, has shown how many dishwasher users rinse their dishes under running hot water before loading them into the machine because they don’t realize that the machine starts its cycle by rinsing as well. Another example is energy-efficient houses, which are equipped with new insulation materials and sophisticated ventilation systems that give an optimal combination of fresh air and heat conservation. Many inhabitants of these houses appear to still open their windows to get fresh air. The strictly technology-oriented design approach followed in the design of these houses appears to have taken too little account the people who actually have to live and work with these devices and houses every night and day.

Environmentally oriented design practices could benefit from an integrated approach to technology and user behavior. Technology influences human behavior, and, conversely, existing patterns in human behavior influence the use and even the functionality of technologies. Developing an integrated approach to technology and behavior can help to prevent the occurrence of rebound effects and can augment information campaigns with

behavior-steering technologies. The mediation approach offers a fruitful framework for developing such an integrated approach as a first step toward articulating the ethical aspects of technology design.

#### MEDIATION THEORY AND THE ETHICS OF DESIGN

As stated in the introduction, there are two ways to take the theory of mediation into the ethics of technology design. First of all, mediation analyses can be used to develop *moral assessments* of technologies, evaluating the quality of their mediating roles in human practices and experiences and their impact on moral actions and decision. Second, mediations can be explicitly designed into a technology. The conclusion that the mediating role of technologies gives them a form of moral significance, after all, expands the realm of ethics from the domain of ideas to that of materiality. When artifacts have moral relevance, ethics cannot occupy itself only with developing conceptual frameworks for moral reflection but should also become engaged in actual development of material environments that help shape moral action and decision making.

The first way to take mediation into ethics is closest to common practices in the ethics of technology. It comes down to an augmentation of the currently predominating focus on risk assessment and disaster prevention. Rather than focusing only on the acceptability of new technologies and on minimizing negative consequences of their introduction, designers could also assess the impact of the mediating capacities of technologies-in-design in their use context, by performing a mediation analysis.

As I will demonstrate more extensively below, such assessments can be informed by various ethical approaches. When an action-ethical approach is followed, moral reflection is directed at the question of whether the actions resulting from specific technological mediations can be morally justified. This reflection can follow the deontological and utilitarian lines that predominate in the field of applied ethics. But in many cases a life-ethical approach—in line with Foucault's ethics of existence, for instance—is at least as fruitful here, because of its focus on the quality of the *practices* that are introduced by the mediating technologies and on the implications of technological mediations for the kinds of lives we are living and the kinds of subjects we become. Not only the impact of technologies on human actions is important, then, but also the ways technologies help to constitute human subjects, the world they experience, and the ways they live their lives.

The second way to augment the ethics of technology with the approach of technological mediation is to actively *shape* them. Designers here take a more

radical step and deliberately design technologies in terms of their mediating roles. When desirable mediating effects are inscribed in technologies, explicitly behavior-influencing or “moralizing” technologies will result. Rather than working from an external standpoint vis-à-vis technology, aiming only to either reject or accept a new technology, the ethics of technology then aims to *accompany* technological developments (cf. Hottois 1996)<sup>2</sup>, experimenting with mediations and looking for ways to discuss and assess how these mediations could fit with the way humans live.

Deliberately building mediations into technological artifacts, however, is a controversial thing to do. Not all behavior-steering technologies are welcomed warmly, as the regular destruction of speed cameras illustrates.<sup>3</sup> Yet since we have seen that *all* technologies inevitably mediate human actions and decisions, such forms of resistance should not keep designers from designing mediations into artifacts. Rather, the inevitability of technological mediation urges ethics to deal with these mediations in a responsible way, informing design practices and contributing to the development of technologies with morally justifiable mediating capacities.

In the Netherlands, the philosopher Hans Achterhuis defended this position. In his 1995 article “De moralisering van de apparaten” (The moralization of devices) he proposed taking Latour’s analysis of “nonhuman morality” into the realm of technology design (Achterhuis 1995; cf. Achterhuis 1998). Inspired by the idea that technologies have “scripts” that help to shape human action, Achterhuis made a plea for an explicit “moralization of technology.” Instead of moralizing only other *people* (“do not shower too long”; “buy a ticket before you enter the subway”), humans should also moralize their *material environment*. To a water-saving showerhead we could delegate the task of seeing to it that not too much water is used when we shower, and to a turnstile the task of making sure that only people who have bought a ticket can enter the train. Such delegations of morality to material objects would free human beings from the burden of some of the decisions that they are increasingly confronted with. Instead of having to continually reflect on the moral quality of our actions, we could hand over our least contested but most frequently occurring moral decisions to our material environment.

The recently developed field of “persuasive technology”—which will play a central role in chapter 6—offers good examples of such “moralized” (and “moralizing”) technologies. Here products are designed that persuade their users to behave in certain ways, like not driving too fast, giving up smoking, or using one’s household appliances in a way that saves energy.

Achterhuis’s plea for the moralization of technology has received severe criticism, however (cf. Achterhuis 1998, 28–31). In the debate that arose

around this issue in the Netherlands, several types of arguments were marshaled against his ideas. First, it was said that human freedom is undermined when human actions are explicitly and consciously steered with the help of technology. This reduction of human freedom was even perceived as a threat to human dignity; if human actions resulted not from deliberate decisions but from steering technologies, critics said, people would be deprived of what makes them human. Second, according to critics, if people are not acting freely their actions cannot be called “moral.” Instead of consciously choosing to act in a certain way, human beings would simply behave in the way desired by the designers of the technology. Third, Achterhuis was accused of attacking the democratic principles of our society, because the development of behavior-steering technology was considered to be an implicit propagation of technocracy. If moral issues were solved through the technological activities of designers instead of the democratic activities of politicians, these critics averred, not humans but technologies would be in control.

These arguments can be countered, though. Anticipating the mediating role of technologies during the design process—with the aim either of signaling undesired forms of mediation or of explicitly “moralizing” technologies—need not be as immoral as it might seem. First of all, human dignity is not necessarily attacked when freedom is limited. A nation’s legal constitution entails a significant limitation of freedom, after all, but this does not make it a threat to our dignity. Human behavior is determined in many ways, and human freedom is limited in many ways too. Few people will protest the legal prohibition of murder, so why protest the material inhibition imposed by a speed bump that prevents us from driving too fast at places where children are often playing on the pavement? Second, now that we have seen that technologies *always* are involved in shaping human actions and decisions, paying deliberate attention to the mediating role of technologies in fact comes down to accepting the responsibility that the analysis of technological mediation implies. When technologies are always influencing human actions and decisions, we had better try to give this influence a desirable and morally justifiable form.

The contested nature of behavior-steering technology, to be sure, does make clear that such “materializations of morality” cannot be left to the responsibility of individual designers. When the actions and decisions of designers have inevitable public consequences, their work should be subject to public decision making. Arrangements should developed, therefore, to democratize technology development. It might be true that technologies do not differ from laws in limiting human freedom, but laws are debated and established in a democratic way, while the moralization of technology is not.

The responsibility for technological mediation should not be left to designers alone—for that would amount to a form of technocracy. In what follows, I will outline a way to find democratic methods for developing desirable forms of “moralizing technology.” I will propose several ways to anticipate, design, and assess the mediating roles of technologies, methods that also open the possibility of making technology design a more democratic activity.

### Anticipating Mediations

An important ingredient of any ethical approach that aims to include the moral significance of technology is the *anticipation* of technological mediations. Any adequate ethical assessment of a technology needs to take into account how this technology will help to shape human practices and perceptions. Just so, a conscious “moralization” of technology requires anticipating the eventual effect of the intended mediations. In all cases, the ethics of design will need to anticipate the future mediating role of the technology-in-design.

This anticipation is a complex task, however. Because of the multistability of technologies, as discussed in chapter 1, there is no unequivocal relationship between the activities of designers and the mediating role of the technologies they are designing. The mediating role of technologies comes about in a complex interplay between technologies and their users. At the very moment human beings *use* them, artifacts change from mere “objects lying around” into artifacts-for-doing-something. And this “for doing something” is determined not entirely by the properties of the technology itself but also by the ways users handle them. Technologies have no fixed identity; they are defined in their context of use and are always “interpreted” and “appropriated” by their users. The telephone, for instance, was originally developed as a hearing aid, and the typewriter was supposed to assist people with poor eyesight in writing texts. If the interpretive step did not exist, accepting the idea of technological mediation would take us back to technological determinism: technologies would simply determine the behavior of their users rather than being part of a sociotechnical network in which entities derive their roles and identities from their relations with each other.

This multistability of technologies makes it complicated to predict the ways given technologies will influence human actions and to evaluate this influence in ethical terms. As the examples of the “rebound effect” in the previous section illustrated, technologies can be used in unforeseen ways and therefore have unforeseen influences on human actions. Another good example is the energy-saving lightbulb. As noted earlier, this promising innovation

has actually resulted in an increase rather than a decrease of energy consumption since its introduction (Steg 1999; Weegink 1996). Revolving doors are another interesting example. They were designed to make it possible for people to enter a building without letting cold air in. Once they were introduced, however, it became apparent that they also kept people in wheelchairs from entering. Technological mediations are not intrinsic qualities of technologies, then, but come to light in complex interactions between designers, users, and the technologies. That is why even though designers play a seminal role in realizing particular forms of mediation, they are not the only source of mediation.

Accordingly, the suggestion that technological scripts result from “inscriptions” (Akrich 1992) or “delegations” (Latour 1992b), which the work of Akrich and Latour offers, does not do justice to the complex way in which mediation comes about. Latour’s examples primarily focus on delegations from humans to nonhumans. Officials have speed bumps installed because they want people to drive slowly; homeowners install door-springs to prevent drafts; hotel owners attach weights to their keys to encourage guests to return them to the reception desk when leaving the hotel. This one-sided focus downplays the fact that nonhuman entities can also delegate tasks to human beings and that technological mediations are the result of specific forms of user appropriation.

Latour’s use of the concept of “inscription” adds to this seemingly asymmetrical approach to human and nonhuman entities. When scripts are the products of “inscriptions,” they are reducible to human activities, and this, to be sure, is not in line with the symmetrical approach Latour intends to set out. The same limitation can be found in Akrich’s initial introduction of the “script” concept: “Designers thus define actors with specific tastes, competencies, motives, aspirations, political prejudices, and the rest. . . . A large part of the work of innovators is that of *‘inscribing’* this vision of . . . the world in the technical content of the new object. I will call the end product of this work a ‘script’ or a ‘scenario’” (1992, 208, emphasis in original). This statement suggests that scripts are only the product of inscriptions—but actually the eventual script is the result of an interaction among the work of designers, who inscribe particular forms of mediation, users, with their interpretations and forms of appropriation, and the technological artifacts themselves, which sometimes give rise to unexpected forms of use and mediation.<sup>4</sup>

These complicated relations between technologies, designers, and users, which form the sources of mediation, are illustrated in figure 1. The figure makes clear that in all mediated human actions and interpretations, three

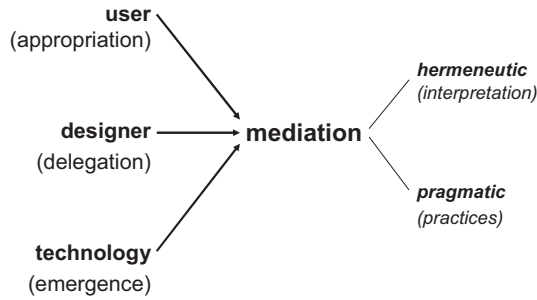


FIGURE 1. Diagram: agency and sources of mediation.

forms of agency are at work: (1) the agency of the human being performing the action or making the moral decision, in interaction with the technology, and appropriating the technological artifact in a specific way; (2) the agency of the designer, who, either unintentionally or in deliberate delegations, gives a shape to the technology and thus helps to shape its eventual mediating role; and (3) the agency of the technology mediating human actions and decisions, sometimes in unforeseen ways.

The fundamental unpredictability of the mediating role of technology that follows from this, however, does not imply that designers are by definition unequipped to deal with it. In order to cope with the complexity of technological mediation, designers should try to establish a connection between the context of design and the context of use. This will enable them to formulate product specifications and moral assessments on the basis not only of the desired functionalities of the product and their possible side effects but also of an informed prediction of its future mediating roles.

Three methods of bringing about such a connection between the contexts of design and use can augment each other in interesting ways. The first option is, simply, prediction by the designer's *imagination*. When designers attempt to imagine what mediating role the technology they are designing might play in the behavior of its users, they can feed back what they anticipate into the design process. An example of this approach is the work done by the Dutch industrial designers collective Eternally Yours, discussed below. A second way to formulate an informed prediction of the future mediating role of technologies consists in an augmentation of the methodology of Constructive Technology Assessment to make it an instrument for a democratically organized moralization of technology. The third possibility, the so-called *scenario method* makes use of a virtual-reality environment to experiment with a virtual version of the product while it is still in design.

## MORAL IMAGINATION

A key tool to establish a link between the design context and the use context, however trivial it may sound, is the designer's moral imagination.<sup>5</sup> A designer can include the product's mediating role in his or her moral assessment during the design phase by trying to imagine the ways the technology-in-design could be used and then shaping user operations and interpretations from that perspective. A mediation analysis can form a good basis for doing this—that is, an analysis of the future role of the technology-in-design in terms of the theory of mediation that was introduced in chapter 1. Designers then try to imagine the various contexts of use in which the technology-in-design could play a role, focusing on how the technology could help to shape specific practices and ways of taking up with reality and how it could shape experiences and ways of interpreting reality. To be sure, it can't be guaranteed that designers will be able to anticipate all relevant mediations, but nevertheless it is an important way in which designers can take responsibility for the mediating roles of their products.

An interesting example of anticipating mediation by imagination is the work of the Dutch industrial designers collective Eternally Yours.<sup>6</sup> Eternally Yours is engaged in ecodesign but works in an unorthodox way (cf. Van Hinte 1997; Van Hinte 2004; Muis 2006). It does not want to address the issue of sustainability only in the usual terms of reducing pollution in production, consumption, and waste. The actual problem, Eternally Yours holds, is that most of our products are thrown away long before actually being worn out. Addressing this problem could be much more effective than reducing pollution in the different stages of the life cycles of various products. For this reason, Eternally Yours focuses on developing ways to create product longevity. It does so by investigating how the attachment between products and users could be stimulated and enhanced.

In order to stimulate longevity, Eternally Yours seeks to design things that invite people to use and cherish them as long as possible. "It's time for a new generation of products, that can age slowly and in a dignified way, become our partners in life and support our memories," as Eternally Yours approvingly says on its letterhead, quoting the Italian designer Ezio Manzini. Eternally Yours investigates which characteristics of products are able to invite their users to form a bond with them. According to Eternally Yours, three dimensions can be discerned in the life span of products: a technical, an economical, and a psychological life span. Products can turn into waste because they simply are broken and cannot be further repaired; because they are outdated by newer models that have come on the market; or because they no longer

fit people's preferences and taste. For Eternally Yours, the psychological life span is the most important. The crucial question for sustainable design is, therefore, how can the psychological lifetime of products be prolonged?

Eternally Yours has developed many ideas to answer this question. For instance, it searched for forms and materials that could stimulate longevity. Materials were investigated that do not lose their attractiveness when aging but have "quality of wear." Leather, for instance, is mostly considered more beautiful when it has been used for some time, whereas a shiny polished chrome surface looks worn out with the first scratch. An interesting example of a design in this context is the upholstery of a couch that was designed by Sigrid Smits. In the velour used for it, a pattern was stitched that is initially invisible. When the couch has been used for a while, the pattern gradually becomes visible. Instead of aging in an unattractive way, this couch renews itself as it gets older. And Eternally Yours does not pay attention only to materials and product surfaces. It also investigated the ways that servicing of products can influence their life span. Ready availability of repair and upgrading services can prevent people from discarding products prematurely.

The most important way to stimulate longevity that should be mentioned in the context of this chapter, however, consists in designing products that establish a bond with their users by engaging users in their functioning. Most technologies are created to require as little attention for themselves as possible when people are using them. Technologies, after all, are often designed to unburden people: a central heating system liberates us from the necessity to gather wood, chop it, fill the hearth, clean it, and so on. We need press a button or slide a lever and our house begins to warm. But this unburdening character also creates a loss of engagement with technological products. Ever fewer interactions are needed to use them (cf. Borgmann 1992). One of the downsides of this development is that this diminishes the attachment between human beings and the technological products they use. The product as a material entity has become less important than the function it fulfills. In many cases, human beings are not invited to interact with the technological artifact they are using but only to consume the commodity it procures.

The work of Eternally Yours shows that this loss of engagement can be countered in a playful way. Technological products could invite users to interact with them without being so demanding that nobody would be prepared to use them. An interesting example in this direction is an engaging electric/ceramic heater that was designed by Sven Adolph. It consists of a heating element surrounded by several concentric, cylindrically shaped ceramic shells of different heights, each having a vertical aperture. The shells can be arranged in several ways so that they radiate their warmth in different

directions. This heater does not withdraw into pure functionality like common radiators, which are installed under the windowsill and are only turned on and off. Adolph's heater is an engaging product that asks for attention and involvement in its functioning, much like a campfire. You cannot hide it under the windowsill but have to put it in the middle of the room. You cannot escape it if you need warmth: you have to sit near it. Its shells have to be arranged if you want it to function. Simply turning the heater on and off is not enough: you must be actually involved in its functioning if you want it to work.

The activities of Eternally Yours can be seen as a form of "anticipating mediation by imagination." Smits's couch and Adolph's heater were designed from the perspective of their possible mediating role in the interactions and affective relationships their owners will have with them. They mediate the behavior of their users in such a way that they are likely to get attached more to these artifacts than to other couches or heaters. These products were designed not only as functional objects but also as artifacts that actively mediate the behavior of their users. The products of Eternally Yours therefore embody an "environmental ethic": they seduce their users to cherish them rather than throw them away prematurely. By anticipating how these products might help to shape experiences and practices of users, their designers took responsibility not only for their functionality but also for their mediating roles.

#### AUGMENTING CONSTRUCTIVE TECHNOLOGY ASSESSMENT

A second way to make an "informed prediction" about the mediating role of a technology-in-design is more systematic. To establish a connection between the context of use and the context of design, designers might employ a method that was developed precisely for making such a connection: Constructive Technology Assessment (CTA; see Schot 1992; Rip, Misa, and Schot 1995). CTA creates a link between the contexts of design and use in a practical way: it aims to involve all relevant stakeholders in the design of technologies. If the CTA methodology is to be applied within the context of technological mediation, however, it needs to be augmented.

CTA is based on an evolutionary view of technology development. The process is seen as generating "variations" that are exposed to a "selection environment," which is formed by entities like the market and government regulations. In this selection environment only the "fittest" variations will survive. There is an important difference, though, between the generation of technologies and the generation of biological species. Contrary to biological

evolution, in technology development there is a connection or “nexus” between variation and selection. After all, designers do not develop technologies blindly. They anticipate the selection environment as they work, in order to avoid putting great effort into developing technologies that will not be accepted by consumers or by government regulators.

CTA is a method for employing this nexus in a systematical way, by feeding back assessments from all relevant actors—users, lobbies, designers, companies, and the like—into the design process. It does so by organizing meetings of these actors in which the aim is to reach consensus about the design of the technology that is “constructively assessed.” This form of technology assessment is called “constructive” because it does not assess technologies after they have been developed but during their development, so that the assessments can be used to modify the original design. CTA can be seen as a democratization of the designing process. When a CTA design methodology is followed, all relevant social actors, not only designers, determine what the technology will look like. Following this method when “inscribing” morality into a technology, therefore, could take away the fear of technocracy discussed above.

Seen from the perspective of technological mediation, however, CTA has limitations that need to be overcome. CTA primarily focuses on *human* actors and pays too little attention to the actively mediating role of the *non-human* entity that is at the center of the activity: the technology-in-design. CTA claims to open up the black box of technology by analyzing the complex dynamics of technology development. In order to do this it relies on the constructivist notion that technologies are not “given” but rather are outcomes of a process in which many actors are involved—different interactions between the actors might have resulted in a different technology. But analyzing the dynamics of *technology development* opens up the black box of technology only halfway. It reveals how technologies emerge from their *design context*, but their role in the *use context* remains blackboxed. Organizing a democratically, domination-free discussion among all relevant actors is not enough to lay bare all relevant aspects of the technology in question. If it is not put explicitly and systematically on the agenda, the mediating role of the technology-in-design is likely to remain hidden during the entire CTA process.

For this reason, participants in CTA processes should be invited not only to integrate assessments of users and social organizations in product specifications but also to anticipate possible mediating roles of the technology-in-design. The vocabulary for analyzing mediation, as presented in the “Designing Mediations” section above, could be helpful for doing this.

When the CTA method is augmented in this way, the method of “anticipation by imagination” can be given a more systematic character. Creating space for all relevant stakeholders to anticipate the possible mediating roles of the technology-in-design enhances the chance that as many mediating roles as possible are taken into account. To be sure, this augmentation of the CTA methodology does not guarantee that all mediating roles of the technology in design will be predicted. The connection it creates between the inscriptions within the context of design and the interpretations or appropriations within the context of use cannot possibly cover all emergent mediating roles of the technology. Still, it offers designers one fruitful way to give shape to their responsibility for the mediating roles of their products.

#### SCENARIOS AND SIMULATIONS

A third way to establish a link between the contexts of design and use is scenario-based product design. This method aims to design products from the point of view of the ways they are used rather than in terms of their intended functionality (cf. Carroll 2000; Rosson and Carroll 2001; Wolters and Steenbekkers 2006; Tideman 2008). Though functionality and use are obviously closely connected, many forms of use do not actually match the intended functionalities of products. This can lead to user dissatisfaction—and this was an important incentive for the development of the scenario-based design approach. In this approach, a scenario is a specific use situation in which users interact with a product in various ways.<sup>7</sup> Anticipating such scenarios can augment the common focus on designing functionalities, because some functionalities might prove to be less relevant in frequently occurring use scenarios while other scenarios might require adaptation of the originally intended functionalities.

Thinking in terms of scenarios, therefore, forces designers to anticipate the use context of the products they are designing. The question is, however, *how* to develop meaningful scenarios. Using one’s imagination—possibly assisted by a form of “mediation analysis”—is one way to do so, along with involving users and other stakeholders as the CTA method does. Another very promising way to build scenarios is offered by virtual-reality technologies. As Martijn Tideman (2008) has shown in the context of the design of a lane change support system for cars, virtual-reality simulations of a product-in-design offer many fruitful points of application for developing use scenarios.

In Tideman’s approach, potential users were given the opportunity to design their own lane change support system by modifying a number of relevant variables of its virtual representation. Combined with a good driving simula-

tor, this setup made it possible to establish a very detailed link between the product-in-design and many possible use scenarios. Tideman used his method primarily to enhance customer satisfaction. He aimed to generate a design that fulfilled as many user preferences that became visible in his approach as possible. What he did not do, however, was use his scenario-simulation combination to anticipate the impact of the product-in-design on the actions and decisions of its users. For the “moralization of technology,” this would be a very fruitful application of the method. Designing virtual representations of the product-in-design, and of relevant aspects of the context in which the product will function, can enable designers to develop more adequate use scenarios for their products. Virtual representations make it possible to use a product before it actually exists, and using such representations in experimental settings permits designers to investigate not only the functionality and the usability of the product but also the ways it will mediate the behavior and experiences of its users. Contrary to Tideman’s approach, the emphasis in developing scenarios is not on the preferences of users but on the interactions between users and technologies as users appropriate the product in various ways and technologies play various mediating roles. A virtual-reality environment in combination with a scenario-based design approach is therefore likely to be able to enable designers to anticipate many facets of the future mediating roles of the technology-in-design.

### Assessing Mediations

Anticipating mediation is only a first step, though, toward responsible technology design. Both for the deliberate “moralization” of technology and for the more modest aim of preventing undesirable forms of mediation, designers will also have to *assess* the quality of the anticipated mediations. Several aspects of this moral assessment of technological mediation bear discussion. First of all, I will explain how designers could use stakeholder analysis—an often-used method in applied ethics—to morally assess the anticipated impact of their designs. In this method, all relevant moral arguments regarding the technology-in-design are gathered and balanced, taking the perspective of all stakeholders involved. Special attention here is given to the technological mediation of moral actions and decisions. Technologies, after all, play a two-fold moral role: not only do they have a societal impact that can be assessed morally, but they also mediate the moral actions and decisions of people.

Beside supplementation of the stakeholder method with mediation analysis, a number of issues that often have central roles in applied ethical analyses of technologies deserve separate discussion. I will focus here on the issues of

responsibility, freedom, and democracy. In line with the amodern approach I am developing in this book, I will not take these issues as criteria in terms of which technologies can be assessed. Rather, I will discuss how technologies actually reorganize what these issues can entail. Because of their role in human actions and decisions, technological mediations have an important impact on the responsibilities of users and designers. Technologies reorganize, take over, and delegate responsibilities—and each of these should be done responsibly. The issues of freedom and democracy are primarily at stake in clear “moralizations” of technology, which, as indicated in the introduction to this chapter, are often feared to diminish human freedom and the democratic quality of society. If we expand our understanding of freedom and democracy, however, new possibilities open up to deal with these issues in design practices.

#### AUGMENTING STAKEHOLDER ANALYSIS

In order to design “moralizing technologies” in a morally responsible way, or to simply morally assess the implicit mediating role of the technology they are designing, designers need a means of moral reflection on the quality of their designs. A common method for such an applied form of doing ethics is *stakeholder analysis*. Integrating this method with the theory of technological mediation is one important way to take seriously the moral significance of technological artifacts in the ethics of design. For that purpose, we need to supplement the method.

The aim of stakeholder analysis is to lay bare all moral arguments that are relevant to a given ethical problem by making an inventory of all stakeholders involved and all arguments that are relevant from their points of view. Some stakeholders, for instance, might suffer from certain negative consequences of using the designed technology. And toward other stakeholders we might have the moral duty to introduce certain forms of mediation, for example when they can help to save lives—like lane-changing assistants in cars, which sound an alarm at unsafe attempts to overtake another vehicle. By weighing all these arguments against each other, we can reach an informed conclusion about the moral quality of a decision.

However, in the context of technology design, such an analysis should not limit itself to *human* stakeholders; it also needs to address the moral significance of the technology-in-design itself. Several aspects deserve separate moral reflection here.<sup>8</sup> First, there is the moral quality of the *intended mediations* that are deliberately inscribed in the technology. Second, there are the *implicit mediations*, insofar as they can be anticipated by means of moral

imagination, constructive technology assessment, and scenarios and simulations. Third, the *forms of mediation* are relevant. These forms can differ widely. As indicated above, in some cases technologies, like a speed bump on a road or a turnstile at a metro station, actually *force* their users to act in certain ways. But in other cases, like an econometer in a car, which gives feedback on the fuel efficiency of one's driving style, technologies *persuade* their users. And other technologies, like the designs of Eternally Yours discussed above, primarily *seduce* their users in noncognitive ways to perform or refrain from specific actions. To what extent and in what circumstances are these forms of mediation morally desirable? Fourth, once a technology is introduced in society, the eventual *outcomes* of the technological mediations—the actions and decisions that eventuate—need to be morally assessed. These can, again, differ radically from the originally intended mediations.

At all of these four levels, stakeholder analysis can be a fruitful method to lay bare the moral relevance of the technology in design. The intended and unintended influences a technology has on the behavior of its users can be subject to moral deliberation; the forms of mediation that are used to evoke this influence should be proportional and acceptable; and the eventual results of all these efforts should be justifiable. Together, these four elements determine the moral quality of the activities of the designer and of the technology in design. Expanding the method of stakeholder analysis along these lines makes it possible to move beyond the predominant focus on risk analysis and whistleblowing in engineering ethics and the ethics of technology. Also, it gives the impact of technologies on the quality of our life and the implicit morality of technological devices a more explicit role in the moral reflection of designers. In chapter 6 I will use this “augmented version” of the method of stakeholder analysis in the context of designing so-called persuasive technologies.

But moral reflection about technology design encompasses more than a method of moral decision making. The question of how we design technologies that will have impacts on our actions and decisions raises several rather pressing ethical issues regarding its implications for human responsibility, freedom, and democracy.

## RESPONSIBILITY

The distributed character of moral agency over human and nonhuman entities has important implications for the relations between technology and responsibility. The design of mediating technologies raises the question to what extent humans can be held responsible for actions that were mediated

or induced by technologies. Does somebody act responsibly, for instance, if he or she keeps to the speed limit because an intelligent speed limitation device enforces this? And who is to be held responsible when an automatic face-recognition system in a security camera wrongly identifies a person as a suspect—which actually appears to happen more often for people with dark skin and for older people, because the software in these systems is attuned to light contrast on young, white skin (Introna 2005)? To what extent can designers and users be held responsible for undesirable forms of mediation?

In order to deal with this complexity in attributing responsibility, we need to first make an elementary distinction between two kinds of responsibility involved here: causal responsibility and moral responsibility. Someone is responsible in the causal sense when he or she is the *cause* of some event or state of affairs. But this can be the case when this person is not considered responsible in a *moral* sense. The event or state of affairs can, for instance, be caused accidentally or under pressure. Only when somebody acts purposively and freely can he or she be held morally responsible for his or her actions. Such freedom and intentionality are two elements of human agency that seem quite complicated in the case of technologically mediated action.

By their impacts on human action—or by their contributions to causal responsibility—technologies also contribute to the moral responsibility of human beings for the actions that come about in human-technological interaction. But as we saw, this does not imply that technologies should be held morally accountable for their mediating roles in human behavior—just as it does not make sense to consider technologies full-fledged moral agents in the way human beings are moral agents.<sup>9</sup> Still, technologies do play a more-than-causal role here. By mediating human interpretations and actions, they actively coshape moral responsibility. Moral decisions regarding preventive mastectomy, for instance, are not simply “causally influenced” by technologies of genetic testing. Rather, the moral questions themselves and the various options available for answering them are coshaped by these technologies; it is an amalgam of humans and technologies that acts morally here and bears moral responsibility.

The fact that responsibility, just like moral agency, is distributed among humans and technologies, however, does not mean that no points of application are available to deal adequately with questions of responsibility in design practices and the ethics of design. This blurring of the boundaries between humans and technologies does not make human being less responsible; rather, it opens up a new realm of responsibility. As discussed in the preceding section of this chapter, the eventual decisions and practices of users are

not only the result of specific impacts of *technologies* but also of specific practices of *use*. Moreover, the impacts of technologies are, to a certain extent, the result of the activities of *designers*. Therefore, including technologies in the realm of responsibility makes visible how both users and designers can have moral responsibility for technologically mediated actions.

For articulating the moral responsibility of users, the analysis of technologically mediated moral subjectivity developed in chapter 4 can be a starting point. In that chapter I argued, elaborating on Foucault, that moral subjectivity in technological contexts should be understood as the active coshaping of one's technologically mediated moral subjectivity. Human beings are not simply determined and controlled by technology. In most cases, they can develop a deliberate relation to the ways in which technologies mediate their actions and interpretations of reality. In this freedom of the human subject—understood in a Foucauldian sense—we find the basis for the moral responsibility of users.

The responsibility of designers is the subject of this chapter. Even though technological mediations can never be fully predicted, several possibilities are available to allow designers to take responsibility for the mediating roles of the technologies they design. They can anticipate and assess the mediating roles of their products, and as we will see, there are ways they might inscribe morality in technologies responsibly.

#### FREEDOM

Another issue surrounding the “moralization of technology” and the design of behavior-steering technology is the implications for human freedom. Technologies that deliberately influence human actions and interpretations, and even moral decisions, sometimes raise the fear that such technologies will take control over our lives. Cars that force us to keep to the speed limit, for instance, seem to leave little room for human freedom and responsibility.

Yet as indicated in the introduction to this chapter, the argument that moralizing technologies are a threat to human freedom is a complicated one. After all, many agreements exist between human beings in which they consciously limit their own freedom. Hardly anybody will find it immoral or beyond human dignity to obey the law. Why, then, accept the legal prohibition to kill while being indignant about building speed bumps that prevent people from driving too fast in neighborhoods where children often play outdoors?

A second reason the moralization of technology does not necessarily form a threat to human freedom is the fact that technological mediation need not

have a character of force or compulsion, as we saw earlier in this chapter. Seductive and persuasive forms of mediation do not steer or determine human behavior but instead inform and suggest human actions. Some technologies can even influence behavior by *enlarging* human freedom, as the “shared space” concept in transport studies shows. This approach aims to increase road safety by deliberately delegating more responsibility to road users and less to road signs and traffic lights. In the Dutch village of Makkinga, all “interventions” in people’s driving behavior (traffic signs, traffic lights, pavements, and sidewalks) were abolished, resulting in a situation where traffic from the right always has priority and where different categories of users are mixed rather than separated.<sup>10</sup> This approach has had very positive results. It has significantly reduced the number of accidents, because people are “forced” to take responsibility instead of merely complying with a prestructured system. Instead of diminishing freedom, the material environment created here influences human behavior by *enlarging* people’s degrees of freedom. Giving people more freedom in complex situations, remarkably enough, may result in the obligation to deal with them in responsible ways.<sup>11</sup>

Third, the analysis of technological mediation above shows that the actions of human beings who are dealing with technologies are *always* mediated. The deliberate moralization of technology, then, amounts to accepting the responsibility this implies. If technologies are always mediating human-world relationships, it seems wise to anticipate this mediation and give it a desirable form, as opposed to rejecting the whole idea of a moralization of technology.

As I pointed out in chapter 4, the fact that technological mediation is always there when technologies are used does not imply that human freedom is permanently under technological attack. Instead, it shows that freedom simply does not exist in an absolute sense—*absolute* being read literally, as “absolved” from “external” influences. Freedom should not be understood as a lack of force and constraints—or as Janis Joplin sang, as “just another word for nothing left to lose”—but as the existential space in which human beings must realize their existence. Humans have a relation to their own existence and to the restraints it encounters in the material culture in which they live. As I argued on the basis of Foucault’s work, this situated character of human existence *creates* specific forms of freedom; it does not *impede* them. Freedom can arise only where possibilities are opened up for human beings to have a relation to the environment in which they live and to which they are bound.

Philip Brey (2006), following Isaiah Berlin (1979), suggests that two forms of freedom need to be distinguished here. A first variant is what Berlin calls

“negative freedom,” consisting in the absence of limits and constraints. Opposite this is “positive freedom,” which indicates human autonomy: mastery over one’s own life. Most moralizing technologies—and explicit behavior-influencing technologies in particular—primarily impact negative freedom, because they make it impossible to act in certain ways or they require users to perform certain actions. This is not necessarily problematic, however, because human freedom is constrained in many other ways too, as we saw: by laws, norms, desires, and more. Reinforcing the prohibition on entering a privately owned building by installing a lock is not likely to receive much resistance.

As soon as technologies start to interfere with positive freedom, though, we will likely require more evidence of their value. Technologies then do not just “externally” constrain our freedom to act but involve themselves “internally” in our intentions. Brey discusses the possibility that “machines like our personal computer, toaster, laundry machine, refrigerator and personal digital assistant, are programmed . . . to make choices for us” that we might not have made otherwise. When technologies “try to determine more overarching goals of activities,” according to Brey, “autonomy is significantly eroded” (Brey 2006, 361).

As we saw in chapter 4, however, the notion of autonomy is highly problematic in a technological context. Many of our actions and decisions are technologically mediated—which does not imply, to be sure, that we are no longer the “authors” of what we do, but does imply that we are not the *only* authors of it. Moral arguments about abortion, decisions of how fast to drive, ways of communicating via e-mail—all of these things are shaped in close interaction with the technologies that make them possible. For this reason, the concept of freedom as Foucault elaborated it appears to be a good alternative to the concept of autonomy. Rather than “positive” or “negative freedom,” this form of liberty could be called “relational freedom.” It comes about in the relations people have with what helps to shape their subjectivity.

The Foucauldian approach to freedom, however, makes visible a limit to technological mediations. Any forms of mediation that make it impossible to develop a relation to them—because they dominate users in such a way that there is no way to appropriate and modify their impact—should be approached very critically. Only under very limited circumstances, which certainly would require democratic legitimization, can we allow technologies to actually *enforce* specific behaviors. When technological mediations do not leave any room for “relational freedom” for human beings to constitute their (moral) subjectivity, they oppress and limit human subjects and must not be permitted to function as a basis for generating forms of subjectivity.

## DEMOCRACY

The last issue in assessing the moral quality of technological mediations is moralizing technologies' alleged threat to the democratic quality of society. Some technologies influence human behavior and install visions of the good life without users' being explicitly aware of this or even being able to opt out. In our liberal democracy, the freedom of the individual is a very important value, but the quest for answers to the question of the good life belongs to the private sphere rather than the public space. We expect our government to pass laws that promote what we hope for in a good life, but not to promulgate detailed answers to the question of what a good life is. Analogously, too much interference from technology in our daily lives, designed by technical engineers rather than democratically elected politicians, could be seen as a direct threat to democracy.

This threat is no science fiction. As I will lay out in chapter 6, the growing attention of companies like Philips to what is called "persuasive technology" might well result in an increasing number of devices that aim to persuade people to change their behavior. The recently developed "persuasive mirror," for example, aims to induce its users to adopt a healthier lifestyle by presenting them with an image of how they will look in the future if they stick to their current pattern of living (Knight 2005). If the state enforced a healthier lifestyle by law, this would cause a lot of consternation, but technologies like the persuasive mirror can introduce similar effects into our lives "through the backdoor."

Such explicit moralization of technology, again, is not wrong or undesirable *per se*. But it does need a more democratic structure. For this reason, it is important to develop democratic procedures for both the evaluation and the design of moralized technology. A highly interesting instrument for such a democratization of design processes is Constructive Technology Assessment, discussed above. When a CTA design methodology is followed, designers alone do not determine what a technology will look like and do; rather, all relevant social actors are involved. Following this method, therefore, could dispel the fear of technocracy discussed above and open a space for deliberative democracy in processes of technology design. This is especially relevant when the design has intentional "moralizing" or "behavior-influencing" aspects.

As said above, organizing a democratic, domination-free discussion between all relevant actors is not enough to illuminate all relevant aspects of the technology in question. The mediating role of the technology-in-design needs to be put on the agenda explicitly and systematically. Whether technologies are designed in merely functional terms or in terms of explicit "moraliza-

tion,” the ways they perform will always involve a process of mediation that requires the designer’s attention. For this reason, participants in the CTA process should be invited not only to integrate assessments of users and relevant social groups in product specifications but also to anticipate possible mediating roles of the technology-in-design.

Latour’s concept of “public things”—as he literally translated the term *res publica*—makes it possible to grasp this integral connection between materiality and democracy (Latour 2005). *Res*, the Latin word for “thing,” also meant “gathering place” or “that which assembles” and even indicated a particular form of parliament. “Things” can thus be interpreted as entities that gather people and other things around them, uniting them and making them differ. Seen in this way, technological artifacts not only help to shape our lives and our subjectivities but should be approached as foci around which humans gather in order to discuss and assess their concerns about the ways these artifacts (Latour’s “things”) contribute to their existence. Technologies are precisely the places where the morality of design should be located, and where democracy is actually in the making.

### Methods of Moralization

Now that I have discussed various aspects of anticipating and assessing moral mediations, the question remains what instruments are available to help designers “moralize” their designs. To answer this question, I will first discuss two existing methods: the “inscription method” proposed by Jaap Jelsma (1999; 2006) and the method of “value-sensitive design” developed by Batya Friedman (1997). After this, I will propose an integrated approach in which the various aspects of anticipating, assessing, and designing mediations come together.

#### MORAL INSCRIPTION

Jaap Jelsma (1999; 2006) has developed a method for “inscribing” morality in technological objects, primarily directed at environmental impact. His method builds on the idea that human behavior not only results from “attitudes, values, and intentions” but is also “embedded in habits and routines,” which he understands as “patterns of unconscious actions guided by material infrastructures” (Jelsma 2006, 222). By adapting these material infrastructures, designers can steer patterns of action in a desirable direction by using the “script” approach developed by Akrich and Latour. This concept, as Jelsma demonstrates, links the contexts of design and use. Scripts, as we

have seen, can be designed into technologies and so help to shape patterns of behavior.

Jelsma makes a distinction between what he calls “user logic” and “script logic.” Scripts in technologies can aim at particular behavior-influencing effects, but if these do not fit with user practices and interpretations, unintended outcomes will result. The art of designing “inscriptions,” therefore, involves anticipating the share that both technologies and users have in producing user behavior. Jelsma has developed an eight-step design method for doing this, focusing primarily on the *redesign* of appliances and devices. In this method, existing scripts are analyzed and “rewritten,” taking into account how users might appropriate the redesigned device.

The method was used, among others, to redesign dishwashers. It laid bare some interesting mismatches between user logic and script logic. As noted earlier, many people rinse their plates under hot running water before loading the machine, even though according to the design logic that was a task for the machine.<sup>12</sup> This mismatch gave rise to several ideas for rewriting the “rinse script,” such as adding a rinse button to highlight that the machine can do the rinsing itself, or having the machine produce a message when it is rinsing, or even giving the machine a transparent front panel to make the rinsing process clearly visible.

Jelsma’s method offers an interesting application of the script theory to the field of design. Yet his method has limitations. The mediation approach that has a central place in this book is broader than the script approach because it also takes into account how technologies help to shape interpretations and moral decisions. Moreover, the method does not include moral reflection on the desirability and quality of the explicitly intended and implicitly operative scripts. I have argued that a broader approach is needed, one that takes up and augments the aspects presented by Jelsma.

#### VALUE SENSITIVE DESIGN

Another existing approach to “moralizing technology” is value-sensitive design (VSD). This method, which was developed by Batya Friedman and others (Friedman 1997; Friedman, Kahn, and Borning 2002), aims to account for human values throughout the design process. In the VSD approach, moral values that need to be supported by the technology-in-design replace the technological functionalities as the primary focus of design activities. The method has been used, for instance, to design a web browser that requires informed consent before saving a cookie (a small file containing personal

information about the person surfing the Internet). Starting from the values of privacy and autonomy, this design resulted in a web browser that not only is functional for surfing the Internet but also respects some important values that are threatened by other web browsers.

VSD uses an iterative methodology that integrates conceptual, empirical, and technical investigations. At the conceptual level, the values that are to be implemented are carefully analyzed in all their facets. For the web browser mentioned, the researchers analyzed what the elements of “informed” and “consent” entail, such as the adequate “disclosure” of the information needed, “comprehension” of this information, and “voluntariness” and “competence” in people’s “agreement” with what they consent to, implying the “clear opportunity to accept or decline,” the actual freedom to do so, and the “capabilities needed to give informed consent” (Friedman, Kahn, and Borning 2002, 4).

The empirical level, subsequently, concerns “the human context in which the technical artifact is situated” (*ibid.*, 3). Investigations here concern, for instance, how stakeholders apprehend different values, how they prioritize competing values, and how much impact these values have on their own behavior. In the case of the privacy-sensitive web browser, this step was conducted only after work was done at the third, technological, level. At the technological level, investigations consider the specific “value suitabilities that follow from properties of the technology” (*ibid.*, 3). The central idea here, just as in mediation theory, is that technologies support certain activities and values while discouraging others. In the VSD method, technical investigations can examine both the “value impacts” of existing technologies and the design of technologies that support given values.

For the privacy-sensitive web browser, technical investigations looked at the development of the technological properties of web browsers in relation to their impact on privacy. What are the default settings for cookie use? How much information is given to users about the benefits and disadvantages of cookies and about the particular information that cookies make available about their surfing behavior? On the basis of this analysis, a redesign of the (Mozilla Firefox) web browser was made. This browser introduced a “peripheral awareness” of cookies, information about individual cookies and cookies in general, and user management of cookies. Empirical investigations were subsequently directed at what users appreciated in this redesign, in order to adapt it in the most desirable direction.

The VSD method offers an interesting possibility for anticipating and designing “moralizing technologies.” Its focus on the social, technological, and

ethical-conceptual aspects of technological designs gives it a broad basis. Yet some important elements of the relations between humans and technologies and the mediation of morality remain underdeveloped in the method. The technological investigations conducted in the context of VSD, for instance, could benefit from a framework to analyze the impact of technologies on human practices and values, like the one offered by mediation analysis. Moreover, the empirical investigations would benefit from some clearer connection with the future use context, as offered by the scenario method, simulations, and involvement of users as in the CTA methodology. The conceptual level, to conclude, rightly focuses on a close analysis of all values involved, but it does not offer sufficient basis for a moral assessment of the eventual (re)design. For that, a method of applied ethics—like stakeholder analysis—would be a first step, augmented with moral reflection on issues like freedom, responsibility, and democracy which are closely connected to the explicit behavior-steering roles of technologies. Moreover, we should find ways to address the more encompassing question of what kinds of mediated subjects we would like to be in a public and democratic way.

Most importantly, though, both the VSD method and the moral inscription method lack an explicit notion of the mediating role of their own outcomes. Any technology, including technologies with inscribed morality and technologies that support certain values, mediates human actions and decisions in new and unexpected ways, which require a reflexive step in design methodologies in order to be adequately addressed. The econometer mentioned above, for instance, might result in a more fuel-efficient driving style—but this might also give users the impression that driving their car has now become an environmentally friendly activity, and that could keep them from using public transport more frequently and induce them to use the car more often than they used to. Such second-order effects of built-in mediations should receive close attention in the ethics of design.

#### TOWARD INTEGRATION

The approach of moral mediation developed in this book can significantly augment existing methods in the ethics of design. From the perspective of mediation, technology design inevitably means intervening in human-technology-world relations where specific “subjectivities” of human beings come into being, as do specific “objectivities” of their world. Moreover, any attempt to design such mediations will start to play a mediating role itself and will derive its impact from the ways it is taken up into human practices and interpretations. So designing mediations can never be seen as a modernist

enterprise in which human subjects “inscribe” morality into technological objects and “influence” human behavior. It rather is a form of engaging in the dynamics of human-technology relations, with careful interventions that always run the risk of having a different impact from what was expected. In order to intervene in a responsible way, the various steps that were elaborated in this chapter can be followed in the design process.

- 1 When designing a technology, a designer first has the choice of whether to moralize the design in an explicit way or only to assess the implicit moralizing roles of the design once it has reached a more mature stage.
- 2 If an explicit moralization of technology is aimed at, a *conceptual analysis* can be made of the values and norms to be designed “into” the technology, in line with the conceptual dimension of the value-sensitive design approach. What kinds of norms and values are embodied and installed by the technologies-in-design? How do they relate to each other, and what other norms and values might they imply or exclude? On the basis of such analysis, the design process can focus on looking for ways to “materialize” these norms and values and to develop *prototypes* of a technology that helps to shape human practices and experiences in ways that support the norms and values at which they aim.
- 3 Next, a *mediation analysis* of the product-in-design should be made, with the intention of anticipating the future mediating role of the technology in design. As discussed in this chapter, the moral imagination of the designer, assisted by a scenario-oriented approach and virtual-reality technologies, can play an important role here, as can involving users and other stakeholders (via methods like CTA). In the terms of Jelsma’s method, such a mediation analysis can reveal both the “script logic” and the “user logic” involved in practices around the technology-in-design, with the “script logic” focusing on the impact of the technology on user behavior, and the “user logic” focusing on interpretations and appropriations by users. And in terms of the VSD method, analyzing this “script logic” requires “technical investigations,” while “user logic” can be investigated by “empirical investigations.” Separate attention is required for what could be called “metamediation”: the eventual mediating effects of the “intended mediations.”
- 4 After this step of anticipation, a *moral assessment* should be made of all mediations involved. As indicated above, a method of applied ethics, such as stakeholder analysis, could be used here, with four points of application standing out: the *intended mediations* that are deliberately inscribed in the technology; the *implicit mediations* evoked by the design, insofar as they can be anticipated; the *forms of mediation* used; and the eventual *outcomes* of the technological mediations. The moral quality of each of these four aspects of the “moralized technology” deserves the attention of designers, augmented with reflection on moral issues involved with influencing human behavior with technology—for example, freedom, responsibility, and democracy.

Special attention should be paid to the questions of what kind of mediated subjects result from the intended mediations and what possibilities exist for human beings to codesign the impact of these mediations on their subjectivity. The various moral issues do not need to be used as given criteria to assess technologies but rather are the dimensions in which technologies play out their moral roles—*mediating* freedom, democracy, responsibility, and the like, rather than possibly *threatening* them.

- 5 On the basis of this moral assessment, a *design* can be chosen. This choice, however, should be seen as an experiment. It can never be guaranteed that the “moral content” will actually work out the way it was anticipated. Technological interventions in society never have the character of “steering” and determining. Unexpected interactions, interpretation, and appropriations will always occur—and these might necessitate adapting the original design. The modernist ideal of manipulability should allow room for careful experimentation and engagement. Moralizing technology is a modest and tentative activity, not a high-handed enterprise for steering human behavior.

## Conclusion

The ethics of design needs to take seriously the moral relevance of technological artifacts. In order to do justice to the profound role of technology in society and in people’s everyday lives, technologies need to be approached as morally relevant entities rather than as mere instruments in the hands of moral human beings. This expansion of ethics to the realm of materiality broadens the locus of ethical activity: it moves from the realm of text to those of materiality and design. If ethics is about the question of how to act, and technologies help to answer this question, technology design is a material form of doing ethics. Designers cannot but help to shape human actions and experiences via the technologies they are designing. Therefore, design processes should be equipped with the means to do this in a desirable, morally justifiable, and democratic way. Designers should focus not only on the functionality of technologies but also on their mediating roles. The fact that technologies always mediate human actions charges designers with the responsibility to anticipate these mediating roles.

This anticipation is complicated, since the mediating role of technologies is not entirely predictable. But even though the future cannot be predicted with full accuracy, ways do exist to develop well-informed and rationally grounded conjectures. To cope with uncertainty regarding the future roles of technologies in their use contexts, design processes include several possibilities for bridging the gap between the context of use and the context of design. Designers can use their (moral) imagination, scenario methods, and

virtual-reality technologies, and they can actively involve users in the design process. All these methods will enable them to develop a mediation analysis of the product in design. Of course, there is no guarantee that these methods will enable designers to predict entirely how the technology they are designing will actually be used, but they will help to identify possible use practices and the forms of mediation that might emerge alongside them.

However, the anticipation of technological mediation introduces new complexities in the design process. Designers, for instance, might have to deal with new trade-offs: in some cases, designing a product with certain desirable mediating characteristics might have negative consequences for the usefulness or attractiveness of the product. Introducing automatic speed influencing in cars will ensure that drivers keep to the speed limit, but at the cost of the experience of freedom—which appears to be rather important to some drivers, judging by the fierce resistance to speed-limiting measures. Also, when designers are anticipating the mediating role of technologies, prototypes might be developed and rejected because they seem likely to bring about undesirable mediations. Dealing with such trade-offs and undesirable spin-offs requires a separate moral decision-making process. Moreover, moral issues regarding the design of behavior-influencing technologies in general will need careful attention. Can the intended mediations, the forms of mediation, and the eventual effects of the intended mediations be morally justified? What are the implications of the technology-in-design for human freedom and responsibility?

Finally, the contested nature of some forms of behavior-steering technology makes clear that not all “materializations of morality” can be left to the responsibility of individual designers. The actions and decisions of designers always have public consequences, and therefore these decisions and their consequences should be subject to public decision making. The products of the designing work then literally become “public things,” in the Latourian sense of *res publica* (Latour 2005)—things that gather human beings around them to discuss and deal with their engaged concerns.

## Moral Environments: An Application

### Introduction

In order to further elaborate the approach to the moral significance of technology developed in this book, this chapter will apply it to a field of technology that is rapidly gaining influence in society: Ambient Intelligence and Persuasive Technology.<sup>1</sup> These technologies embody a fusion of insights from the behavioral sciences with advanced possibilities offered by information technology. The ever-increasing miniaturization of electronic devices and the ever-increasing possibilities for wireless communication between appliances have led to the development of so-called smart environments. Such environments register what is happening in a given space and are able to react to this in intelligent ways. Most of the time, the technology at work here is invisible and carefully attuned to human cognitive processes. Hence the name Ambient Intelligence: these technologies form intelligent environments. Persuasive Technologies add to this intelligence the ability to influence the behavior of their users in particular directions.

These technologies therefore explicitly embody many of the themes this book has taken up: the moral significance of technology, the mediated character of morality, and the hybrid character of agency and morality. In this chapter, after discussing these technologies in more detail, I aim to apply the theory and concepts developed thus far, focusing on the ethical aspects of designing and using such technologies.

## Ambient Intelligence and Persuasive Technology

### AMBIENT INTELLIGENCE

Ambient Intelligence can be seen as the most recent stage in the evolution of computer technology (Wehrens 2007), which is a product of the ongoing miniaturization of electronic devices, the possibilities of wireless communication, and the ever more intelligent interactions between computers and their environment. Ambient Intelligence is a combination of “ubiquitous computing” with intelligent user interfaces (Brey 2005). The concept of “ubiquitous computing” was introduced by Mark Weiser in 1991 to indicate a future of omnipresent information and communication technology functioning invisibly in the background of our existence (Weiser 1991; Bohn et al. 2004). This background does not consist of individual devices like thermostats and electronic clocks but is rather an integrated network in which all kinds of devices communicate with each other.

*Ambient Intelligence*—a term that was introduced by Philips, by the way, but has started to function as a generic term rather than a brand name, like Luxaflex in Europe or Band-Aid in the United States—adds the intelligence of user interfaces able to react to their environment in advanced ways to this omnipresence of technology, for instance by recognizing speech and gestures or patterns in people’s behavior (Aarts and Marzano 2003; Aarts et al. 2001). This makes it possible, for instance, for the doorbell to sound and the lights switch on automatically when somebody is entering the front door; the door could even open automatically when the system recognizes a person who should have access to the house. It also may make possible the development of automatic trip registration and payment in public transport, or intelligent marketing through smart shop windows that recognize passersby and display tailor-made special offers. Places where Ambient Intelligence is available are often termed “smart homes” or “smart environments.”

Ambient Intelligence is not science fiction but a reality that will rapidly gain importance. We are already used to automatic doors, fire detection systems, and cars that switch on the ABS (antilock braking system) when the tires start to slip. The step toward more encompassing systems is only a small one, principally because of the rapid growth of RFID technology. RFID chips are very cheap electronic “labels” that can send their content wirelessly and without a built-in power supply to a reading device. RFID tags can be put into the packaging of products, in identity cards, and subcutaneously in pets in order to be able to identify and track them when they leave home.

Examples of Ambient Intelligence fire the imagination. In geriatric care, for instance, detectors can give alarm when somebody falls from his or her bed or attempts to leave the house at an unusual time. The walls can get “ears,” responding to sounds in the room like a cry for help or a desperate request for assistance in finding a lost key (cf. Schuurman et al. 2007). Toilets can automatically analyze feces and urine in order to detect a health problem at an early stage. The Life Shirt System is an intelligent vest that monitors all kinds of bodily functions and reports these data to a healthcare center (Wehrens 2007).

Outside the realm of health care as well, myriad applications could become reality. With the help of RFID chips in food products, refrigerators can recognize which products they store in order to help people prepare a shopping list. They can even give feedback on people’s eating patterns and offer menu suggestions. To protect public order, cameras have been developed that automatically detect deviant behavior and report it to the police to enable them to act quickly. With the help of cell phones with a global positioning system, parents can locate their children when they are lost or do not come home when expected. And domestic appliances can react to the presence and even to the moods of people in the house, for instance by adapting the intensity of lighting, blocking or passing through phone calls, or making coffee when someone wakes up.

#### PERSUASIVE TECHNOLOGY

The impact of such environments will be even greater when insights from the behavioral sciences are used to design the interactions between user and environment. At the moment, this happens under the rubric of Persuasive Technology—technology that actively “convinces” people to behave in specific ways. Stanford University houses a Persuasive Technology lab, directed by B. J. Fogg, and large companies like Philips are making investments in this type of technology, organizing close cooperation between behavioral scientists and information technologists.

The art of persuasion has a long history. From the rhetoricians and Sophists in classical antiquity to the spin doctors and publicity departments of the moment, human beings have always attempted to develop techniques to persuade other people of certain views, of the need to do certain things, and of the importance of refraining from other things. In the late twentieth century, this art of persuasion became the object of the behavioral sciences. Both the characteristics of the perceiver and the form of the message can be used to influence human behavior. By combining insights regarding ways to influence

human behavior with the possibilities offered by information and communication technology, a new space was discovered in which to design and apply technologies that profoundly intervene in our everyday lives and that help us to act in particular ways and to make particular decisions (cf. Fogg 2003).

Some examples can illustrate this field. As noted earlier, the Persuasive Mirror produces a manipulated image of somebody's face, extrapolating the effects of one's lifestyle to the future and showing the effects on one's visual appearance, thus giving visual feedback on the health risks of one's way of living. The HygieneGuard is an attention system that gives children a reminder if they forget to wash their hands after using the toilet. Not all forms of Persuasive Technology are related to Ambient Intelligence, to be sure. The FoodPhone, for instance uses cell phones with a built-in digital camera to help people to lose weight. When they take pictures of everything they eat and send these to a central number, people can get detailed feedback on the number of calories consumed. And the Baby Think It Over is a doll that can be used in educational programs to prevent teen pregnancy. The doll gives quite a realistic image of how much care and attention a newborn asks for, both during daytime and at night, which might discourage young people from conceiving a child if they are not ready yet to adapt their lives to a little baby.

#### ETHICS

How can we wrestle with the ethical implications of Ambient Intelligence and Persuasive Technologies? Despite the obvious good intentions behind technologies like the FoodPhone (helping to fight obesity), the HygieneGuard (motivating people to wash their hands after using the toilet), and the EconoMeter (helping people to drive their cars more economically), ethical questions abound. Can it be morally justified to deliberately influence human behavior in such specific directions? What methods of persuasion are acceptable? Can persuasive technologies have undesirable consequences or persuade users to do things that cannot be morally justified? What about the implications of technological persuasion for human autonomy? Could Persuasive Technology generate moral laziness so that people delegate all moral decisions to machines, or an antidemocratic force in society, replacing laws set up by the parliament or congress with technologies designed by engineers?

These ethical questions reach much further than the usual discussion of safety risks, reliability, and privacy implications. To be sure, those issues remain in full force, but at the same time these technologies challenge our ideas

about ourselves as moral beings and the subjects of our own lives because they evoke radically new interactions between humans and technologies. Persuasive Technology and Ambient Intelligence make it possible to tailor ways of influencing people's behavior that can be exercised invisibly. Because of this, these technologies embody par excellence all the ethical questions and issues that I have discussed in this book. Issues like agency, responsibility, and freedom, which played a merely theoretical role in the previous chapters, take on an urgent character here. When people are influenced behind their backs, how are we to decide which influences are acceptable and which are not—and *who* is to decide this? How can we keep people responsible for their actions if these are, to an important extent, the results of influences exerted by technology? Can there be democratic forms of developing and applying such technologies? Do people need to have the option to opt out and refuse these influences, or is this simply not possible? How does human existence change under the influence of Ambient Intelligence and Persuasive Technology, and how can we help to shape this influence?

Ambient Intelligence and Persuasive Technology show in a very concrete way why we need to blur the distinction between humans and technologies—though normally this distinction forms an unproblematic and common background for dealing with ethical questions—in order to understand the impact and moral significance of technology. Rather than being used as means for realizing ends, these technologies form a barely noticeable background that actively interferes with human behavior. Agency cannot be understood here exclusively in human terms but becomes a matter of human-technology relations.

Like no other technologies, therefore, Ambient Intelligence and Persuasive Technology show why we need to develop a modern ethical approach that moves beyond the predominant subject-object split and treats ethics as a hybrid affair in which both subjects and objects play an important role. By anticipating and interacting with human behavior, these technologies embody the moral mediation that is a central theme in this book; they create a context that actively helps to shape how human beings act and what decisions they make. More efficient driving behavior induced by an EconoMeter and a different eating pattern that results from using the FoodPhone cannot be understood as purely human actions, but neither are they entirely technologically determined behavior. Without the technologies, human beings would never act the same way—there would not even be a situation of choice. At the same time, human beings are not entirely determined by these technologies but actively incorporate them into their daily lives. Moral action and decision making have become a joint affair between humans and technologies.

Because of this explicit interference with human actions and decisions, Ambient Intelligence and Persuasive Technologies offer interesting insights into the ethics of design and use developed in this book. I will first identify the most important places where moral reflection might occur during the design of persuasive technologies. By approaching Persuasive Technology in terms of moral mediation, I will identify three such applications. Second, I will deal with the process of moral reflection itself, working out a method to facilitate moral decision making during the design of Persuasive Technologies. Third, this chapter will take up a number of ethical issues that are a bit more external to the practice of designing persuasive technologies and mainly concern the moral acceptability of certain persuasive technologies and of the phenomenon of Persuasive Technology as such. And finally, I will use the Foucauldian approach to the ethics of technology, as developed in chapter 4, to discuss the ethics of technology use in the context of Ambient Intelligence and Persuasive Technology.

### Places for Moral Reflection

In 1999, Daniel Berdichevsky and Erik Neuenschwander introduced a framework for evaluating the ethical aspects of persuasive technologies. Central to their framework is the interaction between persuader, persuasive technology, and persuaded person. All elements in this interaction are points where moral reflection might occur: in considering the motives of the designer, the methods of persuasion employed in the technology, and the outcomes of the persuasion. In their model, a “designer with motivations” creates a persuasive technology; this technology employs “persuasive methods” to have an impact on a person; and there will be “intended, reasonably predictable, and unpredictable outcomes of the persuasion” (Berdichevsky and Neuenschwander 1999, 54).

However, this framework could benefit from some modifications if it is to cover all relevant ethical aspects of persuasive technology. First of all, from the postphenomenological approach used in this book, technological persuasion should be seen as part of the more encompassing phenomenon of technological mediation. Augmenting Berdichevsky and Neuenschwander’s framework with insights from the theory of technological mediation, I contend, will result in a broader understanding of the effects of persuasive technologies than “intended persuasions” versus “other, unintended outcomes.”

Second, after “persuasion” has been expanded to “mediation,” more needs to be done to include the unintended effects of technologies in ethical reflection and decision making, as discussed in chapter 5. In Berdichevsky and

Neuenschwander's model, designers' moral responsibility for unintended outcomes remains underdeveloped. Yet these outcomes are highly important, since they are ubiquitous and inevitable. Berdichevsky and Neuenschwander's model will therefore have to be augmented with ways to anticipate unintended outcomes and to incorporate them in moral decision making during the design process.

#### PERSUASION AND MEDIATION

A first step toward analyzing the moral aspects of persuasive technologies is to conceptualize their impact on human beings. As I have said, this impact may well entail more than the behavior that results from the *intended* persuasive effects in the technologies. Many, if not all, persuasive technologies have unintended effects too. These unintended effects can be analyzed with the help of the mediation approach.

Technological persuasion can be seen as one manifestation of the more encompassing phenomenon of technological mediation. Most persuasive technologies actually perform a hermeneutic form of mediation by shaping experiences and interpretations that inform behavior; the FoodPhone, for example, helps to develop new interpretations of food and consequently informs people's eating practice. Not all technological mediations of action, however, take shape as persuasion. In chapter 5 I argued that at least three behavior-steering forms of mediation can be distinguished.

First of all, technologies can *force* people to behave in certain ways, as when a speed bump leaves car drivers hardly any choice regarding their speed. Second, technologies can try to *persuade* users into specific actions, as when the EconoMeter installed in a car gives feedback about the energy efficiency of a driver's driving style. And third, technologies can *seduce* users into a form of behavior, as when elements of road design (such as curves and markings) make it more attractive to drive at a given speed.

There is a second connection between persuasive technology and mediation, though, that is at least equally important. Not only can technological persuasion be seen as a form of mediation, but the persuasive function of technologies can also have a mediating effect *itself*. The FoodPhone, to return to that example, may persuade its users to develop more healthy eating habits; this can be seen as a hermeneutic form of mediation, shaping human interpretations of what they are eating. This persuasive effect itself, however, can also play a mediating role in the relation between humans, their food, and their social environment. Beside having the desired effect of stimulating a more healthy eating pattern, the FoodPhone can, for example, make

eating stressful; it can stimulate humans to interpret their health exclusively in terms of their eating patterns, while neglecting the importance of other factors like getting enough exercise; and for eaters to take pictures of all food consumed will definitely reorganize social relations at the table. To take another example, the Econometer can stimulate people to drive more economically, but this can give them the false impression that driving in this way is actually environmentally *friendly*. Such mediating effects of technological persuasion need to be taken into account in moral decision making regarding persuasive technologies.

#### EXPANDING THE RESPONSIBILITY OF DESIGNERS

Having analyzed the concept of technological persuasion as a manifestation of technological mediation, we need to augment Berdichevsky and Neuenschwander's model for the ethics of persuasive technology. This second step concerns the uncertainty surrounding the eventual effects of persuasive technologies. As we saw in chapter 5, there is no unequivocal relation between the activities of designers and the mediating role of the technologies they are designing. Technologies' so-called multistability makes it difficult to fully predict the ways they will influence human actions or to evaluate this influence in ethical terms. The mediating role of technologies not only results from the activities of the designers, who inscribe scripts or delegate responsibilities, but also depends on the users, who interpret and appropriate technologies, and on the technologies themselves, which can evoke emergent forms of mediation.

This state of affairs has important implications for the ways designers can take responsibility for their designs. It means that the mediating role of technologies in the interpretations and actions of users, which together constitute their behavior, cannot be reduced to design specifications. It also depends on specific user interpretations and the characteristics of the designed technology.

The theory of mediation makes it possible to take this insight beyond the conclusion that persuasive technologies have merely "intended" and "unintended" effects. As we have seen in chapter 5, performing a mediation analysis (cf. Verbeek 2006b; 2006e) can be a good basis for making an informed prediction of the future mediating role of a technology—without claiming, of course, that such predictions are complete and fully adequate. In this way, the moral responsibility of designers can be expanded to also cover the unintended outcomes of technologies, to the extent to which they can reasonably be foreseen. The concept of mediation makes it possible to evaluate new

technologies not only in terms of the quality of their functioning (including the risks connected to them and their unintended consequences) but also in terms of the ways they help to shape new practices and experiences—including technological persuasion, but not limited to that. Deliberate reflection on the possible mediating roles of a technology-in-design should, therefore, be part of the moral responsibility of designers. Moreover, the organizational context in which they function should leave enough room for such reflection and responsibility (cf. Coeckelbergh 2006).

#### AN EXPANDED FRAMEWORK FOR EVALUATING THE ETHICS OF PERSUASIVE TECHNOLOGIES

This elaboration of the connections between persuasive technology and technological mediation makes it possible to expand Berdichevsky and Neuen-schwander's framework and modify it slightly. It has become clear not only that persuasive technologies *persuade* people to make behavior or attitude changes but that these persuasions also *mediate* their behavior in multiple ways. This implies that not only the outcomes of persuasion are relevant here but the outcomes of *all mediations* that arise in the use of the technology. Rather than linking the persuasive technology, via persuasive methods, with a persuaded person, then, I would link it, via mediation, with *behavior*.

As argued above, human behavior results from technological mediation in complex ways, which cannot entirely be reduced to the intentions of designers but which nevertheless require moral anticipation and reflection on the part of designers. Three places for moral reflection can be identified: the *intended persuasions* that are deliberately “built into” the technology; the *forms of mediation* that occur on the basis of this, including the employed method of persuasion; and the *outcomes* of the technological mediations, including both the intended persuasion effects and the concomitant mediation effects.

The next section will take this inventory of sites for moral reflection as a starting point and present a method of moral decision making that can inform moral reflection on these aspects of persuasive technologies.

### Ethics of Design

The theory of technological mediation reveals an inherent moral dimension in technology design. It shows that technologies always help to shape human actions and interpretations on the basis of which (moral) decisions are made. As discussed in chapter 5, design is a form of materializing morality because technologies will inevitably mediate the actions and decisions that constitute

their moral behavior. Every technological object that is used will mediate human actions, and every act of design therefore helps to constitute particular human practices.

In chapter 5, I showed that designers should do at least three things to deal with the morally mediating roles of their designs; they should *anticipate* these mediations, *assess* them, and *(re)design* them if necessary. In order to design persuasive technologies in a morally responsible way, designers should anticipate the mediations involved and perform a moral assessment of the nature, method, and consequences of the persuasions and mediations they are designing. To make an adequate ethical approach to persuasive technology, designers should go beyond the instrumental approach that is implicit in the Berdichewsky and Neuenschwander model. Rather than focusing only on the motivations of designers, the persuasive methods employed, and the outcomes of the persuasion, we will need to focus on how a persuasive technology will mediate human actions and experiences, in both intended and unintended ways. As we saw in chapter 5, performing a *mediation analysis* along the lines set out in the previous section is a first step in this direction.

#### STAKEHOLDER ANALYSIS

Designing persuasive technologies and Ambient Intelligence requires moral reflection on the anticipated technological mediations that will occur when the technology-in-design is used. A common method in applied ethics for doing this, as we saw, is the method of *stakeholder analysis*.<sup>2</sup> The aim of this method is to lay bare all relevant moral arguments from the perspective of the stakeholders involved. Some stakeholders, for instance, might experience some negative consequences of using a persuasive device—like obese people who are suffering from eating disorders and might plunge deeper into anorexia nervosa from using the FoodPhone. Other stakeholders might hold that it is our moral duty to introduce persuasive technologies when they, for example, help to save lives—as when the Persuasive Mirror encourages people to adopt a healthier lifestyle.

If we support stakeholder analysis with mediation analysis, we will improve upon both the Berdichewsky and Neuenschwander model and mainstream stakeholder analysis. Rather than following a purely instrumental approach to technology, focusing on the motivations of designers, the persuasive methods employed, and the outcomes of the persuasion, stakeholder analyses should focus on *all* mediation effects, extending to effects that cannot be reduced to persuasion or to the (intended or unintended) outcomes of persuasion.

Moral reflection on the mediating roles of persuasive technologies can take place along deontological or utilitarian lines. A deontological approach to the ethics of persuasive technology will investigate to what extent the intended persuasions, the forms of mediation used, and the outcomes of the mediations accord with certain moral principles. A utilitarian approach will therefore take inventory of the positive and negative consequences of the intended persuasions, the forms of mediation used, and the outcomes of the mediation; these consequences are then assessed in terms of their contribution to an intrinsic good like happiness, or a combination of intrinsic goods, or their ability to satisfy most of the preferences of the agents involved.

#### ASSESSING PERSUASIVE TECHNOLOGIES

In making a moral assessment of a persuasive technology, analyses should be made of all three sites identified above: the intended persuasion, the form of mediation (including the method of persuasion), and the outcomes of the mediation. The last site also requires that mediation analysis be made. In each of these points of application, typical issues emerge:

- 1 *The intended persuasions of the technology-in-design.* From a utilitarian point of view, we need to balance the desirability of the intended behavior and its costs and negative effects for all stakeholders involved. From a deontological point of view, the intended persuasions need to be in accordance with certain moral principles. Most relevant here seem to be the principle of *no harm* (does the intended persuasion cause no harm to people using it or those affected by its use?); *beneficence* (does the intended persuasion benefit people using the technology or those affected by its use?); and *justice* (is the intended persuasion fair, treating people in equal circumstances equally?).
- 2 *The methods of persuasion used and the emerging forms of mediation.* From a utilitarian point of view, an analysis has to be made of the benefit of persuading people to adopt a desirable form of behavior as compared to the (social and individual) cost of various methods of persuasion and other forms of mediation. From a deontological point of view, moral principles like the following need to be addressed: *respect for autonomy* (do people know they are being persuaded?); *no harm* (is people's privacy respected?); and *justice* (is the technology free from bias against particular social groups?).
- 3 *The outcomes of the mediation.* At this point, mediation analysis comes into play. With the help of one's moral imagination, an inventory has to be made of all possible mediating roles of the technology in both human experiences and human actions. After this, these mediations need to be assessed morally, both in utilitarian and in deontological terms.

To make a moral analysis of the design of a product like the FoodPhone, for instance, we should take a creative inventory of the possible mediating roles of the phone in human actions and experiences. We have to imagine the product as functioning in as many realistic use contexts as possible and focus on its role in the practices and experiences of its users and other stakeholders. As for the domain of *action*, the FoodPhone, for instance, requires its users to take pictures of all food they are eating. This might complicate social interaction. Taking of pictures of everything you eat reveals to other people that you are working on your diet—something you may not want to reveal. More important, it also reveals that you are judging a social meal that may have been prepared for you by someone else mainly in terms of its nutritional aspects, which may be felt as inappropriate. These mediation effects may discourage people from using the FoodPhone at all.

As for the domain of *experience*, the FoodPhone will make its users more aware of what they are actually eating, which will probably result in their losing weight and establishing a more balanced diet. But receiving constant feedback on what you are eating may also result in an unhealthy obsession with food, and in a form of stress that is unhealthy in itself. Moreover, the FoodPhone invites its users to be “observers” of their eating behavior, thus detaching them from their immediate environment.

In order to assess whether the FoodPhone is a morally acceptable technology, we would need to take all these arguments into account and play them against each other. The main question here will be whether the probability of a positive effect on people’s eating habits outweighs the possible negative effects on people’s social life and their attitudes toward eating in general.

#### MORAL ISSUES

Having identified several sites where moral reflection could occur in the design of persuasive technologies, as well as a method for putting such moral reflection into practice, I turn to separating out some moral issues that are a bit more external to the design process. These issues arise not primarily from the perspective of designers but from that of users. They include *trust* (to what extent can consumers trust the creators of persuasive technologies?), *responsibility* (who can be held responsible for the resulting behavior of users?), *reliability* (can we be sure the persuasion will not have undesirable effects?), and the *desirability* or even *legitimacy* of technological persuasion in the first place.

When technologies that aim to influence human behavior on a large scale are introduced, they need to meet certain requirements in order to be morally

acceptable. One of the most important requirements is that users be able to trust the technology they are using. *Trust* in this context means that people can reasonably expect the technology to do what it is supposed to do and that the consequences of using it will not be harmful to them or otherwise undesirable, unless they are adequately informed in advance. This suggests that trust implies both *reliability* of the persuasive technology and *responsibility* on the part of the designers.

The degree to which both of these aspects of trust are actually realized depends on the degree to which the eventual consequences of using a technology can be linked to the activities and intentions of the designers and users. After all, this is the only way to predict the future impact of the technology. If we cannot link the impact of an EconoMeter in a car to what its designers intended it to accomplish and how its users employ it, it is hard to call it “reliable” and feel secure that it was designed in a responsible way.

As noted in chapter 5, predictability of impact is a serious challenge. Causal responsibility for technological mediation needs to be distributed among designers, users, and the technologies themselves, and there will always emerge unforeseen mediations, like energy-saving lightbulbs actually causing an increase in energy use, and cell phones changing patterns of social interaction. Nevertheless, the absence of full predictability regarding the mediating roles of technology does not imply that trust, reliability, and responsibility are incompatible with persuasive technology. After all, several causal responsibilities for mediation can be identified, and they form a good basis for attributing *moral* responsibility to those agents who are capable of taking this responsibility—that is, designers and users of persuasive technologies. Designers need to anticipate the mediation effects of their designs as much as they can, by performing mediation analyses with the help of their moral imagination and using such analyses in moral decision-making processes. Users, in turn, need to anticipate technological mediations as well, to the extent to which such mediations result from their appropriations and interpretations of the technology.

If both users and designers act in a morally responsible way, by not simply designing and using technologies as mere instruments but using their moral imagination to approach them as *mediators*, there is reason enough to trust that technologies will actually do what they were designed for and that few unacceptable consequences will result from using them. At the same time, we can only speak of trust here—not of certainty, because technologies are inevitably surrounded with uncertainties and risks, dependent as they are on the whimsical relations they will develop with human beings.

Another important issue is the *legitimacy* of “technological persuasion” or “behavior-influencing technology” as such. Two main arguments can be identified here, in line with the ideas elaborated in chapter 5. The first is that human freedom might be threatened when human actions are consciously steered with the help of technology. This reduction of human freedom can even be perceived as a threat to human dignity: when human actions do not result from deliberate decisions but emerge from steering technologies, the worry is that people are deprived from what makes them human. Second, developing behavior-influencing technologies can be seen as an implicit move toward technocracy. When moral issues are resolved through the technological activities of designers rather than the democratic actions of politicians, technologists rather than the people will ultimately be in control.

These arguments, however, proved flawed. Given the fact that technologies *always* help to shape human actions, wouldn't we try to give these mediations a desirable shape instead of making futile efforts to resist any influence? Rather than clinging to a view of human freedom as absolute autonomy and sovereignty from technology, it seems wise to reinterpret freedom as a person's *ability to relate to what determines and influences him or her*. Still, this does not address the anxiety that a technocracy would come about if our environment were explicitly “moralized.” After all, if technologies are not “moralized” openly and democratically, responsibility for technological mediation is entirely left to designers, and this would amount to form of technocracy. It is important, therefore, to find democratic ways to design persuasive (and otherwise mediating) technologies. As I explained in chapter 5, the methodology of Constructive Technology Assessment” could contribute to this, because it organizes a domination-free discussion with all stakeholders to anticipate and assess the impact of technologies-in-design and to feed the outcomes back into the design process.

### Ethics of Use

Beside an ethics of design, Ambient Intelligence and Persuasive Technology call for an *ethics of use*. As the “freedom issue” discussed above illustrates, these technologies have impacts on human subjectivity and use practices. As explained in chapter 4, this need not lead to an ethics of “protecting” the subject from technology. The mediation approach starts from the idea that human actions and decisions are always mediated, and from this perspective ethics consists in carefully assessing and experimenting with technological mediations in order to coshape the technological mediation of people's

existence in a technological culture. Technology users are not merely passive objects of technological mediations but active subjects who can develop a relation to these mediations. Even without being able to fully control the impact of technology on their daily lives, users can appropriate the technology and modify the ways it shapes one's existence.

In Foucauldian terms, using and dealing with technology becomes a self practice, a practice in which the self is shaped by relating to the powers that help to shape it (O'Leary 2002, 2–3). As we saw, Foucault distinguishes four aspects of the ways in which human beings constitute themselves as moral subjects: the *ethical substance* (the part of oneself that is subjected to a moral code), the *mode of subjection* (the “authority” on the basis of which the subject aspect is formed), the *practices of the self* (the activities in which the moral subject is formed), and the *teleology* of these practices (the way of existing a subject aspires to by subjecting itself as it does to a given code).

These four aspects of moral subjectivity form a framework for articulating the ethical aspects of using Ambient Intelligence and Persuasive Technologies. If people approach these technologies as powers that help to shape subjectivity and that require active appropriation and experimentation if their impact on one's existence is to be coshaped, the Foucauldian “fourfold” can clarify the various aspects of subject constitution involved.

The *ethical substance* is formed by human intentions and human behavior. These, after all, form the “sites” where mediating technologies like Ambient Intelligence and Persuasive Technology can have a moral impact. On the one hand, our behavior is directly influenced by these technologies, and this reality requires us to find relations to these behavior-influencing effects. This is especially the case for Ambient Intelligence technologies, like Intelligent Speed Adaptation, that interact actively with our behavior and influence it profoundly. On the other hand, Persuasive Technologies, like the Persuasive Mirror that implements normative ideas about a good lifestyle, explicitly “educate” us, which requires that we respond in a sensible and alert way. Only when people consciously take a position with respect to the ways these technologies help to shape their moral substance will it become possible to take responsibility for them. But in order to do that, people do have to see their mediating roles. These technologies are more than interesting new gadgets; they intervene in our existence. When that becomes visible, we are able to incorporate this impact of technology into our everyday lives and to “co-design” how our technologically mediated existence takes shape.

Ambient Intelligence and Persuasive Technology also bring about a *mode of subjection*—a way to invite or stimulate people to recognize a particular moral code at work. In the context of these new technologies, it is not a

divine law or a universally valid principle that gives authority to a specific moral code—here the mode of subjection consists in the tailor-made influences that technologies exert on human beings. In a certain sense, Ambient Intelligence and Persuasive Technologies can be seen as externalizations of human conscience. Norms and principles of behavior are implicitly or explicitly materialized in a technological artifact, which exposes human beings permanently to its moralizing attempts to adjust their behavior according to preprogrammed guidelines.

In the fields of Ambient Intelligence and Persuasive Technology this mode of subjection takes different *Gestalts*. As noted earlier, some forms of Ambient Intelligence *force* people to behave in a certain way, as when an Intelligent Speed Adaptation system matches the speed of people's cars to the speed limit with the help of a GPS system. Persuasive Technologies exhibit a different mode of subjection; they use the force of persuasion, as when users are given feedback on their own behavior by the Persuasive Mirror or the FoodPhone. Other forms of Ambient Intelligence can *seduce* people to act in specific ways, making some actions more attractive than others—like the lamps used on the facades of some Dutch churches in town centers, which automatically switch on when somebody comes very close. Some are even connected to urinals that rise automatically in order to discourage men from urinating against the building (cf. Dorrestijn 2006).

*Self practices*—the third aspect of moral self-constitution in Foucault's approach—consist in consciously giving shape to one's way of using technology and the ways one's existence is impacted by technology. Foucault called this self-designing activity "ascetism"—a broadly construed notion of ascesis in which people develop a relation with the forces that help to shape them. In the context of Persuasive Technology and Ambient Intelligence, ascesis primarily means using technology in a conscious way, realizing that any use practice also shapes one's subjectivity.

An important form that self practices can take involves experimenting with mediation by taking up Ambient Intelligence and Persuasive Technology in specific ways in one's everyday life. Steven Dorrestijn discusses an interesting example of this in the field of Ambient Intelligence. An experiment with Intelligent Speed Adaptation that was conducted in the Dutch city of Tilburg offers interesting points of application for understanding what self practices in a technological culture can entail (Dorrestijn 2004, 100–101). This system, which automatically limits the speed of cars to the maximum speed that applies at the place where the vehicle finds itself, seriously limits the freedom of its users. But contrary to what one might expect, after an initial phase of resistance users began to value the system. The main reason was

that they developed a more relaxed driving style. A rushed style of driving simply became impossible, and eventually for many people this became comfortable rather than annoying (Adviesdienst Verkeer en Vervoer 2001). The users of this system thus gave up a certain form of *autonomy*—understood in terms of the *absence* of factors that influence the subject—but gained a form of *freedom* in return, by developing a relation to these factors and determining how they give shape to users' subjectivity. Freedom here is a *practice* that is coorganized by the technological infrastructure of existence—and within this practice it appears to be possible to take responsibility for one's technologically mediated existence—in interaction with, rather than over and against, technology.

Another good example is e-mail. Many personal and professional forms of interaction have dramatically changed because of the introduction of e-mail. E-mail makes it much easier to contact other people, and it offers the possibility of a quick question-and-reply sequence—while still being an asynchronous form of communication, giving people more time to reflect on their messages than while chatting on the phone or in a face-to-face conversation. But e-mail is not simply a neutral tool that facilitates communication; it also mediates *how* people communicate. A well-known phenomenon in e-mail communication, for instance, is “flaming.” In this pattern in communication, somebody receives an e-mail that contains a remark that sounds unfriendly—in many cases because a lack of verbal and nonverbal context that could help to the recipient interpret the message. A quick and irritated reply then leads to an incensed retort, and without anyone explicitly wanting this, an argument is born (Turnage 2007). Another problematic phenomenon in e-mail communication is the thoughtless use of so-called cc's (“carbon copies”)—digital copies of e-mail messages that are sent to other people besides the main addressee. Because it is so easy to include other e-mail addresses when composing a message, and because it might prevent problems if messages are sent to as many people as possible (then nobody can complain of not having been informed), the cc can become a true plague in professional environments. For these reasons some companies have started to offer their employees courses in e-mail use.<sup>3</sup> Such courses can be seen as self practices: developing a relation to how technologies shape one's subjectivity. Rather than approaching flaming and extensive cc-ing as inevitable phenomena accompanying e-mail, we can approach them as powers that help subjects to come into being and in interaction with which we as e-mail users can “style” our subjectivity.

*Teleology*, to conclude, concerns the question of what kind of subjects we want to be when we behave morally. In the context of Ambient Intelligence

and Persuasive Technology, this question addresses the feelings of uneasiness some people have when imagining a future in which our material environment intervenes quite directly in our daily lives and even our moral choices. Do we want to be such persons? Do we want to be people who delegate important aspects of care for the elderly to smart environments—asking the elderly to literally speak to the walls if they need help and creating an environment in which the walls literally have ears, that can detect it when people fall or are hungry or confused? And do we want to be humans who make moral decisions in interaction with the feedback we receive from technologies? Wouldn't the "instant morality" offered by Persuasive Technologies cause us to lapse into a form of moral laziness?

In the words of the American philosopher of technology Albert Borgmann, Persuasive Technology embodies a "commodification of morality." For Borgmann, commodification is one of the key characteristics of our technological culture: things that used to require effort to acquire have become available with the push of a button (Borgmann 1984). While people in the old days had to walk to the well to get water, we simply open the tap. And while keeping a fire was a very intensive job, now we simply turn the thermostat a bit higher when we feel cold. Warmth and water have become *commodities*: goods of consumption, readily made available by modern technology.

Leaving aside the question of whether Borgmann's diagnosis is adequate in all respects (cf. Verbeek 2005b), the question arises to what extent Persuasive Technologies are a new step in this process of commodification. The ability to reflect morally, which is not the least human faculty, seems to be exchanged here for voluntary exposure to the powers and influences of technology. When the spirit is willing but the flesh is weak, people in this case choose to let not only their flesh be influenced (as is the case in most instances of "moralizing technology") but also their spirit. A part of our conscience is deliberately placed in the material environment, changing this environment from the background of our existence to an active educator. Do we want to be people who make moral decisions in interaction with the feedback received from technology?

Such questions require a public moral debate about the quality of our lives in relation to the technology that we use. When moral discussions about Ambient Intelligence and Persuasive Technology are limited to questions about risks and responsibilities, a crucial point is missed—our ethical subjectivity is at stake here. Precisely because of their subtle interweaving with our daily lives and their sometimes hardly noticeable influence on our actions and decisions, these technologies raise the question of what kind of moral subjects we want to be.

By embedding Ambient Intelligence and Persuasive Technology in society and our daily lives in a careful way, however, the fear of Big Brother scenarios becomes unnecessary. In fact, these technologies reveal an influence that has always been there and has now become more visible. Our material world has always intervened in our lives, and now it is simply doing this more openly. But precisely because the technologically mediated character of our existence is becoming more visible now, it can become the focus of a discussion. From the perspective of the ethics of both use and design, it is tremendously important that we have this discussion and that it reach further than the usual worries about safety, reliability, and privacy. Nothing less than the quality of our lives and the character of our moral subjectivity is at stake.

## Morality beyond Mediation

### Introduction

In order to do justice to the moral significance of technology, this inquiry thus far has aimed to move beyond the “humanist bias” that characterizes many forms of ethics.<sup>1</sup> The humanist framework proved too narrow to do justice to the pervasive moral role of technologies in our culture. From a humanist conception of ethics and morality, as we saw, artifacts have to be excluded from the realm of moral agency, because they lack autonomy and intentionality (cf. Illies and Meijers 2009). From a postphenomenological approach, though, it has become possible to conceptualize morality as a hybrid affair in which human beings make moral decisions in close interaction with the technologies that help them to interpret reality in specific ways and that organize specific practices. The notion of moral mediation has made it possible to develop a closer understanding of the moral significance of technology and to take responsibility for it in practices of use and design.

Yet the way the approach of moral mediation moves beyond the humanist bias in ethics does not cover all aspects of the moral significance of technologies. In order to see that we need to connect to the distinction I made in chapter 2 between a “posthumanist” and a “transhumanist” approach to technology, I characterized the approach to be developed in this book as posthumanist—moving beyond humanism and its focus on the autonomy of the subject but not propagating a transhumanist move beyond the human. The concept of moral mediation reveals that we cannot understand ourselves any longer along humanist lines as autonomous beings, since our lives have become so integrally connected with technologies. Yet not all technologies can be analyzed along these posthumanist lines. In order to explore the limitations of the approach of technological mediation, in this chapter I would

like to discuss two current technological developments that actually move toward either “transhuman” or “nonhuman” forms of intentionality.

First, as chapter 6 showed, ever more technologies are being developed with a built-in form of “technological intentionality.” Rather than mediating human actions and decisions, such technologies—like Intelligent Speed Adaptation and MRI imaging—add their own “intentionality” to that of the humans using them. Second, recent technological developments show a convergence of nanotechnology, biotechnology, and information technology, which makes it possible to intervene in “human nature.” Brain implants can enable deaf people to hear again. “Deep brain stimulation” can mitigate the effects of depression and Parkinson’s disease. Psychopharmacological drugs drastically improve people’s mood. And in the field of genomics, ever more sophisticated interventions in human genetic material are designed. In all of these cases, technologies do not *mediate* human actions and decisions but rather merge with the human subject, resulting in a hybrid entity that has sometimes been called a “cyborg” (cf. Haraway 1991; Hayles 1999; De Mul 2002; Irrgang 2005).

These two forms of human-technology relations take us to the boundary between posthumanism and transhumanism, because they involve a non-human form of intentionality. Interacting with nonhuman agents helps shape human intentionalities in different ways than does using mediating technologies. And merging with nonhuman entities into a new, hybrid entity results in a form of intentionality beyond the human being. How does the moral significance of such “artificially intentional” and cyborglike technologies relate to the phenomenon of moral mediation that is pivotal to the approach set out in this book?

To answer this question, I will examine two extrahuman-technology relations that complement the relation of mediation. First, I will analyze what I will call *composite relations*. Here technologies do not mediate relations between humans and reality but instead add an artificial or “artifactual” intentionality to human intentionality—as is the case with expert systems, Intelligent Speed Adaptation systems, and some forms of imaging technologies. Second, I will analyze what can be called *cyborg relations*. In such relations, the boundaries between technologies and human beings are blurred in a physical way—as is the case with technologies like psychopharmaca and neural implants. An investigation of what these new human-technology relations imply for the ethics of technology will be the focus of this chapter.

### Moral Mediation and Beyond

The approach of moral mediation made it possible to conceptualize the moral significance of technologies in terms of their mediating role in human practices and in the experiences and interpretations on the basis of which human beings make moral decisions. This approach first implied a new status for objects in ethical theory, giving them a mediating role in moral agency. Second, it made it necessary to rethink the moral subject. When technologies so fundamentally help to shape our moral actions and decisions, after all, the autonomy of the subject, which is most often taken as a requirement for moral agency, is seriously mitigated.

In the approach of moral mediation, therefore, I replaced the humanist notion of autonomy with a Foucauldian notion of *freedom*. Rather than indicating the absence of external influences, this notion of freedom focuses on developing a free *relation* to these influences. Human intentionality is most often technologically mediated, but this does not make human beings simply passive products of technological mediations. The ways in which technologies help us to act morally do not have the character of *determination*; technologies help to organize a relation between human beings and reality, which is the basis for certain moral actions and decisions and which depends on characteristics of both the mediating technology and the forms of their appropriation by human beings. The human capacity of reflection enables us to work out an active relationship to these mediations and to modify them in order to “style” and codesign our mediated moral subjectivity.

While the concept of moral mediation covers the vast majority of manifestations of the moral significance of technology, there are also forms of “material morality”—as indicated above—that escape it. Some technologies play moral roles that simply cannot be reduced to the mediation of human-world relations. In order to conceptualize these other forms of moral significance, I will revisit the concept of intentionality. Some technologies appear to bring in “artificial” forms of intentionality instead of merely mediating *human* intentionality. Other technologies merge with human beings in such a way that a form of “joint” intentionality comes about. In order to understand these nonhuman intentionalities and the various types of relations between humans and technologies behind them, I will expand Don Ihde’s account of human-technology relations, augmenting it with two extra relations and using it to analyze the various types of intentionality that are connected to them.

As we saw in chapter 1, Ihde introduced a technological dimension into the phenomenological tradition of understanding human-world relations.

In our technological culture, many of the relations we have with the world around us are either mediated by or directed at technological devices—ranging from looking through a pair of glasses to reading a thermometer, from getting money from an ATM to having a telephone conversation, and from hearing the sound of the air conditioner to having an MRI scan done. Ihde's analysis is a "posthumanist" account of human intentionality because it shows the manifold ways in which intentionality is not "authentic" and "direct" but has a mediated character.

In his well-known analysis of human-technology relations, Ihde distinguishes four types of relations. First, technologies can be *embodied* by their users, like a pair of glasses or the dentist's probe. Second, they can be the *terminus* of our experience, as is the case when taking money from an ATM. Third, technologies can give a *representation* of reality, like a thermometer, which does not produce an actual experience of heat or cold but instead delivers a value that needs to be "read" in order to tell us something about temperature. And fourth, technologies can play a role at the *background* of our experience, creating a context for our perceptions, like the humming of the air conditioner or the automatic switching on and off of the refrigerator. These four human-technology relations, on the basis of which technologies play their mediating roles, are indicated schematically in table 2. In this table, the arrow indicates intentionality.

Table 2 makes explicit several relations between intentionality and technology. Intentionality can work *through* technological artifacts, it can be directed *at* artifacts, and it can take place *against the background* of them. In all these cases, except in the alterity relation, human intentionality is mediated by a technological device. Humans do not experience the world directly here but always *via* a mediating artifact that helps to shape a relation between humans and world. Binoculars, thermometers, and air conditioners help to shape new experiences either by procuring new ways of accessing and disclosing reality or by creating new contexts for experience. These mediated experiences are not exclusively "human"—human beings simply could not have such experiences without these mediating devices. Accounting for experiences like reading a thermometer and having a telephone conversation, therefore, requires a "posthumanist" understanding of intentionality in which intentionality is partly constituted by technology.

But how to understand the intentionality involved in the "artificially intentional" technologies and the cyborgs discussed in the introduction to this chapter? These technologies seem to require a more radical move beyond humanism. As I will elaborate below, beside mediated intentionality two other forms of intentionality need to be distinguished, which are connected

TABLE 2. Human-technology relationships (Ihde 1990)

|                              |                              |
|------------------------------|------------------------------|
| <i>embodiment relation:</i>  | (human–technology) → world   |
| <i>hermeneutic relation:</i> | human → (technology–world)   |
| <i>alterity relation:</i>    | human → technology (–world)  |
| <i>background relation:</i>  | human (– technology – world) |

to two human-technology relations that can be added to the four relations distinguished by Ihde. First I would like to introduce the concept of *cyborg intentionality*, meaning the intentionality of human-technology hybrids in which the human and the technological are merged into a new entity—a cyborg—rather than being interrelated, as in Ihde’s human-technology relations. Second, I will develop the notion of *composite intentionality*. There are situations in which human beings have intentionality, as do the technological artifacts they are using.

Because Ihde’s primary focus is on the relations between humans and technologies rather than the intentionalities involved in these relations, his analysis tends to black-box the various forms of intentionality involved. Precisely by drawing attention to these intentionalities it will be possible to substantially augment his analysis. Ihde’s schematic representations of human-technology relations contain not only arrows, which serve to indicate intentionality, but also dashes, indicating a relation between entities which is not specified further. If we limit ourselves to the embodiment relation and the hermeneutic relation—which are the most relevant relations in the context of intentionality since they ultimately involve relations with the *world*—these dashes indicate a relation between humans and technology or between technology and world. By investigating the nature of these dashes we can develop a closer characterization of a posthumanist, or even a transhumanist, account of intentionality.

First, the dash between *human* and *technology* in the embodiment relation (*human—technology*) → *world* black-boxes the nature of the various relations that can exist between humans and technology and that are extremely relevant in the context of “cyborg intentionality.” The dash is an umbrella under which many types of relations can be hiding; all of these relations can have impacts on the character of the embodiment. Second, the dash between *technology* and *world* in the hermeneutic relation *human* → (*technology—world*) black-boxes the relations that can exist between mediating technologies and the world but does not create enough space to take into account the existence of nonhuman or technological intentionality, which is also highly relevant in the context of a discussion of “cyborg intentionality.”

In what follows, I will elaborate these preliminary thoughts into a more radical interpretation of Ihde's understanding of both the embodiment relation and the hermeneutic relation, expanding these to a "cyborg" and a "composite" relation respectively.

### Cyborg intentionality

Analyzing the nature of the relations between the human and the technological in the embodiment relation makes clear that, in fact, a fifth variant could be added to Ihde's overview of human-technology relations. In Ihde's range of relations, technology moves ever further away from the human—from being "embodied" to being "read," to being "interacted with" and even to being merely a "background." Yet there is a radical variant of the embodiment relation in which technology is even closer to the human being. In this relation technologies actually *merge* with the human body instead of merely being embodied. Such human-technology relations are usually associated with "bionic" beings or cyborgs: half-organic, half-technological. When microchips are implanted to enhance the vision of visually impaired people, when antidepressants help to change people's moods, or when artificial heart valves and pacemakers help to make people's hearts beat, there is no embodiment relation anymore—at least, not a relation that could compare to wearing eyeglasses or using a telephone. In all these cases there is of course an association of a human being and a technological artifact that experiences reality, but the "bionic" or "cyborg" association actually results in a *new entity*. Instead of organizing an interplay between a human and a nonhuman entity, this association physically alters the human. The resulting "cyborg relation" can be indicated thus:

**Cyborg relation:**                      (human / technology) → world

This fifth technology relation is the basis for what can be called *cyborg intentionality*. And this form of intentionality takes us into the realm of the "transhuman," as set out in the introduction to this book. Instead of being a technologically mediated form of human intentionality, this form of intentionality is located beyond the human being. Just as the "being" that experiences reality under the influence of drugs or sees things with the help of an implanted microchip is not entirely human, neither is the intentionality of this being. To be sure, the intentionality involved in the "common" embodiment relation is not entirely human either: the ways that humans are directed toward each other through a mobile phone or that they hear through a hearing aid, for instance, can exist only by virtue of an intimate association of a

human being and a technological artifact. But in embodiment relations a distinction can still be made between the human and the technological element in the mediated experience, while in cyborg relations this is no longer possible.

While the four human-technology relations in Ihde's approach revolve around technologically *mediated* intentionality, in which both (mediated) human beings and (multistable) technological artifacts are constituted, the notion of cyborg intentionality articulates how human-technology relations can also take on a physical character, forming an actual amalgam of the human and the technological, as is the case when pieces of technology actually merge with the human body. Technologies *used*, like telescopes and hearing aids, help to constitute us as different human beings, whereas technologies *incorporated* constitute new, hybrid beings—which could in principle, to be sure, in their turn also use mediating technologies.

### Composite Intentionality

Beside its mediated and hybrid variants, a third form of technology-shaped intentionality, which can be called *composite intentionality*, deserves a closer analysis. In this variant, there is a central role for the “intentionalities” or directedness of technological artifacts themselves, as they interact with the intentionalities of the human beings using these artifacts. “Technological intentionality” here needs to be understood in terms of both experience and action. It can indicate the way in which technologies can be directed at particular aspects of reality and the “purposiveness” that technologies can embody. In the context of experience, for instance, Ihde elaborated the example of the sound recorder as having a different intentionality toward sound than human beings have, recording background noises at a louder volume than perceived by human beings, who focus only on the sounds that are meaningful to them in a given situation (Ihde 1979, 77–78; Ihde 1983, 56; Ihde 1990, 102–3). When this “directedness” of technological devices is added to human intentionality, a *composite intentionality* comes about: a form of intentionality that results from adding technological intentionality to human intentionality.

Composite intentionality, to be sure, also plays a role in what Ihde calls the hermeneutic relation. After all, hermeneutic relations always involve a technologically generated representation of the world, which inevitably is the product of a technological directedness at the world: thermometers focus on temperature, spectrographs on light frequencies, sonograms on how material objects reflect ultrasound. Yet this *representing* intentionality of “nonhuman” perceivers is only one form of composite intentionality, and a specific one at

that. Not all technological intentionalities are directed at actually representing a phenomenon in the world. Some of them instead *construct* reality, like radio telescopes that produce a visible image of a star on the basis of “seeing” forms of radiation that are not visible to the human eye. In this case, one could say the composition of human intentionality and technological intentionality is directed at making accessible ways in which technologies “experience” the world. Looking at the image of a star through a radio telescope comes down to perceiving how the technology “perceives” and makes visible this star.

The concept of composite intentionality, therefore, urges us to augment Ihde’s analysis of the hermeneutic relation. In some situations of technology use, there is a double intentionality involved: one of the technology toward “its” world, and one of human beings toward the result of this technological intentionality. That is, people’s intentionality is directed here at the ways in which a technology is directed at the world. This implies that if we are to conceptualize the basis for composite intentionality, the dash in Ihde’s schematic depiction of the hermeneutic relation *human* → (*technology*—*world*) should be replaced with an arrow. This gives the following scheme:

***Composite relation:***                      **human → (technology → world)**

One good example to give us a clearer understanding of such composite intentionalities is the way that artists experiment with technological intentionalities (cf. Kockelkoren 2003). Two Dutch artists examined below, for instance, explore new regimes of perception with the help of technologies. Their work investigates and demonstrates the intentionalities of technological artifacts in relation to human intentionality. But rather than putting these intentionalities at the service of human relations to the world—as is the case in Ihde’s hermeneutic relation—they analyze technological intentionalities as relevant *in themselves*. By making accessible technological intentionalities to human intentionality, they aim to reveal a reality that can be experienced only by technologies.

#### AUGMENTED INTENTIONALITY

The night pictures of the Dutch photographer Wouter Hooijmans embody the “mildest” form of composite intentionality. Hooijmans makes landscape photographs with shutter times of several hours. This allows him to make use of starlight for exposing his pictures, to stunning effect. Brief incidents, like animals walking through the scene, movements of the leaves on a tree, rippling of the water in a lake, become irrelevant. Only things that last make

it into the picture. Hooijmans's photographs reveal the world as it would look if we did not need to blink. In a sense, his pictures can be seen as the embodiment of Husserl's method of "essential intuition." By imaginatively transforming a phenomenon in various ways, Husserl wanted to determine which aspects were essential to it and which were not. Hooijmans's images seem to accomplish this not in the realm of ideas but in the materiality of a printed photograph.

Hooijmans's photographs embody an extreme mechanical makeover of the intentionality of human vision. Contrary to the most common use of the camera, Hooijmans does not create instantaneous exposures but, rather, "sustained exposures." His photographs blend an infinite number of visual impressions into one single representation of the world, which the human eye could never produce itself. We could call this form of composite intentionality *augmented intentionality*, since it consists in making accessible to the human eye an artificial form of intentionality that functions as an augmentation of human intentionality.

#### CONSTRUCTIVE INTENTIONALITY

The stereophotographic work of De Realisten (The Realists) exemplifies a second form of composite intentionality. As a part of their work, De Realisten have been making stereographic photographs of several sets of identically shaped objects made out of different nonamalgamating materials like wood and bronze. Looking at these photographs with the help of 3D equipment, one is confronted with highly realistic, three-dimensional representations of a reality that cannot exist in everyday experience.

These photographs do not aim to represent reality in any sense but to generate a new reality that can exist for human intentionality only when it is complemented by technological intentionality. The resulting three-dimensional, photorealistic amalgams have no "original" counterpart in everyday reality. The "intentionality" that De Realisten give to their stereographic camera is not directed at making visible an existing reality but at constructing a *new* reality. For this reason, the intentionality involved here can be called *constructive intentionality*.

#### COMPOSITE MORALITY

Composite intentionalities have moral relevance as well. "Intentional" technologies can play a crucial role in moral actions and decisions. Several technologies discussed in this book are in the realm of constructive intentionality.

Obstetric ultrasound, for instance, mediates moral practices regarding abortion, but it does so on the basis of adding its own intentionality (detecting reflected ultrasound and translating this into a visible image) to the intentionality of expectant parents and healthcare professionals. The intentionality involved in some forms of Persuasive Technology can also be seen as a form of composite intentionality. Here the character of technological intentionality is not hermeneutic-perceptual but pragmatic. Persuasive technologies have a built-in “intention” to influence human behavior, persuading us to change our attitudes, views, and practices. When someone uses the FoodPhone or looks in the Persuasive Mirror—to mention two of the examples discussed in chapter 6—a new intentionality comes about which adds the intentions of the device to the intentions of the user.

The examples of ultrasound and the FoodPhone show that composite intentionalities can have a very diverse character and that the technological intentionalities they involve can embody various degrees of “purposiveness” by the designers. While obstetric ultrasound was clearly not designed to help shape moral practices and decisions, persuasive technologies are designed with the intention of changing human behavior. And while ultrasound primarily adds a hermeneutic intentionality to human intentionality, helping to shape *interpretations* of the unborn, the FoodPhone embodies and brings in the intention to alter its user’s *behavior*. In both cases, technologies set up situations of choice that would not have existed without the technology’s being in place. Composite intentionality thus can take many different forms as technologies add their own implicit or explicit intentionality to the intentions and intentionality of their users.

### Moral Intentions and the Limits of Self-Constitution

These last examples show that hybrid and composite forms of intentionality, just like mediated intentionality, have important moral implications. In the intentionality that comes about in “cyborg relations,” for instance, the human element cannot be separated from the nonhuman. When a person with a neuro implant for deep brain stimulation that reduces severe depression makes the decision to act in a certain way, there is no doubt that the implant has contributed to this decision. Yet the decision that is taken cannot be understood as a technologically mediated decision. Rather than being the product of an *association* of human and nonhuman elements, it is made by a blend of both—it is taken by an agent “beyond the human.” Not only is it impossible to separate humans and technologies here, as is the case in situations of mediated morality, but it is impossible to even *make a distinction* between

them. To use the vocabulary introduced in chapter 2, technologies like neuro implants cross the boundary between posthumanism and transhumanism.

Composite relations find themselves at the limits of the posthumanist ethics that is central in this book. When somebody drives a car with a lane-change advice system that gives a warning in unsafe situations, for instance, the eventual driving behavior is not the result of a technologically *mediated* intentionality but rather of the *composition* of two intentionalities—one of the system, the other of the driver. Actually, the example of obstetric ultrasound needs to be included here as well, albeit as a “limit case.” Ultrasound finds itself at the boundary between what Ihde calls a “hermeneutic” relation and what I have identified as a “composite” relation. While obviously producing a representation of reality—which is characteristic of hermeneutic relations—ultrasound imaging also brings in a programmed “directedness” in how it develops these representations. Measuring the nape of the fetus, for instance, is directed at detecting “defects.” This intentionality is added to the intentions of the expecting parents, when a decision about abortion has to be made. Yet contrary to cyborg relations, in composite relations it is still possible to distinguish between a human and a technological constituent of intentionality. But the role of the technology here reaches further than mediating the relation between humans and reality—it has an intentional character of its own, which also exists apart from the human–technology–world relation in which it has a place.

Instead of embodying a form of intentionality that is distributed among humans and nonhumans, cyborg relations and composite relations involve a clearly nonhuman form of intentionality. For this reason, these additional manifestations of the moral significance of technology push the ethical perspective developed in this book to its limits. The position I defended, after all, focused on the ability to develop a free relation to technological mediation, be it in the form of designing moral mediations or in the form of actively coshaping one’s mediated moral subjectivity. In the two new forms of technological morality, the possibilities for developing such a free relation to technology are more limited—either because they take moral self-constitution beyond the human being or because they seriously mitigate the room for freedom. They mitigate it because a close interaction between human intentionalities and technological intentionalities replaces the phenomenon of technologically mediated intentionality.

This book’s focus on moral self-constitution—consisting of materializing morality and styling technologically mediated subjectivity—appears to encounter its limits, then, at the limits of the human being. The ambition of constituting our own moral subjectivity can ultimately result in merging

ourselves physically with technologies or subjecting ourselves to nonhuman intentionalities. What does this imply for the understanding of the moral significance of technology developed in the previous chapters?

For answering this question, composite intentionalities and cyborg intentionalities require separate analyses. The composite relation, as we saw, can be viewed as an advanced continuation of the hermeneutic and alterity relations, although the representation of the world given by the technology, or the interaction that the technology offers with our actions, now involves a specific *technological* intentionality. In the field of experience this implies that a translation is needed from the technological directedness toward reality into a representation that is accessible to human beings—just like ultrasound scanners “perceive” reflected sound that needs to be translated into a visible image. And in the field of action this implies that human beings have to deal with the “intentions” that were built into the technology—just as the Persuasive Mirror explicitly has the built-in intention to influence people’s behavior in a healthy direction.

All aspects of the analysis developed in previous chapters still apply to such composite relations. By presenting specific representations of reality or offering specific forms of interaction, these technologies help to shape human practices and experiences much as mediating technologies do. There is still an interaction between human and nonhuman elements, after all, that results in moral actions and decisions. And because of this, there are still opportunities to design these intentional technologies responsibly and to develop responsible ways of using them and coshaping one’s subjectivity in conjunction with their impact. The only difference here is the explicit intentionality on the part of the technology, which can have a more profound impact than technological mediation. This is especially true for technologies that develop a clear and intelligent interaction with human behavior, as is the case with Ambient Intelligence and Persuasive Technology. But even in cases where technologies have this profound influence, there is still room for self-constitution in relation to the technology, as the example of Intelligent Speed Adaptation showed (cf. Dorrestijn 2004). Even those technologies whose impact can hardly be avoided can be integrated in various ways in people’s everyday lives.

For cyborg relations, a different analysis applies. Here we move even further away from the situation of moral mediation. Because interaction with a technology is now replaced with the formation of a new moral entity, emerging from an actual fusion of humans and technologies, developing a free relation to these technologies becomes a complicated thing. The entity having freedom is modified in its relations to technology. Cyborg technologies help to shape the nature of the relation humans have with them in a more radical

way than mediating technologies do. Cyborg relations open up new ways we might constitute our subjectivity—but now they are physical and organic ways, which makes the impact of the technology designs less reversible. This raises many new and complicated ethical questions, which are currently being explored in the field of the ethics of “human enhancement” and which focus on issues of human dignity, social justice, and respect for the autonomy of (post)human beings.

An interesting example is the technology of deep brain stimulation (DBS). This technology uses a neuro implant to impart electrical signals directly to someone’s brain, activating or deactivating specific parts of the brain. The technology is currently used to treat a wide variety of diseases, ranging from Parkinson’s disease to depression and obsessive-compulsive disorder. A famous case described in the Dutch medical journal *Tijdschrift voor Geneeskunde* recounts how the condition of a patient suffering from Parkinson’s disease improved markedly after DBS (Leentjens et al. 2004). But while the symptoms of Parkinson’s disease were ameliorated, his behavior also changed, and in uninhibited ways that were completely unfamiliar to his family and friends. He took up with a married woman, bought her a second house and a vacation home abroad, bought several cars, was involved in a number of traffic accidents, and eventually had his driving license taken away. The man had no idea that his behavior had changed—until the DBS was switched off. And at that moment his Parkinson’s symptoms returned with such severity that he became entirely bedridden and dependent. There appeared to be no middle way: he would have to choose between a life with Parkinson’s disease, bedridden, or a life without the symptoms but with such loss of inhibition that he would get himself into continual trouble. Eventually he chose—with the DBS switched off!—to be admitted to a psychiatric hospital, where he could switch the DBS on and suffer fewer symptoms of the disease, but he would also be protected against himself.

This case raises all sorts of issues about freedom and responsibility. The man lived as two parallel personalities and was aware of the fact while in only one of them; moreover, he made the eventual choice to go on living in the one that was *not* aware of it. In circumstances like this it is difficult to judge whether a free choice was possible, or for that matter who the authentic person was who was doing the choosing. The posthumanist ambition of self-constitution via careful design and use of technology, which has played a central role in this book, takes on a radically new shape here.

The examples of Ambient Intelligence and deep brain stimulation show, however, that the limits of the human being are not as clear-cut as they might appear. What we are as human beings, after all, has always taken shape in

relation to the technologies that we use. The new feature of recent developments in biotechnology is not the possibility of reshaping ourselves; rather, what is new is the new media available for our technological self-constitution, and their far-reaching impact. At the same time, though, these new technological media for shaping the subject require their users to develop an active relationship with them. But since these media shape the human subject along different lines from those of mediation, they require a separate analysis of how we might understand their constitutive role in human subjectivity and how human beings could develop an free relation to such technologies (cf. Verbeek 2009b).<sup>2</sup>

## Conclusion: Accompanying Technology

### Introduction

What, then, does the approach of moral mediation developed in this book imply for the ethics of technology?<sup>1</sup> Having elaborated an understanding of the moral significance of technology and its implications for the ethics of design and use, and having discussed the moral roles of technology that go beyond mediation, in this concluding chapter I would like to reflect on the place of the mediation approach in the broader field of the ethics and philosophy of technology.

While many ethical discussions of technology, as indicated in chapter 1, focus on the risks and dangers connected to technology and aim to develop criteria for setting limits to technology, this book has taken the interwoven character of human beings and technological artifacts as a starting point for developing an ethics of technology. Connecting to the postphenomenological approach of technological mediation, I explored various ways in which technologies and morality are closely intertwined and investigated their implications for the ethics of technology design and use. From a mediation perspective, technologies should not primarily be approached as invasive powers in need of ethical limits but as morally significant entities that need to be assessed in terms of the quality of their impact on human existence. Technologies help to shape human actions and decisions by mediating our interpretations of the world and the practices we are involved in. Therefore they play a significant role in human morality.

Approaching technologies as morally relevant entities has important implications for our understanding of central ethical notions like moral agency and responsibility. By approaching agency and responsibility as phenomena that are distributed among human beings and nonhuman entities, ethical

theory can do justice to the hybrid character that many actions and practices have acquired. In fact, the humanist bias that has come to characterize ethical theory needs to be abandoned if theory is to take responsibility for the close connections between technological artifacts on the one hand and human actions and decisions on the other. In the ethical approach laid out in this book, therefore, the autonomous subject of modernist ethical theory is replaced with a technologically mediated subject, and the often-found instrumentalist approach to technology is replaced with an approach that focuses on the actively mediating role of technologies in human actions and perceptions.

On the one hand, this reinterpretation of moral agency and responsibility might seem to limit the possibilities for doing ethics. If technological artifacts play such a crucial role in our moral actions, what good can human beings accomplish? On the other hand, though, expanding ethics to the realm of materiality appears to broaden the locus of ethical activity, moving it from the realm of texts and ideas to that of materiality and design. Technology design can be considered a form of “ethics by other means.” Designers cannot but help to shape human actions and experiences via the technologies they are designing. Therefore, design processes should be equipped with the means to do this in a desirable, morally justifiable, and democratic way. Moreover, the phenomenon of moral mediation appears to make technology use a moral activity. Human beings, after all, are not passively subjected to technological mediations but have the ability to actively co-shape their mediated subjectivity by styling and experimenting with the role of technology in their daily lives.

In order to locate the approach of moral mediation in the context of other approaches to technology, I will follow two lines of inquiry in this chapter. First, I will reflect on the type of ethics that results from the approach developed in this book, and on its ability to guide practices of technology development and use. While I have shown the various ways in which ethics and technology are interwoven, and while I have developed ideas about how to take responsibility for this interwoven character in the ethics of design and use, I have not yet developed a moral framework from which this responsibility could take shape. In this chapter, I will provisionally indicate how an “ethics of the good life” could serve as such a framework. After this, I will discuss how analyzing the moral dimensions of technologies can be situated within the philosophy and ethics of technology, in the context of its development over recent decades.

### Ethics of the Good Life

Taking seriously the moral significance of technologies has proven to be quite a complicated thing in the context of current ethical theory. Blurring boundaries between humans and technology and replacing the autonomous moral subject with a technologically mediated subject at first sight seem to leave us no other option than simply accepting that we are slaves to technology, free at most to display some subversive behavior every now and then. What kind of ethics could possibly be the outcome of such an approach? Should not ethics also have the possibility to say no to technological developments? Can we even talk about ethical limits to technology if our minds and bodies are entirely mediated and directed by that technology?

The key message of this book is that delivering ourselves uncritically to technology is the very last thing we should do. It is quite the other way around. Whoever fails to appreciate how technology and humanity are interwoven with each other loses the possibility of taking responsibility for the quality of this interweaving. There is a complex interplay between humans and technologies within which neither technological development nor humans has autonomy. Human beings are products of technology, just like technology is a product of human beings.

This does not mean that we are hapless victims of technology, but neither does it mean that we should try to escape from its influence. As I concluded in chapter 4, in contrast to such a *dialectic* approach, which sees the relationship between humans and technology in terms of oppression and liberation, we need a *hermeneutic* approach. Within such an approach—hermeneutics is the study of meaning and interpretation—technology forms one of the tissues of meaning within which our existence takes shape. We are as autonomous with regard to technology as we are with regard to language, oxygen, or gravity. It is absurd to think that we can rid ourselves of this dependency, because we would remove ourselves in the process. Technology is part of the human condition. We must learn to live with it—in every sense of the word. In other words, we must shape our existence *in relation to* technology. Putting the borderline between human and technology into perspective therefore does not mean that “anything goes.” On the contrary, it means that the aim of the ethics of technology must be to give shape in a sound and responsible way to the relationship between people and technology.

This is no simple matter, however, as became clear in this book. Many discussions in the ethics of technology are dominated by an “externalistic” approach toward technology. Its basic model is often that there are two spheres of reality, one of humanity and one of technology, and that it is the task

of ethics to ensure that technology does not transgress too far into the human sphere. In this model, ethics is a border guard whose job it is to prevent unwanted invasions. In the light of the analysis of the relationship between humans and technology that I have presented in this book, such a model cannot be adequate. It draws a distinction between a “human” domain and a “technological” domain that is ultimately untenable.

Instead of making ethics a border guard that decides to what extent technological objects may be allowed to enter the world of human subjects, ethics should be directed toward the *quality of the interaction* between humans and technology. This does not mean that every form of such interaction is desirable, nor that we should simply develop technologies at random. The crucial question here is not so much where we have to draw the line—for humans or for technologies—but how we are to best shape the interrelatedness between humans and technology that has always been a hallmark of the human condition. We need an ethics that does not stare obsessively at the issue of whether a given technology is morally acceptable but that looks at the quality of life that is lived with technology.

But how can we address this question adequately? What kind of ethical framework can serve as a basis for assessing the quality of the connections between human beings and technological artifacts in practices of use and of design? Which direction should the deliberate shaping of one’s mediated subjectivity and of mediating technologies take? In line with the amodernist approach to the ethics of technology that I have elaborated in this book, and with Michel Foucault’s reading of classical ethics, I would like to connect here to the ethics of the good life as it took shape in classical antiquity. This ethical approach, of course, is amodernist by definition. At the core of classical ethics is not an autonomous subject asking itself how it should act in a world of objects; the ethics of the good life is centered on the question of “how to live.” And as Gerard de Vries has argued (1999), in our technological culture this question is answered not only by human beings but by technologies as well. Human existence takes shape in relation to technologies—and therefore the question of the good life concerns the quality of the ways in which we live with technology.

A good life in classical Aristotelian ethics was directed by *aretè*—a term frequently translated as “virtue” but which is better rendered by the word *excellence*, I think. Ethics, then, was about excellence in living, or mastering the art of living. In a technological culture, an ethics of the good life is about developing forms of excellence in living with technology. And as discussed in chapter 4, Foucault’s reading of the ethics of the good life offers an especially fruitful basis for a amodern ethics of technology. Foucault succeeded in con-

necting a “mediated” account of human beings with an ethical perspective. His ethical work was directed specifically toward the ethics of sexuality, demonstrating that the ethics of sexuality in classical times did not boil down to adherence to commandments and prohibitions but to finding the best way of dealing with lust and passion. Passions impose themselves on us, so to speak, and ethics was about choosing not to follow these passions blindly but to establish a free relationship with them: finding an *appropriate relation* to the passions.

Connecting to the work of Steven Dorrestijn (Dorrestijn 2006), I have argued that an ethics of technology could look similar. When technologies play a profound role in human existence, then the art of living in a technological culture is not about setting limits to the influence of technology but about the art of shaping our own mediated subjectivity by developing responsible forms of technology design and use. Instead of focusing on whether certain technologies are morally acceptable or not, an ethics of the good life asks itself what a good way of living with such technology could be. In fact, a refusal to take the social impact of technologies seriously marginalizes ethics from the outset. The technological developments themselves continue to move forward, and as long as squeaky-clean ethicists grumble on the sidelines, they are missing the opportunity to contribute to the responsible development and responsible use of these technologies. It is high time that ethics moved on from considering simply whether new technologies are acceptable and started addressing the issue of the best way to embed such technologies in our society.

This approach mirrors the classical Aristotelian principle of finding the right middle ground between two extremes—as, for example, courage is the healthy middle ground between cowardice and recklessness. In the ethics of technology there are two extremes that should be avoided. At one extreme there is the conservative desire to safeguard the boundaries of humanity as we know it and to consider any technological mediations or alterations undesirable intrusions. At the other extreme there is the radical desire to improve humanity as much as we can by means of technology, or even the trans-humanist desire to move beyond the human being. These extremes mirror the modernist separation between subject and object, humans and technologies; while the conservative extreme seems to rely primarily on humanity, the technological extreme puts all its trust in technology.

The ethical position that I have developed in this book can be seen as a middle position between the precautionary ethical framework and its technologically fixed counterpart. Neither safeguarding humanity from technology nor merging humanity and technology is its primary concern. It rather

aims at developing ways to take responsibility for our technologically mediated existence. It wants neither to let technology determine humanity nor to protect humanity against technology; its goal is to develop a free relation to technology by learning to understand its mediating roles and to take these into account when designing, implementing, and using technology.

The appropriate middle way in dealing with technology, therefore, is the development of responsible forms of mediation. The principal question in such an ethics of technology is, what is a good way of living with technologies? When we allow technology to be *accompanied by* this ethical question, instead of setting it *at odds* with ethics, it becomes possible to pose questions about those aspects of human existence that are affected by particular technologies and to decide which considerations might be relevant. To ask such questions, ethics cannot restrict itself to scrutinizing ethical theories but will also need to include empirical accounts of technologically mediated human existence and of the social and cultural impacts of technologies. An ethics of the good life in a technological culture needs to address the specific ways that specific technologies have an impact on specific aspects of human existence. On the basis of such analyses, points of application can be found to address the quality of human-technology relations in practices of use, design, and policy making.

Chapter 6's discussion of Ambient Intelligence and Persuasive Technology is an example of such an analysis, asking the types of questions that need to be addressed in an ethics of the good life. It is not the possible technological intrusions in the human lifeworld that have a central place here, but the character of the new interactions that come about between technologies and human existence. How do new forms of Ambient Intelligence give new shape to human freedom? How does Persuasive Technology affect human morality? How are we to understand responsibility when human actions and technological mediations are closely intertwined? How can we assess the quality of such interactions between human beings and technologies? And how can we help to shape the impacts of such technologies in practices of use, design, and policy making?

The further working out of such an ethics of the good life will be one of the most important challenges for the ethics of technology. In fact, various forms of "good life ethics" are currently being developed in relation to technology. The most well-known position is probably Albert Borgmann's "ethics of engagement." This ethical position articulates an account of the good in terms of engaged practices that compensate for the loss of deep relations with reality that, according to Borgmann, is caused by technology. A second

approach, developed by Philip Brey, understands the good life in terms of *well-being* and focuses on assessing to what extent technologies threaten or foster various aspects of human well-being (cf. Brey 2007). Another route focuses on the *capability approach* as formulated by Martha Nussbaum and Amartya Sen. This approach stresses the necessity of having certain capabilities for living a good life. Ilse Oosterlaken is currently exploring how this approach can be connected to technology, focusing on the role of technologies in the development of specific capabilities in specific contexts (Oosterlaken 2009). A third approach connects to classical *virtue ethics*. In her dissertation “Doing Good with Things,” Katinka Waelbers (2010) puts forward a virtue-ethical approach to technology inspired by the work of Alasdair MacIntyre, proposing a number of criteria that need to be met for living a good life.

These analyses are all very fruitful and important. At the same time, though, they should avoid becoming a form of the externalism that I aim to overcome in this book. If we take the thoroughly mediated character of human existence seriously, an ethics of the good life cannot aim to develop a predefined set of criteria that can guide practices of technology development and use. It rather should continually take into account how technological developments themselves contribute to what could be called a good life, and should recognize that ideas and images of the good life change in interaction with the very technologies that we assess in terms of the good life.

One important requirement here is the development of a public space and discourse to raise questions of the good life in a fruitful way. As Tsjalling Swierstra has argued, in many ethical discussions about technology it is hard to use life-ethical arguments; they hardly have a place in our liberal democracy, where issues of the good life have become a private matter (Swierstra 2002). In the public sphere, we have come to restrict ourselves to determining the rules that enable us to realize our own views of the good life. This model has made possible a plurality of visions of the good life, but at the price of a rather meager public debate in which any argument that reverts to the good life is shunted aside immediately as irrelevant or even unacceptable (Valkenburg 2009).

Without seeking to return to a situation in which the question of the good life is answered by a state or a church, the analysis of moral mediation developed in this book urges us to acknowledge that technologies bring the question of the good life back into the public sphere—for the simple reason that they embody and help to shape visions of the good life. When obstetric ultrasound helps to shift norms and responsibilities regarding unborn children, it urges us to address the question of how we want to deal with unborn life in

the public sphere. When Ambient Intelligence in geriatric homes changes the character of the contact between nurses and elderly people, we need to reconsider how we value personal interaction in practices of care. In order to have a full-blown social discussion about technology, we will need to examine the visions of the good life that are at stake, without affecting their plurality. With a broad discussion about various visions of good human-technology associations, human beings can orient themselves regarding their own choices and decisions, and political decision making and technological practices can look for ways to do justice to this plurality as much as possible.

A second important requisite here is developing an adequate basis for having public debates about technology and the good life. In order to accomplish that, participants in these debates—users, designers, policymakers—need to have the capability to “read” technological developments: to understand the social and cultural roles of technologies beyond the ways in which they fulfill their functions. In order for humans to develop a free relation to technology, citizenship in a technological culture requires the competence to understand how technologies help to shape society, including human practices and experiences, social relations, and normative frameworks. This competence, in its turn, requires not only adequate education, science communication, and science journalism but also adequate investigations of the relations between technology and society—investigations that do not shy away from the normative dimensions of these relations. To accomplish this, as I will demonstrate below, the philosophy of technology needs to integrate empirical and ethical analyses of human-technology relations. I consider this integration to be one of the most important challenges for the philosophy and ethics of technology.

### One More Turn after the Ethical Turn

The approach of moral mediation can be placed in the historical context of the development of the philosophy of technology. In retrospect, one could say that in recent decades the philosophy of technology underwent first an “empirical turn” and then an “ethical turn.” While the empirical turn of the 1990s brought the philosophy of technology into closer contact with actual technologies, it also tended to lose the social and political engagement that characterized early philosophy of technology. The subsequent ethical turn, which largely took place in the 2000s, compensated for this but sometimes at the price of reintroducing a separation between technology and society that the empirical turn—or at least its advocates in the field of science and

technology studies (STS)—aimed to overcome. The approach of moral mediation can be said to integrate both turns, as I will explain below.

#### THE EMPIRICAL TURN

While philosophy of technology in the late 1970s was still largely under the sign of “founding fathers” like Martin Heidegger, Jacques Ellul, and Hans Jonas, in the 1980s and 1990s it made an important shift of direction. Partly because of increasing interaction with the empirical field of science and technology studies, and partly because of increased attention for what Carl Mitcham has called “engineering philosophy of technology” (1994), the focus of attention shifted from “Technology” as a broad social and cultural phenomenon toward actual technologies. Against the rather abstract and pessimistic approaches of what has since come to be called “classical” philosophy of technology (Achterhuis 2001), a more empirically informed style of theorizing came into being out of both the analytic and the continental tradition. This shift has sometimes been called an empirical turn—in analogy to the empirical turn in the philosophy of science (cf. Achterhuis 2001; Kroes and Meijers 2000). Its primary aim was to understand actual technologies both in terms of their nature and structure and in terms of their social, cultural, and ethical implications.

Earlier I argued that this empirical turn constituted a radical shift in approaching technology (Verbeek 2005b). It broke away from the predominant focus on the conditions of technology that characterized the early positions. This classical way of thinking can be called “transcendentalism,” because of its kinship to the transcendental-philosophical focus on understanding phenomena in terms of their conditions of possibility. Rather than concentrating on the technological artifacts themselves and their social and cultural impacts, classical positions tended to reduce those artifacts to their conditions, such as the way of disclosing reality they require (Heidegger) or the system of mass production from which they come, which suffocates authentic human existence (Jaspers).

The empirical turn successfully focused the attention of the philosophy of technology on technologies themselves. It did so in both the analytic and the continental traditions in philosophy. On the one hand, engineering philosophy of technology—with a strong presence of analytic philosophers—saw a rapid expansion, developing itself into a small parallel field to the philosophy of science by further analyzing the nature and structure of technological activities and artifacts and of the engineering sciences (Pitt 2000; Kroes and

Meijers 2000). On the other hand, mirroring what happened in the philosophy of science with the rise of science studies, continentally oriented philosophy of technology was influenced by empirical approaches to the relations between technology and society (Latour 1994; Bijker 1995). In these approaches the interwoven character of technology and society plays a central role, and this had major implications for the ways that philosophy of technology came to approach the phenomena of technology (Ihde 1990; Feenberg 1999).

This empirical turn came at a price, though, because it tended to background the critical and sometimes even activist spirit behind the “classical” positions. The increased focus on actual technologies and their relations to society resulted in what has been called a “descriptivist” approach (Light and Roberts 2000). Both STS and empirically oriented “continental” philosophy of technology, therefore, have been criticized for giving up too much on normative analysis—the work of Andrew Feenberg being among the few exceptions. On the one hand, the focus on empirical studies of the relations between technology and society sometimes became an aim in itself, rather than being an “empirical detour” toward answering broader and more normative questions. And on the other hand, showing the interwoven character of technology and society made it more difficult to take a critical stance toward technology. If society, including its normative frameworks, is a product of technology anyway, there seems to be no position left from which to do ethics.

#### THE ETHICAL TURN

The descriptivist orientation that resulted from the empirical turn was compensated for in the first decade of the twenty-first century, which saw an explosion of ethical approaches to technology. A broad variety of ethical subfields emerged, including nanoethics, ethics of information technology, ethics of biotechnology, ethics of engineering design, and more. This rapid growth of applied ethical approaches to technology can partly be seen as a result of the empirical turn. Rather than criticizing “Technology”—as classical philosophers of technology often did, pointing out its potential threat to “humanity”—ethical reflection started to address actual technologies and technological developments.

An interesting side effect of this development was that an increasing number of philosophers became interested in technology as a subject of philosophical analysis. At the same time, though, this development tended to result in a “forgetfulness” about what had been achieved in the empirical turn—more specifically, in the close relations between philosophy of tech-

nology and STS. Despite its focus on actual technologies rather than “Technology,” the new ethical interest in technology often starts from ethical theories, frameworks, and principles, not from analyses of the complex relations between technology and society, including the interwoven character of technology and morality.

From the point of view of an STS-informed philosophy of technology, ethical analyses of technology that focus on values like privacy, safety, or autonomy run the risk of forgetting that the meaning of those values is narrowly intertwined with the technologies they are used to assess. The meaning and importance of privacy, safety, and autonomy are coshaped by the specific ways technologies put those values at stake. From an empirical-philosophical point of view, ethics cannot occupy an external standpoint with respect to technology from which it could assess technologies in terms of pregiven norms and values. Rather, ethics should be aware of the ways in which it is itself a product of technology.

Taking these close relations between technology and morality into account appeared to be a complicated affair, because blurring the boundaries between morality and technology would seem to make ethical technology assessment virtually impossible. As noted before, if we need to see the ethical frameworks we use as products of the technologies we are supposed to assess with their help, we seem to become mere slaves to technology. The whole point of ethical reflection seems to be to have the possibility of saying no to technology. While STS blurs the boundary between humanity and technology, it makes it impossible for ethics to play the role of a boundary guard, preventing technology from invading too deeply the realm of humanity. Should we, then, simply accept that the primary lesson of the empirical turn is that no boundary can be drawn between humans and technology and that for that reason no ethical relation to technology is possible at all?

In fact, this book has shown that things are *more* complicated than this false dilemma suggests. The empirical turn does not make an ethics of technology impossible at all. Giving up the possibility of a privileged external position for ethics, outside the realm of technology, does not imply that we should give up on ethics altogether. Making visible the close intertwining of technology and humanity, at the levels of both society and people’s individual lives, indeed makes it possible to take responsibility for these intertwine-ments and to give them desirable shapes. We should therefore give the idea that morality and technology are closely interwoven a central place in normative reflection. And this, in my opinion, is exactly the challenge that the philosophy and ethics of technology are facing at the moment. In order to see this interwoven character, we need to integrate the empirical-philosophical

approach to technology with normative reflection. We need to make—in other words, and with a nod to Latour—one more turn after the empirical and the ethical turn (cf. Latour 1992a). This is, in fact, what the approach of moral mediation developed in this book aims to realize.

#### BEYOND THE ETHICAL TURN

In order to make this “third turn,” two lines of research need to be developed further, one descriptive, the other normative. First of all, the moral significance of technologies needs to be scrutinized further—and this book aims to take a step in this direction. The moral significance of technologies is a fruitful terrain for empirical-philosophical analysis and for expanding the connections between philosophy of technology on the one hand and ethics and social and political philosophy on the other.

Second, a normative-ethical line needs to be further constructed, one that does not focus only on *analyzing* the moral dimension of technology but also on *doing* ethics of technology. Such a “doubly turned” ethics of technology, to be sure, can no longer exist within technology *assessment*, at least not in the traditional sense of that word, where assessment requires a standpoint external to what is being assessed. Since such a standpoint does not exist, a better concept to describe ethics is *technology accompaniment*, borrowing a concept from the Belgian philosopher Gilbert Hottois (1996—see chapter 5). The crucial question in such an ethics of accompaniment is not so much where we have to draw a boundary between human beings on the one hand and technologies on the other. It is rather how we should give shape to the inter-relatedness between humans and technology that has in fact always been a central characteristic of human existence.

Such an ethics of “accompanying technology” will revive the engaged and sometimes even activist spirit of the early philosophy of technology, albeit in a somewhat different way. Because of its roots in both the empirical and the ethical turns, the ethics of accompaniment is engaged not only with human beings but with technologies as well. Its central aim is to accompany the development, use, and social embedding of technology—as the various chapters of this book have illustrated.

Accompanying technological developments requires engagement with designers and users, identifying points of application for moral reflection, and anticipating the social impact of technologies-in-design. Rather than placing itself *outside* the realm of technology, an ethics of accompaniment will engage directly with technological developments and their social embedding.

Its primary task is to equip users and designers with adequate frameworks to understand, anticipate, and assess the quality of the social and cultural impacts of technologies. This type of ethics therefore requires an integration of the empirical turn and the ethical turn. On the one hand, it involves a further analysis of the mediating roles of specific technologies in human existence, society, and culture, while on the other hand it requires the development of an ethical relation to these mediations. This integration, in my view, is currently one of the foremost tasks for the philosophy of technology.

### Conclusion: Beyond Protagoras

While ethics and technology intuitively might seem to occupy two separate realms, this book aimed to show how thoroughly they are in fact interwoven. In order to grasp the subtle and complex relations between ethics and technology, we moved beyond the externalist and humanist orientation of many ethical theories. Technologies proved to have moral dimensions, just as ethics proved to have a technological dimension. Technological artifacts play central roles in our moral agency; they help to shape human actions and decisions, and therefore they also help to answer the moral questions of how we ought to act and to live. Our moral standards develop in close interaction with technology. Moral decisions, then, are generally not made by autonomous subjects but are coshaped by the material environment in which humans live.

This moral significance of technology charges us with the task of taking responsibility for it. The ethical challenge that technologies impose upon us is to accompany their development in adequate ways. Rather than placing ethics in opposition to technology, giving ethicists the heroic but often also powerless role of setting limits to the interventions of technologies in the human world, we need to develop vocabularies and practices for shaping our lives in interaction with technologies. In order to develop responsible forms of use and design, we need to equip users and designers with frameworks and methods to anticipate, assess, and design the mediating role of technologies in people's lives and in the ways we organize society.

In Protagoras's time, "Man" was "the measure of all things." But in our technological culture, conversely, ethics cannot avoid the conclusion that things are the measure of all human beings too. Material artifacts, and especially the technological devices that increasingly inhabit the world in which we live, deserve a place at the heart of ethics. Just like human beings, albeit in a different way, they belong to the moral community. Ethicists should therefore

count it among their tasks to make explicit the implicit morality of things and to engage in its design and its roles in society. Indeed, only by recognizing that morality has become a matter of both human subjects and nonhuman objects can ethics continue to play a meaningful role in our technological culture.

## Notes

### Chapter One

1. This chapter incorporates reworked excerpts from Verbeek 2006c, Verbeek 2008d, and Verbeek 2009a.
2. The concepts I use here to discuss the phenomenon of technological mediation are developed in more detail in Verbeek 2005b.
3. Ihde also distinguishes two relations that do not directly concern mediation. First, he identifies the “alterity relation,” in which technologies are the terminus of our experience. This relation—which mirrors Heidegger’s “presence at hand”—occurs when we are interacting with a device as if it were another living being, for instance when buying a train ticket at an automatic ticket dispenser. Second, Ihde discerns the “background relation.” In this relation, technologies play a role at the background of our experience, creating a context for it. An example of this relation is the automatic switching on and off of the refrigerator.
4. “War is a continuation of politics by other means” (Von Clausewitz 1976).

### Chapter Two

1. This chapter incorporates reworked excerpts from Verbeek 2008a (in Jan-Kyrre Berg Olsen, Evan Selinger, and Søren Riis (eds.), *New Waves in Philosophy of Technology*, 2008, Palgrave Macmillan; reproduced with permission of Palgrave Macmillan).
2. Other works of Sloterdijk do pay attention to the (technologically) mediated character of humanity. His discussion of humanism in “Rules for the Human Zoo” could have benefited from including these earlier insights in order to develop a broader approach to the technological “cultivation” of humanity.
3. To be sure, in many cases an ultrasound examination does not provide enough certainty to make such a decision; it makes possible only the calculation of a *risk*, while certainty can be provided only by amniocentesis.

### Chapter Three

1. This chapter incorporates reworked excerpts from Verbeek 2006e and 2008c.
2. Unless indicated otherwise, all translations from Dutch to English are mine.

## Chapter Four

1. This chapter incorporates reworked fragments from Verbeek forthcoming b.

## Chapter Five

1. This chapter incorporates reworked excerpts from Verbeek 2006c, 2008c, and 2009a.
2. I use the concept of accompaniment in a different way than Hottois does. Rather than envisioning a symbolic accompaniment of technological developments, I aim to move beyond the distinction between these two spheres and to give ethics the role of reflecting on technocultural developments and developing with it.
3. For a closer analysis of behavior-steering technologies, see Verbeek and Slob 2006.
4. To be sure, this asymmetry is unintended. Implicitly, Latour does discuss delegations that go the other way around: from nonhumans to humans. For instance, in Latour 1992b he expresses admiration for a hydraulic door-closing device, because it easily absorbs the energy of those who open the door, retains it, and then gives it back slowly “with a subtle type of implacable firmness that one could expect from a well-trained butler” (233). This door-closing device delegates to people the delivery of the energy it needs to close the door after it has been opened. Latour’s focus on delegation and inscription remains remarkable, though. If we are to understand the ways in which artifacts mediate, it does not matter all that much how they came to do so. What is important is that they play mediating roles, and the most relevant question for an analysis of technical mediation is *how* they do this. Focusing on the *origins* of these mediating roles of things could be seen as a relic from the early days in science and technology studies (STS), when the ambition was to show that “facts” or “technologies” are actually contingent outcomes of processes of construction in which many actors interact. This deconstructionist approach aimed to unravel how entities come to be what they are. An analysis of the mediating role of artifacts can take for granted the constructed character of this role, however. For the understanding of technical mediation, the inscription processes and delegations from humans to nonhumans may remain black-boxed. Only the mediating role itself is relevant here, not where it comes from.
5. For an extensive analysis of the possible roles of imagination in moral reasoning, see Coeckelbergh 2007.
6. For a more detailed discussion of the work of *Eternally Yours*, see Verbeek and Kockelkoren 1998; Verbeek 2005b, 203–36.
7. The scenario concept can also indicate a specific image of the future, for which design activities can be developed. Klapwijk et al. (2006), for instance, use the concept of “design orienting scenarios” to formulate images of sustainable future practices and follow a back-casting approach to inform design activities in the present. Also in these scenarios, interactions between technologies and human behavior play an important role.
8. The points of application for moral reflection identified here are inspired by, yet different from, Daniel Berdichevsky and Erik Neuenschwander’s approach to the ethics of persuasive technology (Berdichevsky and Neuenschwander 1999), as I will show more extensively in chapter 6. They also mirror the distinctions made by B. J. Fogg between intentions, methods, and outcomes as three categories of ethical issues regarding persuasive technologies (Fogg 2003, 220).
9. Contrary to the approach Aaron Smith takes in Smith 2003.
10. See <http://en.wikipedia.org/wiki/Makkinga> (last visited March 31, 2010).

11. For a detailed analysis of an alternative approach to road safety, see Popkema and Van Schagen 2006.

12. Rinsing is needed only if the machine will be used a few days after loading it, e.g., when people want to wait until the dishwasher is full before running it.

### Chapter Six

1. This chapter incorporates reworked excerpts from Verbeek forthcoming a.

2. This method is also discussed in Fogg 2003, 233–35.

3. Anyone searching the web for references to “e-mail overload” will find myriad examples.

### Chapter Seven

1. This chapter incorporates reworked excerpts from Verbeek 2008b.

2. In my current research project on human-technology relations and philosophical anthropology, these questions play a central role.

### Chapter Eight

1. This chapter incorporates reworked excerpts from Verbeek 2010 and Verbeek forthcoming b.



## References

- Aarts, E. H. L., R. Harwig, and M. F. H. Schuurmans. 2001. Ambient Intelligence. In P. J. Denning (ed.), *The Invisible Future: The Seamless Integration of Technology in Everyday Life*, 235–50. New York: McGraw-Hill.
- Aarts, E., and S. Marzano. 2003. *The New Everyday: Views on Ambient Intelligence*. Rotterdam: 010.
- Achterhuis, H. 1995. “De moralisering van de apparaten.” *Socialisme en Democratie* 52 (1): 3–12.
- . 1998. *De erfenis van de utopie*. Amsterdam: Ambo.
- . 2001. *American Philosophy of Technology: The Empirical Turn*. Translated by R. Crease. Bloomington: Indiana University Press.
- Adviesdienst Verkeer en Vervoer. 2001. *ISA Tilburg: Intelligente Snelheids Aanpassing in de praktijk getest*. Eindrapportage praktijkproef Intelligente Snelheidsaanpassing. The Hague: Ministerie van Verkeer en Waterstaat.
- Akkerman, S. 2002. “Zorg met de dingen.” MA thesis, University of Twente, Enschede, Netherlands.
- Akrich, M. 1992. “The Description of Technical Objects.” In W. E. Bijker and J. Law (eds.), *Shaping Technology / Building Society*, 205–24. Cambridge, MA: MIT Press.
- Anders, G. 1988. *Die Antiquiertheit des Menschen*. Munich: C. H. Beck.
- Aristotle. 1970. *Physics*. Cambridge, MA: Harvard University Press.
- Baudet, H. 1986. *Een vertrouwde wereld: 100 jaar innovatie in Nederland*. Amsterdam: Bert Bakker.
- Berdichevsky, D., and E. Neuenschwander. 1999. “Toward an Ethics of Persuasive Technology.” *Communications of the Association for Computing Machinery (ACM)* 42 (5): 51–58.
- Berlin, I. 1979. “Two Concepts of Liberty.” In *Four Essays on Liberty*, 118–72. Oxford: Oxford University Press.
- Bernauer, J., and M. Mahon. 2005. “Michel Foucault’s Ethical Imagination.” In Gary Gutting (ed.), *The Cambridge Companion to Foucault*. Cambridge: Cambridge University Press.
- Bijker, W. E. 1995. *Of Bicycles, Bakelites and Bulbs: Toward a Theory of Sociotechnical Change*. Cambridge, MA: MIT Press.
- Birsch, D., and J. Fielder. 1994. *The Ford Pinto Case*. Albany: State University of New York Press.

- Boenink, M. 2007. "Genetic Diagnostics for Hereditary Breast Cancer: Displacement of Uncertainty and Responsibility." In Gerard de Vries and Klasien Horstman, eds., *Genetics from the Laboratory to Society*. Houndmills, Basingstoke, UK: Palgrave/Macmillan, 2007.
- Bohn, J., et al. 2004. "Living in a World of Smart Everyday Objects: Social, Economic, and Ethical Implications." *Journal of Human and Ecological Risk Assessment* 10 (5): 763–86.
- Borgmann, A. 1984. *Technology and the Character of Contemporary Life*. Chicago: University of Chicago Press.
- . 1992. *Crossing the Postmodern Divide*. Chicago: University of Chicago Press.
- . 1995. "The Moral Significance of the Material Culture." In A. Feenberg and A. Hannay (eds.), *Technology and the Politics of Knowledge*, 85–93. Bloomington: Indiana University Press.
- . 2006. *Real American Ethics: Taking Responsibility for Our Country*. Chicago: University of Chicago Press.
- Bostrom, N. 2004. "The Future of Human Evolution." In C. Tandy (ed.), *Death and Anti-death: Two Hundred Years after Kant, Fifty Years after Turing*, 339–371. Palo Alto, CA: Ria University Press.
- Boucher, J. 2004. "Ultrasound—a Window to the Womb? Obstetric Ultrasound and the Abortion Rights Debate." *Journal of Medical Humanities* 25 (1): 7–19.
- Brey, P. 2005. "Freedom and Privacy in Ambient Intelligence." *Ethics and Information Technology* 7 (3): 157–66.
- . 2006. "Ethical Aspects of Behavior Steering Technology." In P. P. Verbeek and A. Slob (eds.), *User Behavior and Technology Development*, 357–64. Dordrecht, Netherlands: Springer.
- . 2007. "Theorizing the Cultural Quality of New Media." *Techné: Research in Philosophy and Technology* 11 (1): 1–18.
- Carroll, J. M. 2000. *Making Use: Scenario-Based Design of Human-Computer Interactions*. Cambridge, MA: MIT Press.
- Casert, R. 2004. "Ambient Intelligence: In the Service of Man?" Verslag Workshop. The Hague: Rathenau Instituut.
- Coeckelbergh, M. 2006. "Regulation or Responsibility? Autonomy, Moral Imagination, and Engineering." *Science, Technology and Human Values* 31 (3): 237–60.
- . 2007. *Imagination and Principles: An Essay on the Role of Imagination in Moral Reasoning*. Houndmills, Basingstoke, UK: Palgrave/Macmillan.
- Crystal, D. 2008. "2b or Not 2b?" *Guardian*, July 5, 2008.
- Deleuze, G. 1986. *Foucault*. London: Athlone.
- De Mul, J. 2002. *Cyberspace Odyssee*. Kampen, Netherlands: Klement.
- De Vries, G. 1993. *Gerede twijfel: Over de rol van de medische ethiek in Nederland*. Amsterdam: De Balie.
- . 1999. *Zeppelins: Over filosofie, technologie en cultuur*. Amsterdam: Van Gennep.
- Dorrestijn, S. 2004. "Bestaanskunst in de technologische cultuur: Over de ethiek van door techniek beïnvloed gedrag." MA thesis, University of Twente, Enschede, Netherlands.
- . 2006. "Michel Foucault et l'éthique des techniques: Le cas de la RFID." MA thesis, Université Paris X, Nanterre.
- Feenberg, A. 1999. *Questioning Technology*. London: Routledge.
- Floridi, L., and J. W. Sanders. 2004. "On the Morality of Artificial Agents." *Minds and Machines* 14 (3): 349–79.
- Fogg, B. J. 2003. *Persuasive Technology: Using Computers to Change What We Think and Do*. Amsterdam: Morgan Kaufmann.

- Foucault, M. 1975. *Discipline and Punish: The Birth of the Prison*. New York: Random House.
- . 1984a. *Le souci de soi*. Paris: Gallimard.
- . 1984b. *L'usage des plaisirs*. Paris: Gallimard.
- . [1984] 1990. *The Care of the Self*, vol. 3 of *The History of Sexuality*. London: Penguin.
- . [1984] 1992. *The Use of Pleasure*, vol. 2 of *The History of Sexuality*. London: Penguin.
- . 1997a. *Ethics: Subjectivity and Truth*. Edited by P. Rabinow. New York: New Press.
- . 1997b. "What Is Enlightenment?" In M. Foucault, *Ethics: Subjectivity and Truth*, edited by Paul Rabinow. New York: New Press.
- Friedman, B., ed. 1997. *Human Values and the Design of Computer Technology*. Chicago: University of Chicago Press.
- Friedman, B., P. Kahn, and A. Borning. 2002. *Value Sensitive Design: Theory and Methods*. Computer Science and Engineering Technical Report 02-12-01. Seattle: University of Washington Press.
- Fryslân Province. 2005. *Shared Space—Room for Everyone: A New Vision for Public Spaces*. Leeuwarden, Netherlands: Fryslân Province.
- Gerrie, J. 2003. "Was Foucault a Philosopher of Technology?" *Techné: Research in Philosophy and Technology* 7 (2): 14–26.
- Habermas, J. 1969. *Technik und Wissenschaft als Ideologie*. Frankfurt: Suhrkamp.
- . 1984. *The Theory of Communicative Action*. Boston: Beacon.
- Haraway, D. 1991. "A Cyborg Manifesto: Science, Technology, and Socialist-Feminism in the Late Twentieth Century." In D. Haraway (ed.), *Simians, Cyborgs and Women: The Reinvention of Nature*, 149–81. New York: Routledge.
- Harbers, H. 2005. "Epilogue: Political Materials—Material Politics." In H. Harbers (ed.), *Inside the Politics of Technology*. Amsterdam: Amsterdam University Press.
- Hayles, K. 1999. *How We Became Posthuman*. Chicago: University of Chicago Press.
- Heidegger, M. 1927. *Sein und Zeit*. Tübingen, Germany: Max Niemeyer Verlag.
- . 1951. "Das Ding." In *Vorträge und Aufsätze*. Pfullingen, Germany: Neske.
- . 1969. *Discourse on Thinking*. Translated by J. M. Anderson and E. H. Freund. New York: Harper and Row.
- . [1947] 1976. "Brief über den Humanismus." In *Wegmarken*, complete ed., 9: 313–64. Frankfurt am Main: Klostermann.
- . 1977a. "The Age of the World Picture." In *The Question concerning Technology and Other Essays*. New York: Harper and Row. Translation of "Die Zeit des Weltbildes," in *Holzwege* (Frankfurt am Main: Vittorio Klostermann, 1950).
- . 1977b. "The Question concerning Technology." In *The Question concerning Technology and Other Essays*, trans. W. Lovitt. New York: Harper and Row.
- Hottois, G. 1996. *Symbool en techniek*. Kampen, Netherlands: Kok Agora/Kapellen, Netherlands: Pelckmans.
- Ihde, D. 1979. *Technics and Praxis*. Dordrecht, Netherlands: Reidel.
- . 1983. *Existential Technics*. Albany: State University of New York Press.
- . 1990. *Technology and the Lifeworld*. Bloomington: Indiana University Press.
- . 1993. *Postphenomenology*. Evanston, IL: Northwestern University Press.
- . 1998. *Expanding Hermeneutics*. Evanston, IL: Northwestern University Press.
- Illies, C. F. R., and A. W. M. Meijers. 2009. "Artefacts without Agency." *Monist* 92 (3): 420–40.
- Introna, L. 2005. "Disclosive Ethics and Information Technology: Disclosing Facial Recognition Systems." *Ethics and Information Technology* 7 (2005): 75–86.
- Irrgang, B. 2005. *Posthumanes Menschsein*. Stuttgart: Franz Steiner Verlag.

- ISTAG (Information Society Technologies Advisory Group, European Commission). 2003. *Ambient Intelligence: From Vision to Reality*. Brussels: European Commission.
- Jaspers, K. 1951. *Man in the Modern Age*. Translated by E. Paul and C. Paul. London: Routledge and Kegan Paul.
- Jelsma, J. 1999. "Huishoudelijk energiegebruik: Van beter gedrag naar beter ontwerpen. Een aanzet tot een integrale benadering." Utrecht: NOVEM Gammaprogramma.
- . 2006. "Designing 'Moralized' Products: Theory and Practice." In P. P. Verbeek and A. Slob (eds.), *User Behavior and Technology Development: Shaping Sustainable Relations between Consumers and Technologies*. Dordrecht, Netherlands: Springer.
- Joerges, B. 1999. "Do Politics Have Artefacts?" *Social Studies of Science* 29 (3): 411–31.
- Kant, I. [1785] 2002. *Groundwork for the Metaphysics of Morals*. New Haven, CT: Yale University Press.
- Klapwijk, R., et al. 2006. "Using Design Orienting Scenarios to Analyze the Interaction between Technology, Behavior and Environment in the SusHouse Project." In P. P. Verbeek and A. Slob (eds.), *User Behavior and Technology Development: Shaping Sustainable Relations between Consumers and Technologies*. Dordrecht: Springer.
- Knight, W. 2005. "Mirror That Reflects Your Future Self." *New Scientist*, no. 2485 (February 5, 2005): 23.
- Kockelkoren, P. 2003. *Technology: Art, Fairground and Theatre*. Rotterdam: NAI (Netherlands Architecture Institute).
- Kroes, P., and A. Meijers, eds. 2000. *The Empirical Turn in the Philosophy of Technology*. Amsterdam: JAI.
- Kuijk, L. 2004. "Prenataal Onderzoek: Abortus als logisch vervolg." *Trouw* (Amsterdam), January 3, 2004.
- Landsman, G. H. 1998. "Reconstructing Motherhood in the Age of 'Perfect' Babies: Mothers of Infants and Toddlers with Disabilities." *Signs* 24 (1): 69–99.
- Latour, B. 1988. "Veiligheidsgordel: De verloren massa van de moraliteit." In M. Schwartz and R. Jansma (eds.), *De technologische cultuur*. Amsterdam: De Balie.
- . 1992a. "One More Turn after the Social Turn: Easing Science Studies into the Amodern World." In Ernan McMullin (ed.), *The Social Dimensions of Science*, 272–92. Notre Dame, IN: Notre Dame University Press.
- . 1992b. "Where Are the Missing Masses? The Sociology of a Few Mundane Artifacts." In W. E. Bijker and J. Law (eds.), *Shaping Technology/Building Society*. Cambridge, MA: MIT Press.
- . 1993. *We Have Never Been Modern*. Translated by C. Porter. Cambridge, MA: Harvard University Press. Translation of *Nous n'avons jamais été modernes* (Paris: La Découverte, 1991).
- . 1994. "On Technical Mediation: Philosophy, Sociology, Genealogy." *Common Knowledge* 3: 29–64.
- . 1997. *De Berlijnse sleutel*. Amsterdam: Van Gennep.
- . 1999. *Pandora's Hope*. Cambridge, MA: Harvard University Press.
- . 2002. "Morality and Technology: The End of the Means." *Theory, Culture and Society* 19 (5–6): 247–60.
- . 2004. *Politics of Nature*. Cambridge, MA: Harvard University Press.
- . 2005. "From Realpolitik to Dingpolitik, or How to Make Things Public." In B. Latour and P. Weibel (eds.), *Making Things Public: Atmospheres of Democracy*, 4–31. Cambridge, MA: MIT Press.

- Leentjens, A. F. G., et al. 2004. "Manipuleerbare wilsbekwaamheid: Een ethisch probleem bij elektrostimulatie van de nucleus subthalamicus voor ernstige ziekte van Parkinson." *Nederlands Tijdschrift voor Geneeskunde* 148: 1394–98.
- Lemmens, P. 2008. *Gedreven door techniek: De menselijke conditie en de biotechnologische revolutie*. Oisterwijk, Netherlands: Box Press.
- Light, A., and D. Roberts. 2000. "Toward New Foundations in Philosophy of Technology: Mitcham and Wittgenstein on Descriptions." *Research in Philosophy and Technology* 19: 125–47.
- Lyon, D., ed. 2006. *Theorizing Surveillance: The Panopticon and Beyond*. Devon, UK: Willan.
- Magnani, L. 2007. *Morality in a Technological World: Knowledge as Duty*. Cambridge: Cambridge University Press.
- May, T. 2006. *The Philosophy of Foucault*. Montreal: McGill-Queen's University Press.
- McCalley, L. T., and C. J. H. Midden. 2006. "Making Energy Feedback Work: Goal-Setting and the Roles of Attention and Minimal Justification." In P. P. Verbeek and A. Slob (eds.), *User Behavior and Technology Development: Shaping Sustainable Relations between Consumers and Technologies*. Dordrecht, Netherlands: Springer.
- McWorther, L. 2003. "Subjecting Dasein." In A. Milchman and A. Rosenberg (eds.), *Foucault and Heidegger: Critical Encounters*, 110–26. Minneapolis: University of Minnesota Press.
- Merleau-Ponty, M. 1962. *Phenomenology of Perception*. Translated by Colin Smith. London: Routledge and Kegan Paul.
- Midden, C. 2006. "Sustainable Technology or Sustainable Users?" In P. P. Verbeek and A. Slob (eds.), *User Behavior and Technology Development: Shaping Sustainable Relations between Consumers and Technologies*, 191–200. Dordrecht, Netherlands: Springer.
- Mitcham, C. 1994. *Thinking through Technology: The Path between Engineering and Philosophy*. Chicago: University of Chicago Press.
- Mitchell, L. 2001. *Baby's First Picture: Ultrasound and the Politics of Fetal Subjects*. Toronto: University of Toronto Press.
- Mol, A. 1997. "Wat is kiezen? Een empirisch-filosofische verkenning." Inaugural lecture presented at Universiteit Twente, Enschede, Netherlands.
- Muis, H. 2006. "Eternally Yours: Some Theory and Practice on Cultural Sustainable Products." In P. P. Verbeek and A. Slob (eds.), *User Behavior and Technology Development: Shaping Sustainable Relations between Consumers and Technologies*, 277–93. Dordrecht, Netherlands: Springer.
- Nietzsche, F. [1883] 1969. *Thus Spoke Zarathustra: A Book for Everyone and No One*. Translated by R. J. Hollingdale. London: Penguin.
- Oaks, L. 2000. "Smoke-Filled Wombs and Fragile Fetuses: The Social Politics of Fetal Representation." *Signs* 26 (1): 63–108.
- O'Leary, T. 2002. *Foucault: The Art of Ethics*. London: Continuum.
- Oosterlaken, I. 2009. "Design for Development: A Capability Approach." *Design Issues* 25 (4): 92–102.
- Petchesky, R. P. 1987. "Fetal Images: The Power of Visual Culture in the Politics of Reproduction." *Feminist Studies* 13 (2): 263–92.
- Pitt, J. 2000. *Thinking about Technology*. New York: Seven Bridges.
- Popkema, M., and Schagen, I. van. 2006. "Modifying Behaviour by Smart Design: The Example of the Dutch Sustainable Safe Road System." In P. P. Verbeek and A. Slob (eds.), *User Behavior and Technology Development: Shaping Sustainable Relations between Consumers and Technologies*. Dordrecht, Netherlands: Springer.

- Rajchman, J. 1985. *Michel Foucault: The Freedom of Philosophy*. New York: Columbia University Press.
- Rapp, R. 1998. "Refusing Prenatal Diagnosis: The Meanings of Bioscience in a Multicultural World." *Science, Technology, and Human Values* 23 (1): 45–70.
- Rip, A., T. Misa, and J. Schot, eds. 1995. *Managing Technology in Society: The Approach of Constructive Technology Assessment*. London: Pinter.
- Rosson, M. B., and J. M. Carroll. 2001. *Usability Engineering: Scenario-Based Development of Human-Computer Interaction*. San Francisco: Morgan Kaufmann.
- Safranski, R. 1999. *Martin Heidegger: Between Good and Evil*. Cambridge, MA: Harvard University Press.
- Sandelowski, M. 1994. "Separate, but Less Unequal: Fetal Ultrasonography and the Transformation of Expectant Mother/Fatherhood." *Gender and Society* 8 (2): 230–45.
- Sawicki, J. 2003. "Heidegger and Foucault: Escaping Technological Nihilism." In A. Milchman and A. Rosenberg (eds.), *Foucault and Heidegger: Critical Encounters*, 55–73. Minneapolis: University of Minnesota Press.
- Schmid, W. 1991. *Auf der Suche nach einer neuen Lebenskunst: Die Frage nach dem Grund und die Neubegründung der Ethik bei Foucault*. Frankfurt: Suhrkamp.
- . 1998. *Philosophie der Lebenskunst. Eine Grundlegung*. Frankfurt: Suhrkamp.
- Schot, J. 1992. "Constructive Technology Assessment and Technology Dynamics: The Case of Clean Technologies." *Science, Technology and Human Values* 17 (1): 36–56.
- Schuurman, J., et al. 2007. *Ambient Intelligence: Toekomst van de zorg of zorg van de toekomst?* The Hague: Rathenau Instituut.
- Searle, J. R. 1983. *Intentionality: An Essay in the Philosophy of Mind*. Cambridge: Cambridge University Press.
- Slob, A., and P. P. Verbeek. 2006. "Technology and User Behavior: An Introduction." In P. P. Verbeek and A. Slob (eds.), *User Behavior and Technology Development: Shaping Sustainable Relations between Consumers and Technologies*, 3–12. Dordrecht, Netherlands: Springer.
- Slob, A. F. L., et al. 1996. *Consumption and the Environment: Analysis of Trends*. Report. TNO Knowledge for Business, University of Utrecht, CBS (Centraal Bureau voor de Statistiek).
- Sloterdijk, P. 1999. *Regeln für den Menschenpark ein Antwortschreiben zu Heideggers Brief über den Humanismus*. Frankfurt am Main: Suhrkamp.
- . 2009. "Rules for the Human Zoo: A Response to the Letter on Humanism." *Environment and Planning D: Society and Space* 27: 12–28.
- Smith, A. 2003. "Do You Believe in Ethics? Latour and Ihde in the Trenches of the Science Wars (or Watch Out, Latour, Ihde's Got a Gun)." In D. Ihde and E. Selinger (eds.), *Chasing Technoscience: Matrix for Materiality*. Bloomington: Indiana University Press.
- Steg, L. 1999. *Verspilde energie? Wat doen en laten Nederlanders voor het milieu*. SCP Cahier 156. The Hague: Sociaal en Cultureel Planbureau.
- Stiegler, B. 1998. *Technics and Time*, vol. 1, *The Fault of Epimetheus*. Stanford, CA: Stanford University Press.
- Stormer, N. 2000. "Prenatal Space." *Signs* 26 (1): 109–44.
- Swierstra, Tsj. 1997. "From Critique to Responsibility." *Techné: Research in Philosophy and Technology* 3 (1): 68–74.
- . 1999. "Moeten artefacten moreel gerehabiliteerd?" *K&M: Tijdschrift voor empirische filosofie* 4 (1999): 317–26.
- . 2002. "Moral Vocabularies and Public Debate: The Cases of Cloning and New Reproductive Technologies." In T. E. Swierstra, J. Keulartz, J. M. Korthals, and M. Schermer

- (eds.), *Pragmatist Ethics for a Technological Culture*, 223–40. Deventer, Netherlands: Kluwer Academic.
- Tenner, E. 1996. *Why Things Bite Back: Technology and the Revenge of Unintended Consequences*. New York: Vintage Books.
- Thiele, Leslie Paul. 2003. "The Ethics and Politics of Narrative: Heidegger + Foucault." In A. Milchman and A. Rosenberg (eds.), *Foucault and Heidegger: Critical Encounters*, 206–34. Minneapolis: University of Minnesota Press.
- Tideman, M. 2008. "Scenario Based Product Design." PhD diss., University of Twente, Netherlands.
- Trouw (Amsterdam). 2006. "Abortus om hazenlip komt voor: Onderzoeken of echo's leiden tot meer zwangerschapsafbrekingen." 11 (December 2006).
- Turnage, A. K. 2007. "Email Flaming Behaviors and Organizational Conflict." *Journal of Computer-Mediated Communication* 13 (1): 43–59.
- Valkenburg, G. 2009. *Politics by All Means: An Enquiry into Technological Liberalism*. Simon Stevin Series in Philosophy of Technology. Delft: 3TU. Ethics.
- Van Dijk, P. 2000. *Anthropology in the Age of Technology: The Philosophical Contribution of Günther Anders*. Value Inquiry Book Series 103. Amsterdam: Rodopi.
- Van Hinte, E., ed. 1997. *Eternally Yours: Visions on Product Endurance*. Rotterdam: 010.
- , ed. 2004. *Eternally Yours: Time in Design*. Rotterdam: 010.
- Van Kesteren, N., R. M. Meertens, and M. Fransen. 2006. "Technological Innovations and Energy Conservation: Satisfaction with and Effectiveness of an In-Business Control." In P. P. Verbeek and A. Slob. (eds.), *User Behavior and Technology Development: Shaping Sustainable Relations between Consumers and Technologies*. Dordrecht, Netherlands: Springer.
- Verbeek, P. P. 2000. *De daadkracht der dingen: Over techniek, filosofie en vormgeving*. Amsterdam: Boom.
- . 2002. "Pragmatism and Pragmata: Bioethics and the Technological Mediation of Experience." In J. Keulartz et al. (eds.), *Pragmatist Ethics for a Technological Culture*. Dordrecht, Netherlands: Kluwer.
- . 2004a. "Material Morality." In Ed van Hinte (ed.), *Time in Design*, 198–210. Rotterdam: 010.
- . 2004b. "Stimuleer gedragsbeïnvloedende technologie." *Christen Democratische Verkenningen* 3 (2004): 117–24.
- . 2005a. "De materialiteit van de moraal." *Algemeen Nederlands Tijdschrift voor Wijsbegeerte* 2 (2005): 139–45.
- . 2005b. *What Things Do: Philosophical Reflections on Technology, Agency, and Design*. University Park: Pennsylvania State University Press. (Translation of Verbeek 2000.)
- . 2006a. "Acting Artifacts: The Technological Mediation of Action." In P. P. Verbeek and A. Slob (eds.), *User Behavior and Technology Development: Shaping Sustainable Relations between Consumers and Technologies*. Dordrecht, Netherlands: Springer.
- . 2006b. "Ethiek en technologie: Moreel actorschap en subjectiviteit in een technologische cultuur." *Ethische Perspectieven* 16 (3): 267–89.
- . 2006c. "Materializing Morality: Design Ethics and Technological Mediation." "Ethics and Engineering Design," special issue of *Science, Technology and Human Values* 31 (3): 361–80.
- . 2006d. "Moraliteit voorbij de mens: Over de mogelijkheden van een posthumanistische ethiek." *Krisis* 1 (2006): 42–57.
- . 2006e. "The Morality of Things: A Postphenomenological Inquiry." In E. Selinger

- (ed.), *Postphenomenology: A Critical Companion to Ihde*. New York: State University of New York Press.
- . 2006f. "Persuasive Technology and Moral Responsibility." Paper presented at Persuasive Technology 2006 conference, Eindhoven University of Technology, The Netherlands.
- . 2008a. "Cultivating Humanity: Toward a Non-humanist Ethics of Technology." In Jan-Kyrre Berg Olsen, Evan Selinger, and Søren Riis (eds.), *New Waves in Philosophy of Technology*, 241–66. Houndsmill, Basingstoke, UK: Palgrave/Macmillan.
- . 2008b. "Cyborg Intentionality: Rethinking the Phenomenology of Human-Technology Relations." *Phenomenology and the Cognitive Sciences* 7 (3): 387–95.
- . 2008c. "Morality in Design: Design Ethics and the Morality of Technological Artifacts." In Pieter E. Vermaas, Peter Kroes, Andrew Light, and Steven A. Moore (eds.), *Philosophy and Design: From Engineering to Architecture*, 91–103. Dordrecht, Netherlands: Springer.
- . 2008d. "Obstetric Ultrasound and the Technological Mediation of Morality: A Post-phenomenological Analysis." *Human Studies* 1 (2008): 11–26.
- . 2009a. "The Moral Relevance of Technological Artifacts." In M. Duwell et al. (eds.), *Evaluating New Technologies*, pp. 63–77. Dordrecht, Netherlands: Springer.
- . 2009b. "Technology and the Limits of Humanity: On Technology, Ethics, and Human Nature." Inaugural address presented at University of Twente.
- . 2010. "Accompanying Technology: Philosophy of Technology after the Ethical Turn." *Techné: Research in Philosophy and Technology* 14 (1): 49–54.
- . Forthcoming a. "Persuasive Technology," in H. Zwart et al. (eds.), *Encyclopedia of Applied Ethics*, 2nd ed., Elsevier.
- . Forthcoming b. "Subject to Technology." In Mireille Hildebrandt and Antoinette Rouvroy (eds.), *The Philosophy of Law Meets the Philosophy of Technology: Autonomic Computing and Transformations of Human Agency*. Oxford: Routledge.
- Verbeek, P. P., and A. Slob, eds. 2006. *User Behavior and Technology Design: Shaping Sustainable Relations between Consumers and Technologies*. Dordrecht, Netherlands: Kluwer.
- Verbeek, P. P., and P. Kockelkoren. 1998. "The Things That Matter." *Design Issues* 14 (3): 28–42.
- Von Clausewitz, C. 1976. *On War*. Edited and translated by Michael Howard and Peter Paret. Princeton, NJ: Princeton University Press.
- Waelbers, K. 2010. "Doing Good with Things: Taking Responsibility for the Social Role of Things." PhD diss., University of Twente, Enschede, Netherlands.
- Weegink, R. J. 1996. *Basisonderzoek elektriciteitsverbruik kleinverbruikers BEK '95*. Arnhem, Netherlands: EnergieNed.
- Wehrens, R. 2007. *De Gebruiker Centraal? Een inventarisatie van gebruikersgericht onderzoek op het gebied van Ambient Intelligence en gezondheid*. The Hague: Rathenau Instituut.
- Weiser, M. 1991. "The Computer for the 21st Century." *Scientific American* 265 (3): 94–104.
- Winner, L. 1986. "Do Artifacts Have Politics?" In *The Whale and the Reactor*. Chicago: University of Chicago Press.
- Woolgar, S., and G. Cooper. 1999. "Do Artefacts Have Ambivalence?" *Social Studies of Science* 29 (3): 433–49.
- Wolters, G. W., and L. P. A. Steenbekkers. 2006. "The Scenario Method to Gain Insight into User Actions." In P. P. Verbeek and A. Slob (eds.), *User Behavior and Technology Development*. Dordrecht, Netherlands: Springer.
- Zechmeister, I. 2001. "Foetal Images: The Power of Visual Technology in Antenatal Care and the Implications for Women's Reproductive Freedom." *Health Care Analysis* 9: 387–400.

## Index

- Aarts, Emile, 121  
abortion, 62, 63  
ABS (antilock braking system), 121  
accompanying technology, 95  
Achterhuis, Hans, viii, 2, 50, 95, 161  
action: moral, 13, 38, 88; program of, 10  
Adolph, Sven, 101, 102  
aesthetics, 75–76  
agency, 49, 99, 124; artificial, 50–51; moral, 18, 33, 38, 52–55, 60, 66, 87, 108, 154  
Akkerman, Sijas, 1  
Akrich, Madeleine, 98, 114  
ambient intelligence, 19, 120  
amodern perspective, 28, 36, 39, 64, 88, 156  
amplification, 9  
Anders, Günther, 32  
anesthesia, 88  
*animalitas*, 33  
animal rationale, 33  
animism, 64  
anticipation, 96, 97, 104, 129  
applied ethics, 105  
Aristotle, 79, 157  
artificial intelligence, 61  
ascesis, 78, 135  
assessment, 129; moral, 117  
authenticity, 14, 142  
autonomy, 13, 38, 43–44, 49, 59, 66, 74, 79, 81, 82, 85, 111, 115, 123, 130, 133, 135, 136, 151, 163, 165  
*autopoiesis*, 75  
availability, 47  
  
Baby Think It Over, 123  
background relation, 143  
Baudet, Henri, 57  
  
Beaufret, Jean, 22  
behavior-influencing technologies, 133  
Bentham, Jeremy, 62, 69  
Berdichevsky, Daniel, 125–28  
Berlin, Isaiah, 72, 110  
Bernauer, James, 73  
Big Brother, 138  
Bijker, Wiebe, 162  
biotechnology, 35  
Birsch, Douglas, 4  
birth control pill, 3  
black box, 103  
Bohn, Jürgen, 121  
border guard, 156  
Borgmann, Albert, 6, 7, 18, 47, 48, 52, 53, 137, 158  
Borning, Alan, 114, 115  
Boucher, Joanne, 24, 25, 26  
boundary guard, 163  
breast cancer, 5, 57  
breeding, 36, 39  
Brey, Philip, 110, 111, 121, 159  
Bruulsema, Petra, ix  
  
capability approach, 159  
car, 12, 126, 132, 135, 149, 151  
care of the self, 75, 83  
Carroll, John M., 104  
categorical imperative, 62, 74  
causal responsibility, 42  
cell phone, 4, 132  
Challenger, 4  
chastity, 74  
choice, 25  
citizenship, 160  
Clausewitz, Carl von, 19

- code, moral, 78, 135  
 Coeckelbergh, Mark, 128  
 coming-into-being, 76  
 commodity, 47  
 “communitarian phantasm,” 34  
 community, moral, 41–42  
 composite relation, 140, 146, 149  
 compulsion, 109  
*conceptual analysis*, 117  
 congenital defect, 38  
 conscience, 137  
 consequentialism, 30, 31, 41, 62  
 “craft,” 76  
 “critique,” by Kant, 80  
 Crystal, David, 4  
 Constructive Technology Assessment, 103–4, 112–13, 116–17, 133  
 cyborg, 140, 143, 144  
 cyborg relation, 144, 148–51
- decision, moral, 45, 108  
 deep brain stimulation, 140, 151  
 delegation, 95  
 Deleuze, Gilles, 79  
 deliberation, moral, 107  
 democracy, 106, 112, 159  
 De Mul, Jos, 140  
 deontology, 30, 41, 42, 61, 85  
 Descartes, René, 29  
 descriptivism, 162  
 design, 40, 84, 90, 104, 114, 129, 150, 154  
 determination, 97, 141  
 device, 48  
 de Vries, Gerard, 31, 63, 156  
 dialectic, 72, 155  
 directedness, 55  
*Discipline and Punish* (Foucault), 66, 68  
 divine law, 135  
 domination, 80, 87  
 Dorrestijn, Steven, ix, 73, 82, 135, 150, 157  
 Down syndrome, 23, 25, 27, 32
- EconoMeter, 124, 126, 127, 132  
 Ellul, Jacques, 161  
 e-mail, 136  
 embodiment, 8, 142–44  
 emergent impacts, 44  
 empirical turn, 20, 160–65  
 energy consumption, 93  
 energy-saving lightbulb, 132  
 engagement, 47, 101, 158  
 engineering ethics, 161  
 Enlightenment, 12, 21, 45, 68, 74, 80, 81  
 environment, 102  
 Eternally Yours, 99–102, 107
- ethical substance, 77, 134  
 ethical turn, 20, 160, 164, 165  
 ethics: of design, 128–38; evolutionary, 102; of the good life, 19, 154, 156, 158; of use, 133
- Feenberg, Andrew, 162  
 Fielder, John, 4  
 Floridi, Luciano, 17, 49, 51  
 Fogg, B. J., 122, 123  
 FoodPhone, 123, 126, 129, 131  
 Ford Pinto, 4  
 Foucault, Michel, ix, 17, 18, 27, 66–89, 125, 141, 156  
 Franssen, Mirjam, 93  
 freedom, 43, 58–60, 67, 85, 87, 96, 106, 109, 133, 136, 141, 150  
 Friedman, Batya, 113–15  
 functionality, 91
- Gelassenheit*, 84  
 genetics, 35, 57  
 genomics, 140  
 Gerrie, Jim, 67, 68, 69  
 good life, 31, 48, 53, 67, 85, 112, 154, 156, 158  
 Greek ethics, 18, 28
- Habermas, Jürgen, 80  
 Haraway, Donna, 140  
 Harbers, Hans, 32  
 harelip, 27  
 Hayles, N. Katherine, 140  
 hedonism, 31, 63  
 Heersmink, Richard, ix  
 Heidegger, Martin, 3, 7–8, 15–17, 22–23, 28–35, 68–71, 76, 84, 161  
 hermeneutics, 6, 8, 53, 55, 68, 143, 145, 148, 155  
 heteronomy, 66  
*History of Sexuality* (Foucault), 74  
 Hölderlin, Friedrich, 71  
*Homo Sapiens*, 36, 37  
 Hooijmans, Wouter, 146, 147  
 Hottois, Gilbert, 95, 164  
 humanism, 21, 22, 27, 28, 31, 32, 34, 35, 36, 37, 41, 154  
*humanitas*, 33  
 humans, 14, 35, 60, 155, 156  
 human-technology relation, 124  
 Husserl, Edmund, 15, 147  
 hybrid, 16, 56, 58, 140  
 Hygiene-Guard, 123
- Ihde, Don, 6–10, 24, 70, 141–46, 149, 162  
 Illies, Christian, 139  
 imagination, 99, 100, 104  
 infrared camera, 9  
 inscription, 98, 113; moral, 113

- instrumentalism, 88; moral, 50
- Intelligent Speed Adaptation, 135, 150
- intention, 45, 56, 87, 148; intended effects, 127;
  - intended mediation, 106; intended persuasion, 130
- intentionality, 13–16, 42, 43, 54, 57, 58, 64, 140–44;
  - composite, 143, 145, 146, 147, 150; constructive, 147; hybrid, 58; technological, 9, 55, 58, 145
- interactivity, 49
- Introna, Lucas, 108
- Irrgang, Bernhard, 140
- Jaspers, Karl, 3
- Jelsma, Jaap, 93, 113–14
- Joerges, Bernward, 43–44
- Jonas, Hans, 161
- Joplin, Janis, 110
- Kahn, Peter, 114–15
- kaloskagathos*, 75
- Kant, Immanuel, 12, 30, 39, 61, 74, 77, 80
- Knight, W., 112
- Kockelkoren, Petran, ix, 146
- Kroes, Peter, 161
- Landsman, Gail H., 25
- Latour, Bruno, 1, 6–8, 10–14, 17, 22, 28–30, 45–47,
  - 52–54, 64, 98, 113, 119, 164
- Leentjens, A. F. G., 151
- legitimacy, 131, 133
- Lemmens, Pieter, 36
- liberal democracy, 159
- Light, Andrew, 12, 162
- limit attitude, 80
- limits, of technology, 153
- lingual media, 37
- longevity, 101
- Lyon, David, 71
- Macintyre, Alasdair, 159
- Magnani, Lorenzo, 53, 54
- Mahon, Michael, 73
- Makkinga, 110
- Manzini, Ezio, 100
- Marxist dialectic, 72
- Marzano, Stefano, 121
- mastectomy, 108
- materiality, 63–65
- May, Todd, 77, 78
- McCalley, Teddy, 83
- McWorther, Ladelle, 70, 71
- media: lingual, 37; material, 37
- mediation, 7, 11–15, 20, 40, 46, 56, 59, 70, 94, 100,
  - 104–6, 117, 129, 158; mediation analysis, 130; moral, 50, 52, 53, 54, 116, 139
- medical technologies, 5
- Meertens, Ree M., 93
- Meijers, Anthonie, 139, 161, 162
- Merleau-Ponty, Maurice, 15
- metaphysics, 22, 28, 29
- Midden, Cees, 83
- Mirror, Persuasive, 129
- midwife, 26
- Misa, Tom, 102
- Mitchell, Lisa, 25
- mode of subjection, 83
- modernism, 22, 23, 28, 39, 118
- modernity, 22, 28, 80–81
- Mol, Annemarie, 3
- morality, infrastructure for, 40
- moralization, of technology, viii, 95–99, 105–6,
  - 109–13, 117
- moral significance, 47, 48, 67
- moral subject, 71, 78
- moral substance, 79, 86
- Moses, Robert, 5, 43, 44, 50
- MRI (magnetic resonance imaging), 9
- Muis, Henk, 100
- multistability, 9, 97, 127
- Nazism, 34, 36
- networks, 33, 45
- Neuenschwander, Erik, 125–28
- neuro implant, 148
- nexus, 103
- Nietzsche, Friedrich, 35, 36, 72
- nonhumanism, 17, 19, 21, 23, 33, 36, 37
- nonhumans, 14, 17, 21, 39, 45, 46, 53, 61, 62, 98,
  - 103, 140
- Nussbaum, Martha, 159
- Oaks, Laury, 26
- object, concept of the, 17, 29, 30, 39, 157
- objectivity, 16, 37
- obligations, moral, 77
- O’Leary, Timothy, 75–79, 85, 134
- Olesen, Finn, ix
- ontology, 30, 35, 47
- Oosterlaken, Ilse, 159
- overpasses, 5
- ozone hole, 29
- perception, concept of, 15
- persuasion, 83, 107, 126, 128, 130; methods of, 130
- Persuasive Mirror, 129
- Persuasive Technology, 19, 95, 120, 122, 123, 125, 148
- phenomenology, 7, 14, 16, 17, 53
- Pitt, Joseph, 161
- Plato, 36
- plurality, 160

- politics, 12, 43–44, 160, 164  
 posthumanism, 23, 34, 36, 37, 139, 140, 142, 149, 151  
 postphenomenology, 7, 14–17, 24, 54, 56, 125  
 power, 3, 17–18, 33, 66–74, 78–85, 134–37, 153, 165  
 “present-at-hand,” 7  
 privacy, 115, 163  
 Protagoras, 165  
 public debate, 160  
 “purposiveness,” 145  
  
 quality of life, 156  
  
 Rajchman, John, 71  
 Rapp, Rayna, 86  
 “readiness-to-hand,” 7  
 Realisten, De, 147  
 “rebound effect,” 93  
 reduction, 9  
 “releasement,” 71, 84  
 reliability, 131, 132  
 representation, of reality, 28, 142  
*res cogitans*, 29, 37, 38  
*res extensa*, 29, 32, 37, 38  
 responsibility, 32, 64, 96, 107–8, 111, 124, 131–32, 154–55; moral, 49, 108  
*res publica*, 113, 119  
 RFID, 122  
 Rip, Arie, 102  
 risk, 32, 94, 128  
 Roberts, David, 12, 162  
 Rosson, Mary Beth, 104  
  
 safety, 163  
 Safranski, Rüdiger, 22  
 Sandelowski, Margarete, 24, 25, 26  
 Sanders, J. W., 49, 51  
 Sartre, Jean-Paul, 22, 33, 34  
 Sawicki, Jana, 67, 69, 71  
 scenario, 104–5, 117  
 Schmid, Wilhelm, 66  
 Schot, Johan, 102  
 Schuurman, Jan Gerrit, 122  
 science communication, 160  
 science journalism, 160  
 script, 10, 56, 98, 114, 117  
 Searle, John, 57  
 seduction, 83, 107, 126  
 self-constitution, 151, 152  
 self-mastery, 79  
 self practices, 75, 79, 84, 86, 134, 135  
 sexuality, 75, 76  
 Slob, Adriaan, 92, 93  
 Sloterdijk, Peter, 17, 22, 23, 33, 35, 36, 37, 39  
 smart environment, 19, 121  
 smart home, 121  
 Smith, Aaron, 16  
  
 Smits, Sigrid, 101  
 Sophists, 122  
 speed bump, 12, 47, 52, 98  
 speed limitation, 108  
 spina bifida, 32  
 stakeholder analysis, 106, 107, 116, 129  
 stakeholder method, 105  
 Steenbekkers, Wolter, 104  
 Steg, Linda, 58, 98  
 Stormer, Nathan, 26  
 STS (Science and Technology Studies), 163  
 subject, 17, 28, 29, 30, 31, 39, 40, 155, 157; moral, 71–78, 81–89, 141, 149, 155; subject constitution, 17, 73  
 subjection, mode of, 83  
 subjectivity, 16, 37, 152  
 super-ego, 46  
 Swierstra, Tsjalling, 3, 18, 41, 159  
  
 “taming,” of humanity, 38, 39  
*technè*, 75, 76; *technè tou biou*, 76  
 technocracy, 91, 96, 97  
 Technology Assessment, 20, 102, 164  
 teen pregnancy, 123  
 teleology, 77, 78, 79, 84, 134, 136  
 Tenner, Edward, 51, 93  
 thermometer, 8, 56  
 Thiele, Leslie Paul, 73  
 Tideman, Martijn, 104, 105  
 tools, 7  
 transcendentalism, 161  
 transhumanism, 36, 139, 140, 149  
 translations, of action, 10  
 Turnage, Anna K., 136  
  
*Übermensch*, 33  
 ultrasound (obstetric), vii, 9, 16, 24–27, 32, 37–39, 51, 61, 83–86, 148–49  
 under-ego, 47  
 unintended effects, of technologies, 127  
 “user logic,” 114  
 utilitarianism, 62, 74; act, 31; hedonist, 31; pluralist, 31; preferential, 31; rule, 31  
  
 Valkenburg, Govert, 159  
 value-sensitive design (VSD), 114, 117  
 Van Dijk, Paul, 32  
 van Kesteren, Nicole, 93  
 Van Hinte, Ed, 100  
 Verbeek, Peter-Paul, 1, 7, 18, 92, 93, 127, 137, 152, 161  
 virtual reality, 99, 105  
 virtue ethics, 31, 48, 61, 63  
  
 Waelbers, Katinka, 159  
 Weegink, R. J., 58, 98

- Wehrens, R., 121, 122  
Weiser, Mark, 121  
well-being, 159  
wheelchair, 98  
whistleblowing, 19  
Winner, Langdon, 5, 6, 7, 12, 17, 43, 45, 50  
Wolters, G. W., 104  
Woolgar, Steve, 44  
Zarathustra, 72  
Zechmeister, Ingrid, 25, 26  
*zoon logon echon*, 33

